

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397329
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	McDonnell; Joseph

Drain cleaner apparatus

Abstract

An apparatus configured to dispense a cleaner composition includes an apparatus outlet in fluid communication with an exterior of the apparatus, a dispenser device configured to be operated to selectively dispense an amount of the cleaning composition through the apparatus outlet, and a controller configured to operate the dispenser device to cause the amount of the cleaning composition to be dispensed through the apparatus outlet without manual intervention. The apparatus is configured to be electrically connected with a float switch concurrently with dispensing the cleaning composition, and the apparatus is configured to cause at least a portion of an air conditioning system to shut down in response to receiving a signal generated by the float switch.

Inventors:	McDonnell; Joseph (Miramar, FL)
Applicant:	IFLO LLC (Miramar, FL)
Family ID:	1000008777777
Assignee:	IFLO LLC (Miramar, FL)
Appl. No.:	18/739822
Filed:	June 11, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240335866 A1	Oct. 10, 2024

Related U.S. Application Data

continuation parent-doc US 18510138 20231115 US 12030097 child-doc US 18739822
continuation parent-doc US 17944662 20220914 US 11826799 20231128 child-doc US 18510138
continuation-in-part parent-doc US 17723795 20220419 US 11719464 20230808 child-doc US

Publication Classification

Int. Cl.: **B08B9/032** (20060101); **F24F13/22** (20060101); B08B9/027 (20060101); C02F1/00 (20230101); C02F1/68 (20230101); C02F103/02 (20060101); F24F11/58 (20180101); F24F11/89 (20180101); H01H35/18 (20060101)

U.S. Cl.:

CPC **B08B9/0325** (20130101); **F24F13/222** (20130101); B08B9/027 (20130101); B08B9/032 (20130101); B08B2209/032 (20130101); C02F1/008 (20130101); C02F1/686 (20130101); C02F2103/023 (20130101); C02F2201/006 (20130101); C02F2209/008 (20130101); C02F2303/20 (20130101); F24F11/58 (20180101); F24F11/89 (20180101); F24F2013/227 (20130101); F24F2013/228 (20130101); F24F2221/225 (20130101); H01H35/18 (20130101)

Field of Classification Search

CPC: B08B (9/0325); B08B (2209/032); F24F (13/222); F24F (2013/227); F24F (2013/228); F24F (2221/225); C02F (1/008); C02F (1/686); C02F (2103/023); C02F (2201/006); C02F (2209/008); H01H (35/18)

USPC: 134/57R; 137/238; 137/487.5; 222/325

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3525426	12/1969	Miller	N/A	N/A
3823427	12/1973	Pittet	N/A	N/A
4053955	12/1976	Canham	N/A	N/A
4059858	12/1976	Lambel et al.	N/A	N/A
4376321	12/1982	Dudley	N/A	N/A
4426003	12/1983	Zarov	N/A	N/A
4573029	12/1985	Naimer et al.	N/A	N/A
4626951	12/1985	Nagao	N/A	N/A
4962778	12/1989	Driskill	N/A	N/A
5402813	12/1994	Keen	N/A	N/A
5514344	12/1995	D'Agaro	N/A	N/A
5947432	12/1998	Reddin et al.	N/A	N/A
5996597	12/1998	Gallo	N/A	N/A
6041611	12/1999	Palmer	N/A	N/A
6105638	12/1999	Edwards et al.	N/A	N/A
6487867	12/2001	Herren	N/A	N/A
6526601	12/2002	Kou Hsiao	N/A	N/A
6892907	12/2004	Varney	N/A	N/A
7392658	12/2007	Hardy, III	N/A	N/A

7686034	12/2009	Coogle	N/A	N/A
8277745	12/2011	Mehus et al.	N/A	N/A
8840729	12/2013	Herren et al.	N/A	N/A
9289803	12/2015	Parks	N/A	N/A
9943778	12/2017	Gutierrez et al.	N/A	N/A
9975152	12/2017	McClarren	N/A	N/A
10006656	12/2017	Parks	N/A	N/A
10246358	12/2018	Lohse	N/A	N/A
10514182	12/2018	Oviedo	N/A	N/A
10557644	12/2019	Steele	N/A	N/A
10610907	12/2019	Scaringe et al.	N/A	N/A
D902583	12/2019	Thomas et al.	N/A	N/A
11285520	12/2021	McDonald et al.	N/A	N/A
11320172	12/2021	Walker, Jr.	N/A	N/A
12083238	12/2023	Thompson	N/A	N/A
2003/0010381	12/2002	Kimbrough et al.	N/A	N/A
2005/0269418	12/2004	Fuller	N/A	N/A
2006/0038863	12/2005	Piccinino et al.	N/A	N/A
2006/0222508	12/2005	Cantolino	N/A	N/A
2006/0237806	12/2005	Martin et al.	N/A	N/A
2007/0224050	12/2006	Ward	N/A	N/A
2008/0276359	12/2007	Morgan et al.	N/A	N/A
2009/0049592	12/2008	Kipp	N/A	N/A
2009/0064698	12/2008	Spanger	N/A	N/A
2009/0165868	12/2008	Pearson	N/A	N/A
2010/0147876	12/2009	Mehus et al.	N/A	N/A
2013/0146094	12/2012	Riefenstein et al.	N/A	N/A
2013/0239315	12/2012	Kipp	N/A	N/A
2014/0075967	12/2013	Arensmeier et al.	N/A	N/A
2014/0109988	12/2013	Oakner et al.	N/A	N/A
2014/0130888	12/2013	Kaiser et al.	N/A	N/A
2015/0048034	12/2014	Reinolds	N/A	N/A
2015/0153055	12/2014	Oakner et al.	N/A	N/A
2015/0323241	12/2014	Oakner et al.	N/A	N/A
2016/0325981	12/2015	Diaz et al.	N/A	N/A
2017/0045282	12/2016	Thornberry, Jr.	N/A	N/A
2017/0130443	12/2016	Mills	N/A	N/A
2018/0112889	12/2017	Barker et al.	N/A	N/A
2018/0230689	12/2017	Schaafsma	N/A	N/A
2018/0259222	12/2017	Murphy	N/A	N/A
2019/0337027	12/2018	Raynes, II	N/A	N/A
2019/0353367	12/2018	Atchison	N/A	N/A
2020/0181898	12/2019	Huitron	N/A	N/A
2020/0254495	12/2019	Rechlin	N/A	N/A
2020/0261949	12/2019	Hollars et al.	N/A	N/A
2020/0269289	12/2019	Hollars et al.	N/A	N/A
2020/0370772	12/2019	Inoue et al.	N/A	N/A
2021/0239356	12/2020	Thompson	N/A	N/A
2021/0341173	12/2020	Akiyama et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202019010560	12/2019	BR	N/A
2675090	12/2010	CA	N/A
2995074	12/2017	CA	N/A
206793163	12/2016	CN	N/A
207306840	12/2017	CN	N/A
208329123	12/2018	CN	N/A
215654197	12/2021	CN	N/A
3831381	12/1989	DE	N/A
2281112	12/1994	GB	N/A
2398849	12/2003	GB	N/A
20210140916	12/2020	KR	N/A
2011/075324	12/2010	WO	N/A
2012/164289	12/2011	WO	N/A

OTHER PUBLICATIONS

Office Action dated Oct. 11, 2022 issued in corresponding U.S. Appl. No. 17/723,795. cited by applicant

U.S. Notice of Allowance dated Feb. 3, 2023 for corresponding U.S. Appl. No. 17/723,795. cited by applicant

Partial International Search Report and Provisional Written Opinion and Invitation to Pay Additional Fees dated Feb. 2, 2023 for corresponding International Application No. PCT/US2022/078747. cited by applicant

International Search Report and Written Opinion thereof dated Mar. 23, 2023 for corresponding International Application No. PCT/US2022/078747. cited by applicant

U.S. Office Action dated Apr. 28, 2023 for corresponding U.S. Appl. No. 17/944,662. cited by applicant

Extended European Search Report dated Jun. 5, 2023 for corresponding European Application No. 22204020.6. cited by applicant

U.S. Notice of Allowance dated Jun. 16, 2023 for corresponding U.S. Appl. No. 17/723,795. cited by applicant

U.S. Office Action dated Sep. 6, 2023 for corresponding U.S. Appl. No. 17/944,662. cited by applicant

U.S. Notice of Allowance dated Oct. 16, 2023 for corresponding U.S. Appl. No. 17/944,662. cited by applicant

U.S. Office Action dated Feb. 15, 2024 for corresponding U.S. Appl. No. 18/510,138. cited by applicant

U.S. Notice of Allowance dated Mar. 6, 2024 for corresponding U.S. Appl. No. 18/360,173. cited by applicant

U.S. Notice of Allowance dated Apr. 2, 2024 for corresponding U.S. Appl. No. 18/510,138. cited by applicant

“Kollea Reliable Automatic Watering System, Plant Self Watering System Automatic Drip Irrigation Kit with 60-Day Programmable Timer, LED Display & USB Power, Indoor Irrigation System for Potted Plants, and Selected Customer Reviews,” webpage <<https://a.co/d/iCKKroZ>>, 17 pages, available on Internet as of at least Dec. 21, 2020, retrieved from Internet on May 7, 2024. cited by applicant

International Preliminary Report on Patentability dated May 2, 2024 for corresponding International Application No. PCT/US2022/078747. cited by applicant

U.S. Office Action dated Oct. 9, 2024 for corresponding Design U.S. Appl. No. 29/871,185. cited by applicant
Drain Line Cleaner, posted Jun. 2, 2017 [online], [retrieved Sep. 23, 2024], Retrieved from internet, <https://www.amazon.com/SDNselect-Drain-Cleaner-Condensate-Lines/dp/B072C3PYFB> (Year: 2017). cited by applicant
Mini Pump, posted Mar. 16, 2019 [online], [retrieved Sep. 23, 2024], Retrieved from internet, <https://www.amazon.com/Condensate-FUJIWARA-Condensation-Automatic-Dehumidifier/dp/B07PR365TV> (Year: 2019). cited by applicant
Request for Ex Parte Reexamination of U.S. Pat. No. 11,719,464, filed on Nov. 21, 2024 in U.S. Appl. No. 90/019,736. cited by applicant
U.S. Notice of Allowance dated Jan. 22, 2025 for corresponding U.S. Appl. No. 29/871,185. cited by applicant
U.S. Office Action dated Mar. 31, 2025 for corresponding U.S. Appl. No. 90/019,736. cited by applicant
Notice of Acceptance for Corresponding United Arab Emirates Application No. P6002172/2022 dated Mar. 25, 2025. cited by applicant
Notice of Acceptance for Corresponding United Arab Emirates Application No. P6000292/2024 dated May 13, 2025. cited by applicant
U.S. Office Action dated Jun. 17, 2025 for corresponding U.S. Appl. No. 90/019,736. cited by applicant

Primary Examiner: Le; Minh Q

Attorney, Agent or Firm: Harness, Dickey & Pierce, P.L.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/510,138, filed in the United States Patent and Trademark Office on Nov. 15, 2023, which is a continuation of U.S. patent application Ser. No. 17/944,662, filed in the United States Patent and Trademark Office on Sep. 14, 2022, which is a continuation in part of U.S. patent application Ser. No. 17/723,795 filed in the United States Patent and Trademark Office on Apr. 19, 2022, which claims priority to and the benefit of U.S. Provisional Patent Application No. 63/277,323 filed in the United States Patent and Trademark Office on Nov. 9, 2021, the entire contents of each of which are incorporated herein by reference.

BACKGROUND

Field

(1) The present disclosure relates generally to air-conditioning systems, and more particularly to providing cleaner chemical compositions into condensate drain lines of air handlers of air-conditioning systems without manual intervention.

Description of Related Art

(2) Air-conditioning systems may include an air handler, also referred to as an air handling unit (AHU) that may circulate and cool air within a space and/or structure. An air handler may move air, via operation of an air mover such as a blower or fan, to flow in thermal communication with a heat exchanger such as an air coil. The air handler may circulate a refrigerant through the heat exchanger to absorb (e.g., remove) heat from the flow of air to cool the air, and the air-conditioning system may circulate the refrigerant through a heat exchanger to discharge the absorbed heat into a heat sink (e.g., the ambient environment).

(3) In some cases, cooling air due to the heat exchanger absorbing heat from the air may result in condensation of moisture (e.g., condensate) out of the cooled air at the heat exchanger. The condensate may be collected and discharged from the air handler via a condensate drain line.

SUMMARY

(4) According to some example embodiments, a drain cleaner apparatus for dispensing a cleaning composition into a condensate drain line of an air handler of an air conditioning system may include an apparatus outlet in fluid communication with an exterior of the drain cleaner apparatus, a dispenser device configured to be actuated to selectively dispense an amount of the cleaning composition through the apparatus outlet, a connector interface configured to detachably couple with a complementary connector interface of a cartridge having a cartridge reservoir configured to hold the cleaning composition, to establish flow communication from the cartridge reservoir to the dispenser device, such that the dispenser device is between the connector interface and the apparatus outlet and the dispenser device is configured to be actuated to selectively dispense the amount of the cleaning composition from the cartridge reservoir and through the apparatus outlet, and a controller configured to actuate the dispenser device to cause the amount of the cleaning composition to be dispensed through the apparatus outlet without manual intervention.

(5) The connector interface of the drain cleaner apparatus or the complementary connector interface of the cartridge may include a check valve that is configured to open in response to the connector interface of the drain cleaner apparatus coupling with the complementary connector interface of the cartridge to establish the fluid communication between the cartridge reservoir and the dispenser device.

(6) The dispenser device may include at least one valve that is configured to be selectively opened based on a control signal generated by the controller to establish a flow path through the at least one valve to the apparatus outlet. The drain cleaner apparatus may include a dispenser reservoir that is between the check valve and the at least one valve, such that the connector interface is configured to detachably couple with the complementary connector interface of the cartridge to establish flow communication from the cartridge reservoir to the dispenser reservoir, and the dispenser device is configured to be actuated to selectively dispense the amount of the cleaning composition from the dispenser reservoir and through the apparatus outlet. The controller may be configured to actuate the dispenser device based on causing the at least one valve to open the flow path to enable at least a portion of the cleaning composition held in the dispenser reservoir to flow from the dispenser reservoir to the apparatus outlet.

(7) The drain cleaner apparatus may further include a structure connector that is configured to removably couple with an outer housing of the drain cleaner apparatus, the structure connector configured to connect the drain cleaner apparatus to an external structure to at least partially hold the drain cleaner apparatus in place in relation to an opening of the condensate drain line.

(8) The structure connector may include a magnet configured to magnetically attach the structure connector to a metal surface of the external structure.

(9) The controller may be configured to actuate the dispenser device in response to an elapse of a particular period of time.

(10) The controller may be configured to repeatedly actuate the dispenser device at a fixed time interval that is the particular period of time, based on monitoring a timer that increments a timer value at a fixed frequency, actuating the dispenser device in response to the timer value reaching a particular time value corresponding to the elapse of the particular period of time, and resetting the timer value to an initial timer value in response to actuating the dispenser device.

(11) The controller may be configured to monitor a counter that increments a counter value in response to each actuation of the dispenser device, and generate a depletion signal in response to the counter value reaching a particular counter value that corresponds to at least partial depletion of a fixed reservoir of the cleaning composition.

(12) The controller may be configured to determine a volume of the cartridge reservoir in response

to receiving a command signal indicating the volume of the cartridge reservoir, and adjust the particular counter value based on the determination of the volume of the cartridge reservoir.

(13) The drain cleaner apparatus may further include a network communication interface device that is configured to establish a network communication link with a remote computing device. The controller may be configured to cause the depletion signal to be transmitted to the remote computing device via the network communication link.

(14) The drain cleaner apparatus may further include a network communication interface device that is configured to establish a network communication link with a remote computing device. The controller may be configured to cause the counter value to be reset to an initial counter value in response to receiving a reset signal from the remote computing device via the network communication link.

(15) The drain cleaner apparatus may further include a network communication interface device that is configured to establish a network communication link with a remote computing device. The controller may be configured to cause the air conditioning system to shut down, in response to receiving a shutdown command signal from the remote computing device via the network communication link.

(16) The drain cleaner apparatus may be configured to cause at least a portion of the air conditioning system to shut down in response to receiving a signal generated by a float switch apparatus.

(17) The drain cleaner apparatus may be configured to cause a float switch of the air handler to actuate to cause at least the portion of the air conditioning system to shut down in response to receiving the signal generated by the float switch apparatus.

(18) The drain cleaner apparatus may be configured to actuate an actuator to cause the float switch of the air handler to actuate.

(19) According to some example embodiments, a system may be configured to control dispensation of a cleaning composition into a condensate drain line of an air handler of an air conditioning system, where the air handler includes an air handler float switch, where the air handler is configured to shut down in response to actuation of the air handler float switch. The system may include the drain cleaner apparatus, and a float switch apparatus configured to be coupled to the condensate drain line. The float switch apparatus may be configured to enable the drain cleaner apparatus to supply the cleaning composition into the condensate drain line. The float switch apparatus may include a drain cleaner float switch. The drain cleaner float switch may be configured to be electrically coupled to the drain cleaner apparatus such that the drain cleaner float switch is configured to transmit a float switch signal to the drain cleaner apparatus in response to a presence of fluid in the condensate drain line. The controller of the drain cleaner apparatus may be configured to transmit an electrical signal to the air handler to cause at least a portion of the air conditioning system to shut down in response to receiving the float switch signal from the drain cleaner float switch.

(20) According to some example embodiments, a system may be configured to control dispensation of a cleaning composition into a condensate drain line of an air handler of an air conditioning system, where the air handler includes an air handler float switch, where the air handler is configured to shut down in response to actuation of the air handler float switch. The system may include the drain cleaner apparatus, and an actuator apparatus configured to be electrically coupled to the drain cleaner apparatus. The actuator apparatus may include an actuator. The actuator apparatus may be configured to position the air handler float switch in relation to the actuator, such that the actuator apparatus is configured to cause the actuator to actuate the air handler float switch in response to receiving an actuator command signal from the drain cleaner apparatus. The controller of the drain cleaner apparatus may be configured to transmit the actuator command signal to the actuator apparatus to cause the actuator to actuate the air handler float switch.

(21) The system may further include a float switch apparatus configured to be coupled to the

condensate drain line. The float switch apparatus may be configured to enable the drain cleaner apparatus to supply the cleaning composition into the condensate drain line. The float switch apparatus may include a drain cleaner float switch. The drain cleaner float switch may be configured to be electrically coupled to the drain cleaner apparatus such that the drain cleaner float switch is configured to transmit a float switch signal to the drain cleaner apparatus in response to a presence of fluid in the condensate drain line. The controller of the drain cleaner apparatus may be configured to transmit the actuator command signal to the actuator apparatus to cause the actuator to actuate the air handler float switch in response to receiving the float switch signal from the drain cleaner float switch.

(22) The apparatus outlet of the drain cleaner apparatus may be coupled to a first end of a dispenser conduit, the dispenser conduit having an opposite, second end in fluid communication with the condensate drain line, such that the apparatus outlet of the drain cleaner apparatus is in fluid communication with the condensate drain line through at least the dispenser conduit.

(23) The float switch apparatus may include a support housing configured to couple with an opening of the condensate drain line, a supply conduit extending through the support housing, a first end of the supply conduit configured to be coupled with the second end of the dispenser conduit, the second end of the supply conduit configured to be in fluid communication with the condensate drain line, such that the supply conduit is configured to establish the fluid communication of the apparatus outlet of the drain cleaner apparatus with the condensate drain line through the dispenser conduit and the supply conduit. The drain cleaner float switch may be attached to the support housing. The drain cleaner float switch and the supply conduit may be offset from a central axis of the support housing.

(24) The actuator may include an actuator piston and a servomotor configured to cause the actuator piston to move along a first axis. The actuator apparatus may be configured to hold the air handler float switch in place in relation to the actuator piston. The actuator apparatus may be configured to actuate the air handler float switch based on causing a float of the air handler float switch to move in relation to a remainder of the air handler float switch based on the actuator piston moving along the first axis.

(25) The actuator apparatus may include a cup structure coupled to the actuator and further configured engage the float of the air handler float switch to move the float along the first axis based on movement of the actuator piston along the first axis.

(26) The actuator apparatus may include a conduit structure having an inner surface defining a conduit space extending along the first axis and having opposite first and second openings, the conduit structure configured to receive the air handler float switch into the conduit space through the first opening, the conduit structure further configured to receive at least the cup structure into the conduit space through the second opening.

(27) The controller of the drain cleaner apparatus may be configured to transmit the actuator command signal to the actuator apparatus to cause the actuator to actuate the air handler float switch based on processing a signal received from a remote computing device via a network communication interface of the drain cleaner apparatus.

(28) According to some example embodiments, an actuator apparatus may be configured to actuate an air handler float switch of an air handler of an air conditioning system. The actuator apparatus may include an actuator and one or more support structures configured to position the air handler float switch in relation to the actuator, such that the actuator apparatus is configured to cause the actuator to actuate the air handler float switch based on causing at least a float of the air handler float switch to move in relation to a remainder of the air handler float switch.

(29) The actuator may include an actuator piston and a servomotor configured to cause the actuator piston to move along a first axis. The actuator apparatus may be configured to hold the air handler float switch in place in relation to the actuator piston. The actuator apparatus may be configured to actuate the air handler float switch based on causing a float of the air handler float switch to move

in relation to a remainder of the air handler float switch based on the actuator piston moving along the first axis.

(30) The actuator apparatus may include a cup structure coupled to the actuator and further configured engage the float of the air handler float switch to move the float along the first axis based on movement of the actuator piston along the first axis.

(31) The one or more support structures may include a conduit structure having an inner surface defining a conduit space extending along the first axis and having opposite first and second openings, the conduit structure configured to receive the air handler float switch into the conduit space through the first opening, the conduit structure further configured to receive at least the cup structure into the conduit space through the second opening.

(32) According to some example embodiments, a float switch apparatus configured to be coupled to a condensate drain line of an air conditioning system may include a support housing configured to couple with an opening of the condensate drain line, a drain cleaner float switch attached to the support housing such that the drain cleaner float switch is configured to be positioned in the condensate drain line in response to the support housing being coupled with the opening of the condensate drain line, the drain cleaner float switch configured to be actuated to transmit a float switch signal in response to a presence of fluid in the condensate drain line, and a supply conduit extending through the support housing, a first end of the supply conduit configured to be coupled in fluid communication with an apparatus outlet of a drain cleaner apparatus, a second end of the supply conduit configured to be in fluid communication with the condensate drain line, such that the supply conduit is configured to establish fluid communication of the apparatus outlet of the drain cleaner apparatus with the condensate drain line through at least the supply conduit to enable a supply of cleaning composition from the drain cleaner apparatus to the condensate drain line through the float switch apparatus. The drain cleaner float switch and the supply conduit may be offset from a central axis of the support housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings may have been exaggerated.

(2) FIG. 1 is a schematic view of an air-conditioning system according to some example embodiments.

(3) FIGS. 2A and 2B are schematic views of a drain cleaner apparatus according to some example embodiments.

(4) FIGS. 3A and 3B are schematic views of a drain cleaner apparatus and a cartridge according to some example embodiments.

(5) FIG. 4 is a schematic view of a drain cleaner apparatus including a dispenser device that further includes first and second valves and a dispenser reservoir according to some example embodiments.

(6) FIG. 5 is a schematic view of a drain cleaner apparatus including a moisture sensor according to some example embodiments.

(7) FIG. 6 is a schematic view of a drain cleaner apparatus including a structure connector according to some example embodiments.

(8) FIG. 7 is a schematic view of a drain cleaner apparatus and a remote computing device

communicatively coupled via a network communication link according to some example embodiments.

(9) FIG. 8 is a flowchart illustrating a method of operation of the drain cleaner apparatus according to some example embodiments.

(10) FIG. 9 is a flowchart illustrating a method of operation of the drain cleaner apparatus according to some example embodiments.

(11) FIG. 10 is a schematic view of a computing device according to some example embodiments.

(12) FIG. 11A is a perspective top-front-right view of a drain cleaner apparatus system according to some example embodiments.

(13) FIG. 11B is a perspective bottom-rear-left view of the drain cleaner apparatus system of FIG. 11A according to some example embodiments.

(14) FIG. 11C is a perspective cross-sectional view of the drain cleaner apparatus system along cross-sectional view line XIC-XIC' of FIG. 11A according to some example embodiments.

(15) FIG. 11D is a plan cross-sectional view of the drain cleaner apparatus system along cross-sectional view line XIC-XIC' of FIG. 11A according to some example embodiments.

(16) FIG. 11E is a perspective cross-sectional view of the drain cleaner apparatus system along cross-sectional view line XIE-XIE' of FIG. 11A according to some example embodiments.

(17) FIG. 11F is a plan cross-sectional view of the drain cleaner apparatus system along cross-sectional view line XIE-XIE' of FIG. 11A according to some example embodiments.

(18) FIG. 12A is a perspective top-front-right view of the drain cleaner apparatus shown in FIG. 11A according to some example embodiments.

(19) FIG. 12B is a plan cross-sectional view of the drain cleaner apparatus along cross-sectional view line XIIB-XIIB' of FIG. 12A according to some example embodiments.

(20) FIG. 12C is a plan cross-sectional view of the drain cleaner apparatus along cross-sectional view line XIIC-XIIC' of FIG. 12A.

(21) FIG. 12D is a plan top view of the of the drain cleaner apparatus of FIG. 12A according to some example embodiments.

(22) FIG. 13A is a perspective top-front-right view of the cartridge shown in FIG. 11A according to some example embodiments.

(23) FIG. 13B is a perspective bottom-rear-left view of the cartridge shown in FIG. 13A according to some example embodiments.

(24) FIG. 13C is a plan cross-sectional view of the cartridge along cross-sectional view line XIIIIC-XIIIIC' of FIG. 13A according to some example embodiments.

(25) FIG. 13D is a plan cross-sectional view of the cartridge along cross-sectional view line XIIID-XIIID' of FIG. 13A according to some example embodiments.

(26) FIG. 14A is a perspective bottom-rear-left view of the structure connector shown in FIG. 11A according to some example embodiments.

(27) FIG. 14B is a perspective top-front-right view of the structure connector shown in FIG. 14A according to some example embodiments.

(28) FIG. 14C is a perspective view of the drain cleaner apparatus according to some example embodiments.

(29) FIG. 14D is a plan bottom view of the drain cleaner apparatus according to some example embodiments.

(30) FIG. 15A is a schematic view of a system including a drain cleaner apparatus system, a float switch apparatus, and an actuator apparatus, according to some example embodiments.

(31) FIG. 15B is a schematic view of a system including a drain cleaner apparatus system and a float switch apparatus, according to some example embodiments.

(32) FIG. 16A is a perspective top-front-right view of a float switch apparatus according to some example embodiments.

(33) FIG. 16B is a perspective bottom-rear-left view of the float switch apparatus of FIG. 16A

according to some example embodiments.

(34) FIG. 16C is a perspective cross-sectional view of the float switch apparatus along cross-sectional view line XVIC-XVIC' of FIG. 16A according to some example embodiments.

(35) FIG. 16D is a plan cross-sectional view of the float switch apparatus along cross-sectional view line XVIC-XVIC' of FIG. 16A according to some example embodiments.

(36) FIG. 16E is a plan top view of the float switch apparatus of FIG. 16A according to some example embodiments.

(37) FIG. 17A is a perspective top-front-right view of an actuator apparatus according to some example embodiments.

(38) FIG. 17B is a perspective bottom-rear-left view of the actuator apparatus of FIG. 17A according to some example embodiments.

(39) FIG. 17C is a perspective bottom-rear-right view of the actuator apparatus of FIG. 17A according to some example embodiments.

(40) FIG. 18A is a perspective top-front-right view of an actuator apparatus according to some example embodiments.

(41) FIG. 18B is a perspective cross-sectional view of the actuator apparatus along cross-sectional view line XVIIIIB-XVIIIIB' of FIG. 18A according to some example embodiments.

(42) FIG. 18C is a plan cross-sectional view of the actuator apparatus along cross-sectional view line XVIIIIB-XVIIIIB' of FIG. 18A according to some example embodiments.

(43) FIG. 18D is a perspective cross-sectional view of the actuator apparatus along cross-sectional view line XVIIIID-XVIIIID' of FIG. 18A according to some example embodiments.

(44) FIG. 18E is a plan cross-sectional view of the actuator apparatus along cross-sectional view line XVIIIID-XVIIIID' of FIG. 18A according to some example embodiments.

(45) FIG. 19A is a perspective top-front-right view of an actuator apparatus according to some example embodiments.

(46) FIG. 19B is a perspective cross-sectional view of the actuator apparatus along cross-sectional view line XIXB-XIXB' of FIG. 19A according to some example embodiments.

(47) FIG. 19C is a perspective cross-sectional view of the actuator apparatus along cross-sectional view line XIXC-XIXC' of FIG. 19A according to some example embodiments.

(48) FIG. 20 is a perspective view of elements of an actuator apparatus according to some example embodiments.

(49) FIG. 21A is a perspective view of a containment apparatus according to some example embodiments.

(50) FIG. 21B is a perspective cross-sectional view of the containment apparatus along cross-sectional view line XXIB-XXIB' of FIG. 21A according to some example embodiments.

(51) FIG. 21C is a perspective cross-sectional view of the containment apparatus along cross-sectional view line XXIC-XXIC' of FIG. 21A according to some example embodiments.

(52) FIG. 22 is a perspective view of outer shells and hinge connection of a containment apparatus according to some example embodiments.

(53) FIG. 23A is a perspective view of an adaptor sleeve structure of a containment apparatus according to some example embodiments.

(54) FIG. 23B is a perspective cross-sectional view of the adaptor sleeve structure along cross-sectional view line XXIIIB-XXIIIB' of FIG. 23A according to some example embodiments.

(55) FIG. 24A is a perspective view of an adaptor sleeve structure of a containment apparatus according to some example embodiments.

(56) FIG. 24B is a perspective cross-sectional view of the adaptor sleeve structure along cross-sectional view line XXIVB-XXIVB' of FIG. 24A according to some example embodiments.

(57) FIG. 25A is a plan cross-sectional view of the actuator apparatus along cross-sectional view line XVIIIIB-XVIIIIB' of FIG. 18A in which an air handler float switch is positioned according to some example embodiments.

- (58) FIG. 25B is a plan cross-sectional view of the actuator apparatus along cross-sectional view line XVIIIID-XVIIIID' of FIG. 18A in which an air handler float switch is positioned according to some example embodiments.
- (59) FIG. 26A is a perspective top-front-left view of a drain cleaner apparatus system according to some example embodiments.
- (60) FIG. 26B is a perspective bottom-rear-left view of the drain cleaner apparatus system of FIG. 26A according to some example embodiments.
- (61) FIG. 26C is a perspective bottom-rear-left view of the drain cleaner apparatus system of FIG. 26A according to some example embodiments.
- (62) FIG. 26D is a perspective view of an actuator holster according to some example embodiments.
- (63) FIG. 26E is a perspective cross-sectional view of the actuator holster along cross-sectional view line XXVII-XXVII' in FIG. 26D according to some example embodiments.
- (64) FIG. 27 is a perspective top-front-right view of a drain cleaner apparatus system according to some example embodiments.
- (65) FIG. 28 is a flowchart showing a method of operation of a system according to some example embodiments.

DETAILED DESCRIPTION

- (66) Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.
- (67) Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of example embodiments of the inventive concepts.
- (68) Example embodiments are described herein with reference to cross-sectional illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of example embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, example embodiments should not be construed as limited to the shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.
- (69) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.
- (70) It will be understood that elements and/or properties thereof (e.g., structures, surfaces, directions, or the like), which may be referred to as being “perpendicular,” “parallel,” “flush,” or the like with regard to other elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) may be “perpendicular,” “parallel,” “flush,” or the like or may be “substantially perpendicular,” “substantially parallel,” “substantially flush,” respectively, with regard to the other elements and/or properties thereof.
- (71) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially perpendicular” with regard to other elements and/or properties thereof will be understood to be “perpendicular” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or

angle from “perpendicular,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a. tolerance of $\pm 10\%$).

(72) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially parallel” with regard to other elements and/or properties thereof will be understood to be “parallel” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “parallel,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a. tolerance of $\pm 10\%$).

(73) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially flush” with regard to other elements and/or properties thereof will be understood to be “flush” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “flush,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a. tolerance of $\pm 10\%$).

(74) It will be understood that elements and/or properties thereof may be recited herein as being “the same” or “equal” as other elements, and it will be further understood that elements and/or properties thereof recited herein as being “identical” to, “the same” as, or “equal” to other elements may be “identical” to, “the same” as, or “equal” to or “substantially identical” to, “substantially the same” as or “substantially equal” to the other elements and/or properties thereof. Elements and/or properties thereof that are “substantially identical” to, “substantially the same” as or “substantially equal” to other elements and/or properties thereof will be understood to include elements and/or properties thereof that are identical to, the same as, or equal to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances. Elements and/or properties thereof that are identical or substantially identical to and/or the same or substantially the same as other elements and/or properties thereof may be structurally the same or substantially the same, functionally the same or substantially the same, and/or compositionally the same or substantially the same.

(75) It will be understood that elements and/or properties thereof described herein as being the “substantially” the same and/or identical encompasses elements and/or properties thereof that have a relative difference in magnitude that is equal to or less than 10%. Further, regardless of whether elements and/or properties thereof are modified as “substantially,” it will be understood that these elements and/or properties thereof should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated elements and/or properties thereof.

(76) When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

(77) FIG. 1 is a schematic view of an air conditioning system **100** according to some example embodiments. The air conditioning system **100**, which may be interchangeably referred to as an air conditioning system, air conditioner, or the like, may be configured to provide cooling of air within an interior of a structure **1** and may be at least partially located within the structure **1**, but example embodiments are not limited thereto. The air conditioning system **100** may be included as a part of a Heating, ventilation, and air conditioning (HVAC) system, but example embodiments are not limited thereto, and in some example embodiments the air conditioning system **100** may be separate from any heating system.

(78) Referring to FIG. 1, the air conditioning system **100** may include an air handler **102** and a condenser assembly **104** that are configured to draw return air **106** from an interior of the structure **1**, cool (e.g., absorb heat from) the drawn return air **106** into conditioned air **114**, and discharge (e.g., supply) the conditioned air **114** back into the interior of the structure **1**. The air handler **102** may include, within a housing **101** that may at least partially comprise metal (e.g., steel), an air

intake **103**, an air filter **105**, an air mover **108** (e.g., fan, blower, etc.), a heat exchanger **110** (e.g., evaporator coil), an expansion valve **111**, a drip pan **122**, a condensate drain line **124** (also referred to herein as a condensate drain conduit, condensate drain pipe, etc.), a controller **140**, a float switch **160**, and an air outlet **112**. The condenser assembly **104** may include a compressor **150**, a second heat exchanger **152** (e.g., condenser coil), and an air mover **154** (e.g., fan, blower, etc.).

(79) It will be understood that example embodiments of an air conditioning system, air handler, condenser assembly, or the like may have different arrangements of devices therein and may omit or add to the aforementioned elements of the air conditioning system **100** as shown in FIG. **1**. It will be understood, for example, that elements shown as being included in the air handler **102** may in some example embodiments be located in the condenser assembly **104** (e.g., the controller **140** may be located in the condenser assembly **104** instead of the air handler **102**). As shown, the condenser assembly **104** may be located external to the structure **1** while the air handler **102** is located internal to the structure **1**, but example embodiments are not limited thereto.

(80) In some example embodiments, the air conditioning system **100** may draw return air **106** into the air handler **102** via the air intake **103** and through the air filter **105**, where the air filter **105** may be any known air filter that is configured to remove some matter (e.g., particulate matter, including dust) from the return air **106**. The air mover **108** (e.g., blower) may induce the flow of air into, through, and out of, the air handler **102**. The air mover **108** may cause return air **106** to be drawn through the air filter **105** to remove some matter and may move (e.g., blow) the return air **106** through the air mover **108** and to the heat exchanger **110**. The return air **106** may flow in thermal communication with (e.g., in contact with outer surfaces of) one or more coils of the heat exchanger **110** so that heat is removed from the return air **106** to cool the return air **106** into conditioned air **114**. The air handler **102** may move the conditioned air **114** out of the air handler **102** and back into an interior space of the structure **1** via the air outlet **112**.

(81) The air conditioning system **100** may circulate a working fluid (e.g., a refrigerant, including known R22 refrigerant, R410A refrigerant, or any known refrigerant) between the heat exchangers **110** and **152** to remove heat from the return air **106** when the return air **106** flows in thermal communication (e.g., through and/or in contact with one or more coils of) the heat exchanger **110**. The heat exchanger **110** may include any known heat exchanger used for an air conditioning system, for example an evaporator coil exchanger that includes one or more coils of one or more tubes through which the working fluid flows (e.g., as a cooled liquid). The heat exchanger **110** may cause heat to be transferred from the return air **106** and into the working fluid when the return air **106** is caused to flow across (e.g., in contact with, in thermal communication with, etc.) the one or more coils, thereby resulting in the working fluid becoming heated (e.g., heated into a low-pressure gas). The heated working fluid may be drawn, via fluid line **116** (e.g., fluid conduit, pipe, etc.) into the condenser assembly **104**.

(82) The air conditioning system **100** may include, in the condenser assembly **104**, a compressor **150** (which may be any known compressor) that induces flow of the working fluid through the air conditioning system **100**. The compressor **150** may draw the heated working fluid from the fluid line **118** and may compress the heated working fluid into a high-pressure gas. The heated working fluid may pass (e.g., flow), for example as the high-pressure gas, from the compressor **150** to the heat exchanger **152** (which may be any known heat exchanger and may be referred to as a condenser coil). The air mover **154** may cause ambient air **192** from the ambient environment **190** to be drawn across (e.g., in thermal communication with) one or more tubes of the heat exchanger **152** to remove heat from the heated working fluid passing through the one or more tubes of the heat exchanger **152**, thereby discharging the heat originally removed from the return air **106** into the ambient environment **190** which serves as a heat sink for the air conditioning system **100**. As a result, the working fluid passing through the heat exchanger **152** may be cooled back into a liquid. The working fluid may then pass (e.g., flow, circulate, etc.) back to the air handler **102** via a fluid line **118**, where the working fluid may pass through an expansion valve **111** (which may be any

known expansion valve) to cool the working fluid which then passes into the heat exchanger **110** to remove additional heat from return air **106**.

(83) As noted above, the circulation of working fluid through the heat exchanger **110**, heat exchanger **152**, fluid lines **116** and **118**, and expansion valve **111** may be induced by operation of the compressor **150**.

(84) As further shown, the air conditioning system **100** may include a controller **140** that is configured to control elements of the air conditioning system **100**, including for example controlling operation of the air handler **102**, condenser assembly **104**, or any part thereof. As described further below, the controller **140** may be implemented by a computing device, including a memory storing a program of instructions and a processor configured to execute the program of instructions. While the controller **140** is shown as being included within the housing **101** of the air handler **102**, it will be understood that the controller **140** may be located external to the housing **101** and, in some example embodiments, may be located within the condenser assembly **104** or may be attached to an exterior of the air handler **102** for ease of manual access.

(85) Still referring to FIG. **1**, when heat is removed from the return air **106** based on the return air **106** passing in thermal communication with the heat exchanger **110**, water may condense out of the cooled return air as condensate **120** at the heat exchanger **110**. The air handler **102** may include a drip pan **122** located beneath the heat exchanger **110**, and the condensate **120** may fall under gravity to collect in the drip pan **122**. The air handler **102** may further include a condensate drain line **124** having an inlet opening **128** coupled to the drip pan **122** (e.g., a bottom surface where the drip pan **122** has an inclined surface that is angled downwards towards the inlet opening **128** of the condensate drain line **124**) and an outlet opening **130** that is external to the structure **1** and open to the ambient environment **190**, as shown. Condensate **120** collected in the drip pan **122** may pass under gravity to the inlet opening **128** of the condensate drain line **124**, and the condensate drain line **124** may direct the condensate **120** to flow out of the air handler **102** and out of the structure **1** to the ambient environment **190** via the outlet opening **130** of the condensate drain line **124**.

(86) In some example embodiments, the condensate drain line **124** may become clogged due to buildup of various substances within the condensate drain line. Such substances may include, for example, mold, algae, mildew, bacteria, and/or fungi. When the condensate drain line becomes clogged, backflow and/or overflow of condensate **120** out of the condensate drain line **124** may occur. For example, condensate **120** may accumulate in the drip pan **122** due to the clogging and may eventually overflow over the sides of the drip pan **122**. Such overflow of condensate **120** out of the drip pan **122** may cause damage to the air handler **102** and/or to the structure **1**, including water damage to structural members of the structure **1**, water damage to elements of the air handler **102**, flooding of the structure **1** and/or the air handler **102**, or the like.

(87) As shown in FIG. **1**, the air conditioning system **100** may include a float switch **160** that is located in the drip pan **122** and/or in the condensate drain line **124** (as shown). The float switch **160** may be a switch that is configured to be actuated based on backflow and/or overflow of condensate **120** in the condensate drain line **124**. For example, the float switch **160** may be any known float switch and may be configured to be closed or opened (e.g., actuated) based on accumulation of condensate **120** in the drip pan **122** to at least a threshold volume held therein. The float switch **160** may be communicatively (e.g., electrically) coupled to the controller **140**, and the controller **140** may be configured to shut down some or all of the air conditioning system **100** (e.g., shut down the air mover **108**, the compressor **150**, and/or the air mover **154**) in response to the float switch **160** being actuated, thereby reducing or preventing damage being caused in the structure and/or air conditioning system **100** due to the condensate **120** accumulation.

(88) Still referring to FIG. **1**, in some example embodiments a drain cleaner apparatus **200** may be coupled to the condensate drain line **124** at an opening **125** into the condensate drain line **124** (e.g., a cleanout opening of the condensate drain line **124**), where the drain cleaner apparatus **200** is configured to dispense a cleaning composition into the condensate drain line **124**. As described

herein, the drain cleaner apparatus **200** may be configured to dispense a cleaning composition into the condensate drain line **124** to reduce, remove, and/or prevent clogging of the condensate drain line **124** due to the presence of various potential clogging substances (e.g., mold, algae, mildew, bacteria, and/or fungi) therein.

(89) In some example embodiments, the drain cleaner apparatus **200** may be configured to dispense the cleaning composition into the condensate drain line **124** without human intervention (e.g., automatically), for example to dispense discrete amounts (e.g., a particular amount, which may be a particular volume and/or particular mass) of the cleaning composition at a particular (or, alternatively, predetermined) fixed time interval, thereby reducing or preventing clogging of the condensate drain line **124** due to the presence of various potential clogging substances (e.g., mold, algae, mildew, bacteria, and/or fungi) therein while reducing or minimizing human intervention and/or effort expended to implement the dispensing. Because the drain cleaner apparatus **200** is configured to dispense the cleaning composition (e.g., repeatedly at a fixed time interval) without human intervention, the buildup of potential clogging substances (e.g., mold, algae, mildew, bacteria, and/or fungi) in the condensate drain line **124** may be reduced, removed, or prevented. This may thereby reduce or prevent the likelihood of condensate **120** backup and/or overflow which might otherwise result in shutdown of at least the air handler **102** and/or air conditioning system **100**, flooding damage to the air handler **102** and/or structure **1**, or the like. Because human intervention is not required to implement the dispensing of the cleaning composition, particularly dispensing of the cleaning composition repeatedly at a fixed time interval, the likelihood of condensate drain line **124** clogging due to a missed or forgotten manual dispensing of cleaning composition by a human operator is reduced or prevented, thereby improving operational performance of the air conditioning system **100** and reducing workload by a human operator.

(90) FIGS. 2A and 2B are schematic views of a drain cleaner apparatus **200** according to some example embodiments. Referring to FIGS. 2A and 2B in reference to FIG. 1, the drain cleaner apparatus **200** is configured to dispense a cleaning composition **230** into a condensate drain line **124** of the air handler **102** shown in FIG. 1.

(91) Referring to FIGS. 2A and 2B, the drain cleaner apparatus **200** may include an apparatus reservoir **202** configured to hold the cleaning composition **230**, an apparatus outlet **206** (e.g., opening), and a dispenser device **204** that is configured to be actuated (e.g., operated) to selectively dispense an amount (e.g., a particular amount, which may be a particular volume and/or a particular mass) of the cleaning composition **230** from the apparatus reservoir **202** and through the apparatus outlet **206**. The drain cleaner apparatus **200** may further include a connector interface **208** that is configured to couple with the condensate drain line **124** to cause the apparatus outlet **206** of the drain cleaner apparatus **200** to be in fluid communication with (e.g., open to) the opening **125** (e.g., cleanout opening) of the condensate drain line **124**.

(92) As shown in FIGS. 2A and 2B, the apparatus reservoir **202** may include an inner surface **202S** defining an interior volume space in which cleaning composition **230** may be held within a housing **201** of the drain cleaner apparatus **200**. The apparatus reservoir **202** may further include an outlet **202A** that is configured to be in fluid communication with the dispenser device **204** to enable cleaning composition **230** to flow from the apparatus reservoir **202** to the dispenser device **204**. The apparatus reservoir **202** may further include a cover **203** (e.g., a hatch) that may be opened or removed to enable filling or refilling of the apparatus reservoir **202** with cleaning composition **230**. However, it will be understood that in some example embodiments, the cleaning composition **230** may be provided within a cartridge container (e.g., “cartridge”) that may be received into and held within the apparatus reservoir **202** instead of being poured directly into the apparatus reservoir **202** from outside the drain cleaner apparatus **200**.

(93) Still referring to FIGS. 2A and 2B, the dispenser device **204** is a device that may be actuated (e.g., operated, based on an electrical control signal) to selectively open or close at least one fluid path from the apparatus reservoir **202** (e.g., via outlet **202A**) to the apparatus outlet **206** to enable at

least an amount of the cleaning composition **230** to be dispensed through the apparatus outlet **206**.
(94) The dispenser device **204** may be configured to dispense an amount of cleaning composition **230** that is a particular amount (e.g., a particular volume, particular mass, etc.) so that the drain cleaner apparatus **200** may dispense a particular amount of cleaning composition **230** (e.g., repeatedly at a fixed time interval). For example, in some example embodiments, the amount of cleaning composition **230** as described herein that is dispensed when the dispenser device **204** is actuated once may be 3 oz of cleaning composition **230**, and the dispenser device **204** may be configured to be actuated to cause the particular amount of cleaning composition **230** from the apparatus reservoir **202** to the apparatus outlet **206**.

(95) The connector interface **208** is configured to couple (e.g., removably couple, detachably couple, reversibly couple, etc.) the drain cleaner apparatus **200** with the condensate drain line **124** so that the apparatus outlet **206** is in fluid communication with the opening **125** into the condensate drain line **124**, for example as shown in FIG. 2B. As shown, the connector interface **208** is configured to couple with the opening end of the condensate drain line **124** to cause the apparatus outlet **206** to be directly adjacent to, and directly open to, the opening **125** into the condensate drain line **124**, so that the actuation of the dispenser device **204** to dispense an amount of the cleaning composition **230** from the apparatus reservoir **202** to the apparatus outlet **206** further causes the amount of the cleaning composition **230** to flow into the condensate drain line **124** through the apparatus outlet **206** and the opening **125** into the condensate drain line **124**.

(96) In some example embodiments, the connector interface **208** may be any connector that is configured to couple at least the housing **201** of the drain cleaner apparatus **200** with the condensate drain line **124**. In some example embodiments, the connector interface **208** may be a friction fit connector interface that includes an inner surface having an inner diameter that corresponds to the outer diameter of the opening end of the condensate drain line **124**, so that the connector interface **208** is configured to establish a friction fit connection with the opening **125**. The connector interface **208** may further include a seal, O-ring, or the like along the inner surface of the connector interface **208** to further establish a connection with the opening **125**. In some example embodiments, the connector interface **208** includes a threaded connector, bayonet connector, or the like that is configured to be coupled with a complementary connector interface of the condensate drain line **124** (e.g., a threaded connector, bayonet connector, or the like at the opening **125** of the condensate drain line **124**). In some example embodiments, the connector interface **208** may include an adaptor (e.g., a variable inner diameter connector) that is configured to couple the drain cleaner apparatus **200** to various condensate drain lines **124** having various outer diameters. In some example embodiments, the connector interface **208** is configured to at least partially transfer a structural load (e.g., weight) of the drain cleaner apparatus **200** to the condensate drain line **124**, so that the drain cleaner apparatus **200** is configured to be at least partially structurally supported in place on the condensate drain line **124**.

(97) In some example embodiments, the drain cleaner apparatus **200** includes a structure connector **220** that is configured to connect the drain cleaner apparatus **200** to an external structure (e.g., a housing **101** of the air handler **102** as shown) to at least partially hold the drain cleaner apparatus **200** in place in relation to the opening **125** of the condensate drain line **124** (e.g., at least partially structurally support the drain cleaner apparatus **200** on the opening **125**). As described further herein, the structure connector **220** may have various structures. For example, the structure connector **220** may include an adhesive connector, a magnet, or the like to couple with the housing **101** of the air handler **102**.

(98) In some example embodiments, the dispenser device **204** may include at least one valve that is configured to be actuated to be selectively opened (e.g., to selectively open a flow path **204A** through the at least one valve) based on a control signal generated by the controller **210** to establish a flow path **204A** through the at least one valve and through which the cleaning composition **230** may flow (e.g., a flow path **204A** from the apparatus reservoir **202** to the apparatus outlet **206**). For

example, a valve of the dispenser device **204** as described herein may include an electromechanically operated valve, including a solenoid valve, which may be selectively actuated based on a control signal from the controller **210**.

(99) In some example embodiments, the dispenser device **204** may include a pump (e.g., any known positive displacement pump) that is configured to operate for a particular period of time to move the amount of the cleaning composition **230** from the apparatus reservoir **202** to the apparatus outlet **206**, based on a control signal generated by the controller **210**.

(100) As described herein, a cleaning composition **230** may be any known chemical composition (e.g., solution, liquid, fluid, etc.) that may be configured to clean (e.g., remove) potential clogging substances (e.g., mold, algae, mildew, bacteria, and/or fungi) from an inner surface of the condensate drain line **124**. In some example embodiments, the cleaning composition **230** may be a chemical substance that is or includes a chelating agent (e.g., chelant) including, for example, sodium hexametaphosphate, that is configured to remove potential clogging substances from the inner surface of the condensate drain line based on chelation upon contact with the potential clogging substances. For example, the cleaning composition **230** may be a liquid solution that includes 3%-7% sodium hexametaphosphate, by weight of the total weight of the cleaning composition **230**. Based on the drain cleaner apparatus **200** being configured to dispense cleaning composition **230** through the apparatus outlet **206**, where the cleaning composition **230** is dispensed into the condensate drain line **124**, the drain cleaner apparatus **200** may be configured to enable removal of potential clogging substances (e.g., mold, algae, mildew, bacteria, and/or fungi) from an inner surface of the condensate drain line **124** by the cleaning composition **230**, which may thereby reduce or prevent the occurrence of backflow and/or overflow of the condensate drain line **124** due to clogging.

(101) As shown in FIGS. 2A and 2B, the drain cleaner apparatus **200** may include a power supply **212** that is configured to supply electrical power to devices included therein, including the controller **210**, the dispenser device **204**, a network communication interface **224**, a sensor (not shown in FIGS. 2A and 2B, shown in FIG. 5), or the like. As shown, the power supply **212** may include a battery **214**, which may include any known rechargeable battery (e.g., a lithium ion battery). As further shown, in some example embodiments the power supply **212** may include a wired power connection **216** which may be configured to couple to a power outlet provided at the structure **1** and/or the air handler **102**. The power supply **212** may further include a charging circuit **218** that may be configured to recharge the battery **214** from the wired power connection **216** and may be configured to enable the battery **214** to supply power to operate the drain cleaner apparatus **200** in the absence of electrical power being received via the wired power connection **216**.

(102) As shown in FIGS. 2A and 2B, the controller **210** may be configured to actuate the dispenser device **204** to cause a particular amount of the cleaning composition **230** to be dispensed from the apparatus reservoir **202** and through the apparatus outlet **206** without manual intervention. For example, the controller **210** may be configured to cause an electrical signal to be generated and transmitted to the dispenser device **204** to cause the dispenser device **204** to actuate, selectively opening or closing a flow path **204A** therethrough, to thus cause a particular amount of the cleaning composition **230** to be dispensed.

(103) The controller **210** may include a memory (e.g., a solid state drive, or SSD) storing a program of instructions, and the controller **210** may include a processor (e.g., a Central Processing Unit, or CPU) configured to execute the program of instructions to implement any functionality of the controller **210** according to any example embodiments. However, example embodiments are not limited thereto. For example, in some example embodiments, the controller **210** may include circuitry that is configured to implement a timer circuit (e.g., a clock, timer, or any combination thereof) and is configured to generate a signal to actuate the dispenser device **204** based on the timer circuit counting a particular time interval.

(104) In some example embodiments, the controller **210** is configured to actuate the dispenser

device **204** (e.g., actuate at least one valve, pump, or the like therein) to cause the dispenser device **204** to dispense an amount of cleaning composition **230** through the apparatus outlet **206** to be dispensed into the condensate drain line **124**. In some example embodiments, the controller **210** may be configured to generate a signal to cause at least a portion of the dispenser device **204** (e.g., a valve, pump, etc.) to be operated (e.g., a valve opened, a pump operating) for a particular period of time that is associated, at the controller **210**, with causing a particular amount of cleaning composition **230** to be dispensed by the dispenser device **204**. The controller **210** may cause a particular amount of cleaning composition **230** to be dispensed based on accessing a look-up-table that is stored in a memory of the controller **210**, where the look-up-table is empirically generated and associates a period of time of actuation of at least a portion of the dispensing device (e.g., a period of time of generation of a control signal) with dispensing of a corresponding amount of cleaning composition **230** by the dispenser device **204**. The controller **210** may determine a particular amount of cleaning composition **230** to be dispensed, access the look-up-table to determine a corresponding duration or period of applied control signal to the dispenser device **204**, and then generate a control signal that is transmitted to the dispenser device **204** to cause at least a portion of the dispenser device **204** to be actuated for the corresponding duration or period.

(105) In some example embodiments, the controller **210** is configured to actuate the dispenser device **204** to cause an amount of cleaning composition **230** (e.g., 3 oz) to be dispensed in response to an elapse of a particular period of time (e.g., 7 days, or 168 hours). The controller **210** may be configured to actuate the dispenser device **204** repeatedly upon repeated elapse of the particular period of time, which may be referred to as a “fixed time interval” (e.g., a fixed time interval of 7 days). In some example embodiments, the apparatus reservoir **202** may be configured to hold a total volume of 36 oz, so that the drain cleaner apparatus **200** may be configured to dispense 3 oz of cleaning composition **230** every 7 days for a period of 12 weeks (84 days).

(106) The controller **210** may be configured to repeatedly actuate the dispenser device **204** at a fixed time interval (e.g., 7 days), based on monitoring a timer that increments a timer value at a fixed frequency, actuating the dispenser device **204** in response to the timer value reaching a particular time value corresponding to the elapse of the particular period of time, and resetting the timer value to an initial timer value (e.g., 0 days) in response to actuating the dispenser device **204**. For example, the controller **210** may include and/or implement a clock and/or timer that counts a period of elapsed time from an initial timer value (e.g., increments from 0 days) at a fixed frequency (e.g., counts days, hours, minutes and/or seconds at a fixed frequency of days, hours, minutes and/or seconds). In response to determining that a threshold timer value is reached (e.g., a timer value corresponding to the particular period of time and/or fixed time interval of 7 days), the controller **210** may generate a signal to cause the dispenser device **204** to actuate to cause an amount of the cleaning composition **230** to be dispensed through the apparatus outlet **206** and further re-set the timer value so that the controller **210** may subsequently cause the dispenser device **204** to dispense another amount of the cleaning composition **230** upon a re-elapse of the particular period of time. The controller **210** may be configured to perform this process repeatedly so long as electrical power is supplied to the controller **210** (e.g., from power supply **212**), so that the process may be performed (e.g., repeatedly at a fixed time interval) without human intervention.

(107) In some example embodiments, the controller **210** is configured to implement a counter that increments a counter value, starting from an initial value (e.g., 0), in response to each actuation of the dispenser device **204**. As a result, where the controller **210** repeatedly actuates the dispenser device **204** at a fixed time interval, the controller **210** may track the number (e.g., quantity) of dispensings of an amount of cleaning composition **230** (e.g., the number of actuations of the dispenser device **204**) over time. Therefore, where the drain cleaner apparatus **200** is configured to hold a particular total amount of cleaning composition **230** (e.g., 36 oz), the controller **210** may track the counter value to determine when the total amount of cleaning composition **230** available

to be dispensed is about to be depleted or is depleted and may generate a signal (e.g., a depletion signal) in response to the counter value reaching a value that corresponds to partial or complete (e.g., total, final, etc.) depletion of the cleaning composition **230** held by the drain cleaner apparatus **200**.

(108) For example, where the drain cleaner apparatus **200** is configured to hold a particular total amount of cleaning composition **230** that is 36 oz, and where the controller **210** is configured to cause the dispenser device **204** to dispense an amount of 3 oz of cleaning composition **230** at a fixed time interval of 7 days, the total amount of cleaning composition **230** may be depleted upon completion of 12 dispensings. The controller **210** may store a threshold counter value of 10, 11, or 12 that corresponds to partial depletion, near-depletion, or total depletion of the total amount of cleaning composition **230** held in the drain cleaner apparatus **200**. The controller **210** may implement and/or monitor a counter that increments a counter value in response to each actuation of the dispenser device **204**, and generate a depletion signal in response to the counter value reaching a particular counter value that corresponds to at least partial depletion of a fixed reservoir of the cleaning composition (e.g., 10, 11, or 12). As described herein, the controller **210** may transmit the depletion signal to a display interface (e.g., an LED, an audio speaker), which may be included in the drain cleaner apparatus **200** or may be included in a remote computing device, to provide a depletion warning. The controller **210** may further or alternatively be configured to cause the depletion signal to a remote computing device (e.g., via a network communication interface **224** as described herein) in order to inform a remote human user supported by the remote computing device of the partial or complete depletion (e.g., final depletion) of the total amount of cleaning composition **230** held in the drain cleaner apparatus **200**. The human user may then be informed of the partial or complete depletion so that the human user may take action to replenish the cleaning composition held in the drain cleaner apparatus **200**.

(109) Additionally, the drain cleaner apparatus **200** may include a counter reset interface **222** (e.g., a button) that is configured to cause the counter value to be reset to an initial counter value (e.g., 0) in response to human interaction with the counter reset interface **222** (e.g., in response to a human user pushing the button after replenishing the total amount of cleaning composition **230** held in the drain cleaner apparatus **200**).

(110) Still referring to FIGS. 2A and 2B, the drain cleaner apparatus **200** may include a network communication interface **224** that is communicatively coupled to the controller **210**. It will be understood that the network communication interface **224** may be separate from the controller **210** as shown or may be included in and/or implemented by the controller **210**. The network communication interface **224** may be any known network communication transceiver, including a wireless network communication transceiver such as a WI-FI transceiver, 5G cellular network communication transceiver, an ad hoc network communication transceiver such as a Bluetooth® transceiver, any combination thereof, or the like.

(111) The controller **210** may be configured to establish a network communication link (which may be a wired network communication link, a wireless network communication link, an ad hoc wireless network communication link, or the like) with a remote computing device as described herein and may engage in one-way or two-way communication with the remote computing device via the network communication link.

(112) In some example embodiments, the controller **210** may communicate signals over the network communication link that indicate operations of the controller **210** (e.g., indicating actuation of the dispenser device **204** at particular points in time, a present timer value, a present counter value, etc.). In some example embodiments, the controller **210** may communicate the depletion signal (generated in response to the counter value reaching a threshold value) to the remote computing device via the network communication link.

(113) In some example embodiments, the controller **210** may be configured to perform operations in response to receiving signals from the remote computing device via the network communication

link. For example, the controller **210** may be configured to cause the counter value of the counter to be reset to an initial counter value (e.g., 0) in response to receiving a reset signal from the remote computing device via the network communication link (which may be transmitted by the remote computing device in response to a human user replenishing the total amount of cleaning composition **230** held in the drain cleaner apparatus **200**).

(114) FIGS. **3A** and **3B** are schematic views of a drain cleaner apparatus **200** and a cartridge **300**, also referred to interchangeably as a “cleaner cartridge,” “cleaning composition cartridge,” or the like according to some example embodiments. Referring to FIGS. **3A** and **3B** in reference to FIG. **1**, the drain cleaner apparatus **200** is configured to dispense a cleaning composition **230** into a condensate drain line **124** of the air handler **102** shown in FIG. **1**. The drain cleaner apparatus **200** shown in FIGS. **3A** and **3B** may include some or all of the same elements as the drain cleaner apparatus of any of the example embodiments.

(115) In some example embodiments, the drain cleaner apparatus **200** may be configured to receive and couple with a cartridge **300** that contains (e.g., holds) the cleaning composition **230** within a cartridge reservoir **304** such that a flow path is established between the cartridge reservoir **304** and the dispenser device **204**. The cartridge **300** may be provided instead of the cleaning composition **230** being poured into, and directly held within, the apparatus reservoir **202** in contact with the inner surface **202S** thereof, for example as shown in FIGS. **2A** and **2B**. Replenishment of the cleaning composition **230** held in the drain cleaner apparatus **200** may be simplified based on the cleaning composition **230** being held in the cartridge **300**, as replenishment of the total cleaning composition **230** held in the drain cleaner apparatus **200** may involve replacing a cartridge **300** that is coupled to the drain cleaner apparatus **200** based on being inserted into the apparatus reservoir **202** instead of directly pouring the cleaning composition **230** directly into the apparatus reservoir **202**. Such simplification may include reducing or preventing inadvertent spilling of cleaning composition **230** during the replenishment process.

(116) As shown in FIGS. **3A** and **3B**, the cartridge **300** may include a cartridge housing **302** that has at least an inner surface **3021** defining a cartridge reservoir **304** which may hold the cleaning composition **230** therein. In some example embodiments, the cartridge reservoir **304** may have a particular volume, for example 36 oz and thus may be configured to hold the particular volume (e.g., 36 oz) of cleaning composition **230**.

(117) As further shown, the apparatus reservoir **202** and the cartridge **300** may be sized and shaped so that the cartridge **300** may be received at least partially into the apparatus reservoir **202** to establish a sliding contact fit between the outer surface **302S** of the cartridge housing **302** and the inner surface **202S** of the apparatus reservoir **202**, for example so that the cartridge **300** occupies all or substantially all of the internal volume space of the apparatus reservoir **202** when the cartridge **300** is coupled to the drain cleaner apparatus **200**.

(118) As shown in FIGS. **3A** and **3B**, the cartridge **300** may have a greater volume than the apparatus reservoir **202** and may protrude out of the opening **2020** of the apparatus reservoir **202** when the cartridge **300** is received into the apparatus reservoir **202** and coupled with the drain cleaner apparatus **200**. Such protrusion of the cartridge **300** may enable easier human access to grasp the cartridge **300** to simplify replacement of cartridges **300**, but example embodiments are not limited thereto: in some example embodiments the cartridge **300** may be located entirely within the apparatus reservoir **202** when the cartridge **300** is coupled to the drain cleaner apparatus **200**.

(119) As shown in FIGS. **3A** and **3B**, the drain cleaner apparatus **200** may include the apparatus reservoir **202** which is configured to receive the cartridge **300** to enable the cartridge **300** to be coupled with the drain cleaner apparatus **200**, but example embodiments are not limited thereto. For example, in some example embodiments, the apparatus reservoir **202** may be entirely absent from the drain cleaner apparatus **200**, and the cartridge **300** may couple with a port that is exposed at the outer surface of the housing **201** of the drain cleaner apparatus **200** to put the cartridge reservoir **304** in fluid communication with the dispenser device **204**.

(120) As shown, the cartridge **300** may have a cartridge housing **302** that defines a cartridge outlet **302A** through which the cleaning composition **230** may exit the cartridge reservoir **304** when a flow path is established between the cartridge reservoir **304** and the dispenser device **204**.

(121) The cartridge outlet **302A** may include a connector interface configured to establish a connection with the dispenser device **204**, and the dispenser device **204** or the apparatus reservoir **202** may further include a complementary connector interface to enable a complementary connection with the cartridge **300**. Such connector interfaces may include any known connector interface, for example a friction fit connector, a threaded connector, a bayonet connector, any combination thereof, or the like.

(122) As further shown, at least one of the cartridge **300** or the drain cleaner apparatus **200** may include a check valve **306** that is configured to be opened based on the drain cleaner apparatus **200** being coupled with the cartridge **300** (e.g., in response to establishing a threaded connection, bayonet connection, friction fit connection, or the like between the drain cleaner apparatus **200** and the cartridge **300**). The check valve **306** may be configured to actuate to open a flow path between the cartridge reservoir **304** and the apparatus reservoir **202** and/or between the cartridge reservoir **304** and the dispenser device **204** in response to the drain cleaner apparatus **200** being coupled with the cartridge **300**, so that the cartridge reservoir **304** is in fluid communication with the dispenser device **204** via the cartridge outlet **302A**.

(123) While, in FIGS. **3A** and **3B**, the check valve **306** is shown as being a part of the cartridge **300** such that the check valve **306** is fixed to the cartridge housing **302** (e.g., via adhesive and/or the cartridge housing **302** being a plastic material (e.g., high density polyethylene or HDPE) that is formed to at least partially enclose the check valve **306**), example embodiments are not limited thereto. For example, in some example embodiments, the check valve **306** may be fixed to the apparatus reservoir **202** and/or the dispenser device **204**. The check valve **306** may be included in a connector that is configured to couple with the cartridge **300** to establish the coupling between the drain cleaner apparatus **200** and the cartridge **300**. For example, the check valve **306** may be included in a threaded connector, bayonet connector, friction fit connector, or the like. In another example, the check valve **306** may be removably (e.g., detachably) coupled to the apparatus reservoir **202** and/or the dispenser device **204** via a set of complementary connectors (e.g., threaded, bayonet, etc.), and the check valve **306** may be detached from the drain cleaner apparatus **200** and coupled to the cartridge **300** prior to coupling of the drain cleaner apparatus **200** with the cartridge **300**, and the check valve **306** may be detached from the cartridge **300** subsequent to removal of an empty cartridge **300** from the drain cleaner apparatus **200** and then attached to a new, full cartridge **300** prior to coupling of the full cartridge **300** to the drain cleaner apparatus **200**, such that a check valve **306** may be re-used between separate cartridges **300**.

(124) Accordingly, in some example embodiments, the apparatus reservoir **202** may be configured to receive a cartridge **300** that includes a cartridge reservoir **304** configured to hold the cleaning composition **230**, and a cartridge outlet **302A**, and the drain cleaner apparatus **200** may be configured to couple with the cartridge **300** so that the cartridge reservoir **304** is in fluid communication (e.g., via an open flow channel) with the dispenser device **204** via the cartridge outlet **302A**. Additionally, in some example embodiments, the drain cleaner apparatus **200** or the cartridge **300** may include a check valve **306** that is configured to open in response to the drain cleaner apparatus **200** coupling with the cartridge **300** to establish the fluid communication between the cartridge reservoir **304** and the dispenser device **204** via the cartridge outlet **302A**.

(125) It will be understood that the dispenser device **204**, the controller **210**, the power supply **212**, and/or the network communication interface **224** of the drain cleaner apparatus **200** of FIGS. **3A** and **3B** may be configured to operate similarly to the described operation thereof as presented herein with reference to the example embodiments shown in FIGS. **2A** and **2B**, except that replenishment of cleaning composition **230** held in the drain cleaner apparatus **200** is implemented via replacing the cartridge **300** coupled to the drain cleaner apparatus **200** instead of directly

pouring cleaning composition **230** into the apparatus reservoir **202**. It will further be understood that the dispenser device **204**, the controller **210**, the power supply **212**, and/or the network communication interface **224** of the drain cleaner apparatus **200** of any of the example embodiments may be configured to operate similarly to the described operation thereof as presented herein with reference to the example embodiments shown in FIGS. 2A and 2B.

(126) FIG. 4 is a schematic view of a drain cleaner apparatus **200** including a dispenser device **204** that further includes first and second valves **402** and **404** and a dispenser reservoir **406** according to some example embodiments. Referring to FIG. 4 in reference to FIG. 1, the drain cleaner apparatus **200** is configured to dispense a cleaning composition **230** into a condensate drain line **124** of the air handler **102** shown in FIG. 1.

(127) The drain cleaner apparatus **200** shown in FIG. 4 may include some or all of the same elements as the drain cleaner apparatus of any of the example embodiments. For example, the example embodiments shown in FIG. 4 include an apparatus reservoir **202** configured to directly hold cleaning composition **230**, similarly to the example embodiments shown in FIGS. 2A and 2B, but it will be understood that the drain cleaner apparatus **200** shown in FIG. 4 may be configured to couple with a cartridge **300** as shown in FIGS. 3A and 3B instead of cleaning composition **230** being directly held (e.g., poured into) the apparatus reservoir **202** and/or the apparatus reservoir **202** may be entirely absent (e.g., where the dispenser device **204** is configured to couple with a cartridge **300** that is external to housing **201**). Conversely, it will be understood that the drain cleaner apparatus **200** according to any of the example embodiments (e.g., the example embodiments shown in FIGS. 2A and 3B, the example embodiments shown in FIGS. 3A and 3B, or the like) may include the dispenser device **204** as shown in FIG. 4.

(128) Referring to FIG. 4, in some example embodiments, the dispenser device **204** may include a dispenser reservoir **406** that is configured to hold the particular amount of the cleaning composition **230** that is to be dispensed when the dispenser device **204** is actuated. For example, the dispenser reservoir (which may be a container having two openings **406A** and **406B** as shown) may have an internal volume of exactly or about 3 oz.

(129) The dispenser device **204** may include a first valve **402** between the apparatus reservoir **202** and the dispenser reservoir **406**. The dispenser device **204** may further include a second valve **404** between the dispenser reservoir **406** and the apparatus outlet **206**. As shown, the dispenser reservoir **406** may be directly between the first and second valves **402** and **404**, where a first opening **406A** of the dispenser reservoir **406** is connected to an outlet of the first valve **402** and the second opening **406B** of the dispenser reservoir **406** is connected to an inlet of the second valve **404**. The first and second valves **402** and **404** may each be any known type of valve, including for example a solenoid valve.

(130) In some example embodiments, the first valve **402** is configured to be actuated (e.g., based on a control signal generated by the controller **210**) to selectively open or close a first flow path **402A** between the apparatus reservoir **202** and the dispenser reservoir **406**, and the second valve **404** may be configured to be actuated (e.g., based on a separate control signal generated by the controller **210**) to selectively open or close a second flow path **404A** between the dispenser reservoir **406** and the apparatus outlet **206**.

(131) In some example embodiments, the controller **210** may be configured to actuate the dispenser device **204** based on causing the first valve **402** to open the first flow path **402A** for a first period of time, to enable the dispenser reservoir **406** to be filled with an amount of the cleaning composition **230** from the apparatus reservoir **202**. The controller **210** may cause the first valve **402** to remain open for a first period of time that is sufficiently long to fill the dispenser reservoir **406** from the apparatus reservoir **202** (and/or cartridge **300** in example embodiments where the drain cleaner apparatus **200** is configured to be coupled to a cartridge **300** as described with regard to FIGS. 3A and 3B) regardless of the amount of cleaning composition **230** held in the apparatus reservoir **202** (directly and/or via a cartridge **300** coupled to the drain cleaner apparatus **200**), so that the

dispenser reservoir **406** holds an amount of cleaning composition **230** that corresponds to (e.g., matches) the internal volume of the dispenser reservoir **406**.

(132) In some example embodiments, the controller **210** may be configured to, in response to an elapse of the first period of time, cause the first valve **402** to close the first flow path **402A** to isolate the dispenser reservoir **406** from the apparatus reservoir **202**, and cause the second valve **404** to open the second flow path **404A** to enable the amount of the cleaning composition **230** held in the dispenser reservoir **406** to flow from the dispenser reservoir **406** to the apparatus outlet **408**. As a result, the dispenser device **204** may be configured to cause an amount of cleaning composition **230** that is dispensed at each actuation of the dispenser device **204** to be controlled to be a particular amount which corresponds to the specific internal volume of the dispenser reservoir **406**, so that the drain cleaner apparatus **200** is configured to improve the uniformity of the amount of cleaning composition **230** dispensed at each actuation of the dispenser device **204**.

(133) FIG. **5** is a schematic view of a drain cleaner apparatus **200** including a moisture sensor **500** according to some example embodiments. Referring to FIG. **5** in reference to FIG. **1**, the drain cleaner apparatus **200** is configured to dispense a cleaning composition **230** into a condensate drain line **124** of the air handler **102** shown in FIG. **1**.

(134) The drain cleaner apparatus **200** shown in FIG. **5** may include some or all of the same elements as the drain cleaner apparatus of any of the example embodiments. For example, the example embodiments shown in FIG. **5** include an apparatus reservoir **202** configured to directly hold cleaning composition **230**, similarly to the example embodiments shown in FIGS. **2A** and **2B**, but it will be understood that the drain cleaner apparatus **200** shown in FIG. **5** may be configured to couple with a cartridge **300** as shown in FIGS. **3A** and **3B** instead of cleaning composition **230** being directly held (e.g., poured into) the apparatus reservoir **202** and/or the apparatus reservoir **202** may be entirely absent (e.g., where the dispenser device **204** is configured to couple with a cartridge **300** that is external to housing **201**). Additionally, the drain cleaner apparatus **200** shown in FIG. **5** may include the dispenser device **204** shown in FIG. **4**. Conversely, it will be understood that the drain cleaner apparatus **200** according to any of the example embodiments (e.g., the example embodiments shown in FIGS. **2A** and **3B**, the example embodiments shown in FIGS. **3A** and **3B**, the example embodiments shown in FIG. **4**, or the like) may include some or all of the elements of the drain cleaner apparatus **200** as shown in FIG. **5**.

(135) Referring to FIG. **5**, in some example embodiments, the drain cleaner apparatus **200** may include a moisture sensor **502** configured to extend through the opening **125** into the condensate drain line **124** based on the connector interface **208** being connected to the condensate drain line **124**. The moisture sensor **502** may be any known moisture sensor, for example a sensor device that is configured to receive electrical power from power supply **212** (either directly or via controller **210** and including a switch that is closed in response to contact with a liquid such as water). The moisture sensor **502** may thus be configured to generate a signal based on contacting condensate backup in the condensate drain line **124**.

(136) Such a signal may be used (e.g., may be processed by controller **210**) to make a determination that a backflow and/or overflow of condensate **120** in the condensate drain line **124** is occurring and/or is about to occur. The signal may be used to prompt a shutdown of at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**, including shutdown of at least one of the air mover **108**, compressor **150**, and/or air mover **154**) which may reduce or stop accumulation of condensate **120** in the drip pan **122** and the condensate drain line **124**, which may therefore reduce or prevent damage to the air handler **102** and/or structure **1** that may otherwise result from the backflow and/or overflow of condensate **120** in the condensate drain line **124**.

(137) In some example embodiments, the drain cleaner apparatus **200** may include a bypass device **506** that is configured to be actuated to cause at least the air handler **102** to shut down based on the signal generated by the moisture sensor **502**. Such a bypass device may be a float switch bypass device that, when actuated, generates a signal that is transmitted to the controller **140** of the air

conditioning system **100** and bypasses the float switch **160** of the air conditioning system **100** to serve as a float switch signal and thus cause the controller **140** to shut down some or all of the air conditioning system **100** (e.g., at least the air handler **102**), which may include shutting down at least one of the air mover **108**, compressor **150**, and/or air mover **154**.

(138) In the example embodiments shown in FIG. 5, the bypass device **506** is a separate device in a housing **508** that is attached to the housing **201** of the drain cleaner apparatus **200**, but example embodiments are not limited thereto. For example, the bypass device **506** may be included in and/or may be implemented by the controller **210**, such that the controller **210** may generate a signal that causes the controller **140** to shut down some or all of the air conditioning system **100** (e.g., at least the air handler **102**), which may include shutting down at least one of the air mover **108**, compressor **150**, and/or air mover **154**. In some example embodiments, the controller **210** may be communicatively coupled between the moisture sensor **502** and the bypass device **506** (e.g., switch), and the controller **210** may be configured to actuate the bypass device **506** in response to the controller **210** processing a signal generated by the moisture sensor **502** to determine that the bypass device **506** is to be actuated.

(139) In some example embodiments, the bypass device **506**, the controller **210**, and/or the network communication interface **224** may be communicatively coupled to the controller **140** of the air conditioning system to enable communication of a shutdown signal to the controller **140** in response to the signal generated by the moisture sensor **502**. Such a communication coupling may be a wired communication link between the drain cleaner apparatus **200** and the controller **140**, a wireless network communication link between the drain cleaner apparatus **200** and the controller **140**. For example, the air conditioning system **100** may include a network communication interface **142** separate from, included in, and/or implemented by controller **140**, and the controller **210**, and/or the network communication interface **224** may be communicatively coupled to the controller **140** of the air conditioning system via a network communication link (e.g., wireless network communication link) between network communication interface **224** and a corresponding network communication interface **142** coupled to, included in, and/or implemented by controller **140** of the air conditioning system **100**.

(140) Still referring to FIG. 5, the drain cleaner apparatus **200** may include a containment tube **504** configured to extend through the opening **125** into the condensate drain line **124** based on the connector interface **208** being connected to the condensate drain line **124**. As shown, the moisture sensor **502** may be located within an interior of the containment tube **504**, and the containment tube **504** may have an open end **503** that is exposed to the interior of the condensate drain line **124**. As a result, the containment tube **504** may be configured to isolate the moisture sensor **502** from generating a signal based on the cleaning composition **230** being dispensed by the dispenser device **204** through the apparatus outlet **206**, thereby reducing or preventing the risk of a false-positive signal being generated by the moisture sensor **502**. The containment tube **504** may further be configured to expose the moisture sensor **502** to the condensate drain line **124** through the open end **503** of the containment tube **504**, to enable a condensate **120** backup in the condensate drain line **124** to pass into the interior of the containment tube **504** to contact the moisture sensor **502** and thus enable the moisture sensor **502** to generate the signal indicating condensate **120** backflow/overflow.

(141) While FIG. 5 shows the bypass device **506**, in some example embodiments the bypass device **506** and housing **508** may be omitted and the controller **210** may be communicatively coupled to the float switch **160** of the air handler **102** and may be configured to cause the float switch **160** to actuate to cause some or all of the air conditioning system **100** to shut down (e.g., based on operation of the controller **140** in response to float switch **160** actuation) based on the signal generated by the moisture sensor **502**.

(142) In some example embodiments, the drain cleaner apparatus **200** may include a network communication interface **224** that is configured to establish a network communication link with a

remote computing device, as described herein, and the controller **210** may be configured to generate and transmit a warning signal to the remote computing device via the network communication link in response to detection of the signal generated by the moisture sensor **502**. As a result, the drain cleaner apparatus **200** may be configured to warn a human user supported by the remote computing device of the occurrence of the detected backflow/overflow of condensate **120** in the condensate drain line **124**.

(143) FIG. **6** is a schematic view of a drain cleaner apparatus **200** including a structure connector **220** according to some example embodiments. Referring to FIG. **6** in reference to FIG. **1**, the drain cleaner apparatus **200** is configured to dispense a cleaning composition **230** into a condensate drain line **124** of the air handler **102** shown in FIG. **1**.

(144) The drain cleaner apparatus **200** shown in FIG. **6** may include some or all of the same elements as the drain cleaner apparatus of any of the example embodiments. For example, the example embodiments shown in FIG. **6** include an apparatus reservoir **202** configured to directly hold cleaning composition **230**, similarly to the example embodiments shown in FIGS. **2A** and **2B**, but it will be understood that the drain cleaner apparatus **200** shown in FIG. **6** may be configured to couple with a cartridge **300** as shown in FIGS. **3A** and **3B** instead of cleaning composition **230** being directly held (e.g., poured into) the apparatus reservoir **202** and/or the apparatus reservoir **202** may be entirely absent (e.g., where the dispenser device **204** is configured to couple with a cartridge **300** that is external to housing **201**). Additionally, the drain cleaner apparatus **200** shown in FIG. **6** may include the dispenser device **204** shown in FIG. **4**. Additionally, the drain cleaner apparatus **200** shown in FIG. **6** may include the moisture sensor **502**, containment tube **504**, and/or bypass device **506** as shown in FIG. **5**. Conversely, it will be understood that the drain cleaner apparatus **200** according to any of the example embodiments (e.g., the example embodiments shown in FIGS. **2A** and **3B**, the example embodiments shown in FIGS. **3A** and **3B**, the example embodiments shown in FIG. **4**, the example embodiments shown in FIG. **5**, or the like) may include some or all of the elements of the drain cleaner apparatus **200** as shown in FIG. **6**.

(145) In some example embodiments, the drain cleaner apparatus **200** may include a structure connector **220** that includes a coupler **602** that is configured to attach to an outer surface of an external structure, such as an outer surface of a housing **101** of the air handler **102**. The coupler **602** may include a magnetic bracket (e.g., any known magnet) that is configured to magnetically attach to a metal surface of the external structure (e.g., a metal surface of the housing **101**). The coupler **602** may enable the structure connector **220** to couple to the external structure to hold the drain cleaner apparatus **200** in place in relation to the condensate drain line **124**.

(146) In some example embodiments, the structure connector **220** may include a set of lateral and vertical adjustable brackets **604A** and **604B**, respectively. The lateral and vertical adjustable brackets **604A** and **604B** may each be an adjustable actuator and/or an adjustable bracket (e.g., adjustable mounting bracket), including for example an adjustable tooth bracket (e.g., an adjustable tooth gear, adjustable worm screw and/or worm gear, adjustable rack and pinion, etc.) that is configured to adjustably position the coupler **602**, in both a horizontal direction and a vertical direction, respectively, in relation to a remainder of the drain cleaner apparatus **200**. As a result, the set of lateral and vertical adjustable brackets **604A** and **604B**, together with the coupler **602**, may enable adjustable positioning of the drain cleaner apparatus **200** in relation to the external structure (e.g., air handler **102**) to which the coupler **602** is attached and/or in relation to the condensate drain line **124**.

(147) FIG. **7** is a schematic view of a drain cleaner apparatus **200** and a remote computing device **700** communicatively coupled via a network communication link **702** according to some example embodiments. Referring to FIG. **7** in reference to FIG. **1**, the drain cleaner apparatus **200** is configured to dispense a cleaning composition **230** into a condensate drain line **124** of the air handler **102** shown in FIG. **1**.

(148) The drain cleaner apparatus **200** shown in FIG. **7** may include some or all of the same

elements as the drain cleaner apparatus of any of the example embodiments. For example, the example embodiments shown in FIG. 7 include an apparatus reservoir **202** configured to directly hold cleaning composition **230**, similarly to the example embodiments shown in FIGS. 2A and 2B, but it will be understood that the drain cleaner apparatus **200** shown in FIG. 7 may be configured to couple with a cartridge **300** as shown in FIGS. 3A and 3B instead of cleaning composition **230** being directly held (e.g., poured into) the apparatus reservoir **202** and/or the apparatus reservoir **202** may be entirely absent (e.g., where the dispenser device **204** is configured to couple with a cartridge **300** that is external to housing **201**). Additionally, the drain cleaner apparatus **200** shown in FIG. 7 may include the dispenser device **204** shown in FIG. 4. Additionally, the drain cleaner apparatus **200** shown in FIG. 7 may include the moisture sensor **502**, containment tube **504**, and/or bypass device **506** as shown in FIG. 5. Additionally, the drain cleaner apparatus **200** shown in FIG. 7 may include the structure connector **220** as shown in FIG. 6. Conversely, it will be understood that the drain cleaner apparatus **200** according to any of the example embodiments (e.g., the example embodiments shown in FIGS. 2A and 3B, the example embodiments shown in FIGS. 3A and 3B, the example embodiments shown in FIG. 4, the example embodiments shown in FIG. 5, the example embodiments shown in FIG. 6, or the like) may include some or all of the elements of the drain cleaner apparatus **200** as shown in FIG. 7.

(149) In some example embodiments, the drain cleaner apparatus **200** includes a network communication interface **224** (e.g., a wireless network communication transceiver) that is configured to establish a network communication link with a remote computing device **700**. The remote computing device **700** may be configured to support a human user.

(150) As shown, the remote computing device **700** may include a processor **720** (e.g., a CPU), a memory **730** (e.g., a SSD), a power supply **740** (e.g., a rechargeable battery), a network communication interface **750** (e.g., a wireless network communication transceiver), and an interface **760** that may include a display device (e.g., an LED display panel, an OLED display panel, or the like) a button, a touchscreen display device, any combination thereof, or the like that are communicatively and/or electrically coupled via a bus connection **710**.

(151) At least some of the remote computing device **700**, including for example the processor **720**, the memory **730**, the network communication interface **750**, or any combination thereof, may be included in, and/or may be implemented by one or more instances (e.g., articles, pieces, units, etc.) of processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), or any other device or devices capable of responding to and executing instructions in a defined manner. It will be understood that any type of non-transitory computer readable storage device may be used as the memory **730** in addition or alternative to an SSD. In some example embodiments, the processing circuitry may include a non-transitory computer readable storage device, or memory (e.g., memory **730**), for example a solid state drive (SSD), storing a program of instructions, and a processor (e.g., processor **720**) that is communicatively coupled to the non-transitory computer readable storage device (e.g., via a bus connection **710**) and configured to execute the program of instructions to implement the functionality of some or all of any of the devices and/or mechanisms of any of the example embodiments and/or to implement some or all of any of the methods of any of the example embodiments. It will be understood that, as described herein, an element (e.g., processing circuitry, digital circuits, any part of the remote computing device **700**) will be understood to implement the functionality of said implemented element (e.g., the functionality of the remote computing device **700**).

(152) As shown, the network communication interface **224** of the drain cleaner apparatus **200** may

be configured to establish a network communication link **702** with the remote computing device **700** (e.g., with network communication interface **750**) and may be configured to implement one-way or two-way communication between the drain cleaner apparatus **200** and the remote computing device **700**.

(153) In some example embodiments, the controller **210** is configured to generate and transmit signals to the remote computing device **700** via the wireless network communication link **702**.

(154) In some example embodiments, the controller **210** may communicate signals over the network communication link **702** that indicate operations of the controller **210** (e.g., indicating actuation of the dispenser device **204** at particular points in time, a present timer value, a present counter value, etc.). In some example embodiments, the controller **210** may communicate the depletion signal (generated in response to the counter value reaching a threshold value) to the remote computing device **700** via the network communication link **702**.

(155) In some example embodiments, the controller **210** may be configured to perform operations in response to receiving signals from the remote computing device **700** via the network communication link **702**. Such signals may be generated at the remote computing device **700** based on operation of at least a portion of the remote computing device **700** (e.g., based on operation of the processor **720**), which may be based on human user interaction with at least a portion of an interface of the remote computing device **700** (e.g., the display screen interface **760**, which may be a touchscreen display). For example, the remote computing device **700** may generate a reset signal based on human interaction with a display screen interface **760** to indicate that the amount of cleaning composition **230** held in the drain cleaner apparatus **200** has been replenished (e.g., via replacement of a cartridge **300** coupled to the drain cleaner apparatus **200**). The remote computing device **700** may transmit the reset signal to the drain cleaner apparatus **200** via the network communication link **702**, and the controller **210** may be configured to cause the counter value of the counter value to be reset to an initial counter value (e.g., 0) in response to receiving the reset signal from the remote computing device **700** via the network communication link **702**. As a result, a human user may be able to remotely reset the counter value used by the drain cleaner apparatus **200** in response to cleaning composition **230** replenishment without direct interaction with the drain cleaner apparatus (e.g., via a button on the drain cleaner interface).

(156) Referring to FIGS. 5 and 7, in some example embodiments, the controller **210** may be configured to generate and transmit a warning signal to the remote computing device **700** via the network communication link **702** in response to detection of a signal generated by the moisture sensor **502**. As a result, the drain cleaner apparatus **200** may be configured to warn a human user supported by the remote computing device **700** of the occurrence of the detected backflow/overflow of condensate **120** in the condensate drain line **124**.

(157) In some example embodiments, the controller **210** may be configured to cause some or all of the air conditioning system **100** to shut down in response to receiving a shutdown command signal from the remote computing device **700** via the network communication link **702**. For example, the remote computing device **700** may display a warning notification to a supported user (e.g., via display screen interface **760**) in response to receiving the warning signal to the remote computing device **700**. The remote computing device **700** may enable the human user to interact with the interface **760** (e.g., a touchscreen display) to command the remote computing device **700** to transmit a shutdown signal to the drain cleaner apparatus **200** in response to the warning signal via the network communication link **702**. The remote computing device **700** may transmit the shutdown signal to the drain cleaner apparatus **200** via the network communication link **702**. The controller **210** may generate a signal to cause some or all of the air conditioning system **100** to shut down (e.g., transmit a signal to the controller **140** via a network communication link **790** with a network communication interface of the air conditioning system **100** that may be included in and/or implemented by controller **140** to cause the controller **140** to shut down some or all of the air conditioning system **100**, actuate the bypass device **506** and/or the float switch **160**, etc.) in

response to receiving the shutdown signal.

(158) In some example embodiments, the remote computing device **700** may enable the human user to interact with the interface **760** (e.g., via a touchscreen display) to command the remote computing device **700** to transmit a dispensing signal to the drain cleaner apparatus **200** to cause the controller **210** to implement an immediate actuation of the dispenser device **204** to immediately dispense an amount of the cleaning composition **230**, thereby allowing more frequent or user-commanded dispensings of cleaning composition. The remote computing device may transmit the dispensing signal to the drain cleaner apparatus **200** via the network communication link **702**, and the controller **210** may actuate the dispenser device **204** in response to receiving the dispensing signal.

(159) FIG. **8** is a flowchart illustrating a method of operation of the drain cleaner apparatus according to some example embodiments. The method shown in FIG. **8** may be implemented by any example embodiment of the drain cleaner apparatus **200** according to any example embodiments.

(160) It will be understood that operations of the method shown in FIG. **8** may be changed in order relative to what is shown in FIG. **8**. It will further be understood that one or more operations of the method shown in FIG. **8** may be omitted from the method shown in FIG. **8**. It will further be understood that one or more operations may be added to the method shown in FIG. **8**.

(161) The method shown in FIG. **8** includes a method for operating a drain cleaner apparatus **200** according to any of the example embodiments to dispense a cleaning composition **230** into a condensate drain line **124** of an air handler **102**, where the drain cleaner apparatus **200** is coupled with the condensate drain line **124** such that an apparatus outlet **206** of the drain cleaner apparatus **200** is in fluid communication with an opening **125** of the condensate drain line **124**. As shown, the method of FIG. **8** includes controlling a dispenser device **204** of the drain cleaner apparatus **200** to cause the dispenser device **204** to selectively dispense an amount (e.g., 3 oz) of the cleaning composition **230** from an apparatus reservoir **202** of the drain cleaner apparatus **200** and through the apparatus outlet **206** without manual intervention (e.g., without human intervention). It will be understood that some or any of the operations shown in FIG. **8** may be performed (e.g., performed by controller **210**) without human intervention (e.g., some or any operations may be performed by controller **210** based on programming of the controller **210** and may be performed independently of any commands or signals received at the controller **210** based on human interaction with an interface (e.g., button, touchscreen display, etc.).

(162) At **S802** and **S804**, a timer of the controller **210** may count (e.g., increment a timer value at a fixed frequency) from an initial timer value (e.g., 0). At **S806**, the controller **210** compares the timer value with a threshold (e.g., particular) timer value (e.g., 7 days) that may be stored at the controller **210** and determines whether the present timer value has reached (e.g., is equal to or greater than) the threshold timer value. If not, the controller **210** permits the timer to continue to increment at **S804**. If so, at **S808**, the controller **210** actuates the dispenser device **204** in response to cause the dispenser device **204** to dispense a particular amount of cleaning composition **230** (e.g., 3 oz), thereby actuating the dispenser device **204** in response to an elapse of a particular period of time.

(163) The amount of cleaning composition **230** that is dispensed at **S808** may be based on structural features of the dispenser device **204** and control thereof. For example, referring to FIG. **4**, in example embodiments where the dispenser device **204** includes a dispenser reservoir **406** that is configured to hold the amount of the cleaning composition (e.g., 3 oz internal value), a first valve **402** between the apparatus reservoir **202** and the dispenser reservoir **406** and configured to be actuated to selectively open or close a first flow path **402A** between the apparatus reservoir **202** and the dispenser reservoir **406**, and a second valve **404** between the dispenser reservoir **406** and the apparatus outlet **206** and configured to be actuated to selectively open or close a second flow path **404A** between the dispenser reservoir **406** and the apparatus outlet **206**, the actuating of the

dispenser device at **S808** may include generating a signal to cause the first valve **402** to open the first flow path **402A** for a first period of time (e.g., 5 seconds), to enable the dispenser reservoir **406** to be filled (e.g., completely filled) with the amount of the cleaning composition **230** (e.g., an amount corresponding to the internal volume of the dispenser reservoir **406**) from the apparatus reservoir **202**, and, in response to an elapse of the first period of time, causing the first valve **402** to close the first flow path **402A** to isolate the dispenser reservoir **406** from the apparatus reservoir **202** and causing the second valve **404** to open the second flow path **404A** to enable the amount of the cleaning composition to flow from the dispenser reservoir **406** to the apparatus outlet **206** and thus to be dispensed through opening **125** into the condensate drain line **124**.

(164) At **S810**, in response to the actuating at **S808**, the controller **210** causes the timer to reset to the initial timer value (0) and resume counting to enable a repeated performance of **S802-S808** (at least partially depending upon an outcome of the determination at **S826**, described further below), thereby repeatedly actuating the dispenser device **204** at a fixed time interval that is the particular period of time, based on monitoring a timer that increments a timer value at a fixed frequency at **S802-S806**, actuating the dispenser device **204** at **S808** in response to the timer value reaching a particular time value corresponding to the elapse of the particular period of time, and resetting the timer value to an initial timer value at **S810** in response to actuating the dispenser device at **S808**.

(165) At **S812**, in response to the actuating at **S808**, the controller **210** causes a counter to count (e.g., increment) a counter value from an initial counter value (e.g., 0), thereby tracking a quantity of actuations (**S808**) and thus a cumulative amount of cleaning composition **230** dispensed.

(166) At **S814** the controller **210** compares the counter value with a threshold (e.g., particular) counter value (e.g., 10, 11, 12, etc.) that may be stored at the controller **210** and determines whether the present counter value has reached (e.g., is equal to or greater than) the threshold counter value. If not, the controller **210** returns to **S802** and continues the method. If so, at **S816**, the controller **210** generates a warning signal. The controller **210** may monitor multiple possible threshold values, including a partial depletion threshold counter value (e.g., 10 and/or 11) and a final depletion threshold counter value (e.g., 12) and the controller **210** may generate a particular warning signal (e.g., indicating partial depletion or final depletion (e.g., complete depletion) of cleaning composition **230** held in the drain cleaner apparatus **200**) based on which threshold is determined to be reached at **S814**.

(167) At **S818**, a determination is made regarding whether to reset the counter to the initial counter value. The determination may include a determination of whether a reset signal that indicates a command to reset the counter value is received. Such a determination may be based upon receiving a reset signal, which may be received from a counter reset interface **222** of the drain cleaner apparatus **200** (e.g., a button) and/or from a remote computing device **700** via a network communication link **702** (e.g., via network communication interface **224**). If a reset is determined to be commanded at **S818** (e.g., a reset signal is determined to be received at **S818**), at **S820** the controller **210** resets the counter value to the initial counter value. If not, at **S822** a further determination is made regarding whether the threshold determined to be reached at **S814** is a final depletion threshold (e.g., **12**) that indicates complete depletion (e.g., final depletion) of cleaning composition **230** in the drain cleaner apparatus **200**. If not, (e.g., a partial depletion threshold of **11** was determined to be reached at **S814**), then the method returns to **S802**. If so, at **S824** the controller **210** may inhibit further operation of the dispenser device **204** (e.g., disable the dispenser device **204**) until a determination is made at **S818** to perform a reset at **S820** (e.g., until a reset signal is determined to be received at **S818**). Such operations at **S822** and **S824** may reduce or prevent the likelihood of the drain cleaner apparatus **200** continuing to actuate the dispenser device **204** in the absence of cleaning composition **230** in the drain cleaner apparatus **200**. At **S824**, the controller **210** may further generate another warning signal indicating that the dispenser device **204** is inhibited (e.g., disabled). Additionally or alternatively, such an indication may be included in the warning signal generated at **S816** in response to a determination at **S814** that a final threshold

counter value is reached.

(168) At **S826**, a determination is made regarding whether a dispensing command is received, for example based on human interaction with an interface (e.g., button) of the drain cleaner apparatus **200** and/or based on a dispensing signal being received from a remote computing device **700** via a network communication link based on a dispensing of cleaning composition **230** being commanded at the remote computing device **700**. If not, the method continues at **S802**. If so, the method moves to **S808** and the controller **210** actuates the dispenser device **204**.

(169) FIG. **9** is a flowchart illustrating a method of operation of the drain cleaner apparatus according to some example embodiments. The method shown in FIG. **9** may be implemented by any example embodiment of the drain cleaner apparatus **200** according to any example embodiments.

(170) It will be understood that operations of the method shown in FIG. **9** may be changed in order relative to what is shown in FIG. **9**. It will further be understood that one or more operations of the method shown in FIG. **9** may be omitted from the method shown in FIG. **9**. It will further be understood that one or more operations may be added to the method shown in FIG. **9**.

(171) At **S902**, a moisture sensor **502** of the drain cleaner apparatus **200**, which is coupled to the condensate drain line **124** such that the moisture sensor **502** is within the condensate drain line **124**, generates a signal in response to contact thereof with moisture (e.g., liquid, including water) within a condensate drain line **124**. Such moisture (e.g., liquid) may contact the moisture sensor **502** based on entering an open end **503** of a containment tube **504** in which the moisture sensor **502** is located.

(172) At **S904**, the controller **210** generates a warning signal in response to receiving and processing the signal generated by the moisture sensor **502** at **S902**. The controller **210** may cause the warning signal to be transmitted to a remote computing device **700** via a network communication link **702** therewith according to any example embodiments.

(173) At **S906**, the controller **210** may generate a shutdown signal that causes some or all of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down in response to receiving and processing the signal generated by the moisture sensor **502** at **S902**. The controller **210** may transmit the signal to a bypass device **506** to actuate the bypass device **506** which causes the controller **140** of the air conditioning system **100** to partially or completely shut down the air conditioning system **100** as described herein. At **S906**, the controller **210** may transmit the signal to a float switch **160** of the air conditioning system **100**, additionally or alternatively to transmitting the signal to the bypass device **506**, to actuate the float switch **160** which causes the controller **140** of the air conditioning system **100** to partially or completely shut down the air conditioning system **100** as described herein.

(174) At **S908**, the controller **210** may generate a shutdown signal that causes at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down in response to receiving and processing the signal generated by the moisture sensor **502** at **S902**. The controller **210** may transmit the shutdown signal directly to the controller of the air conditioning system where the signal is processed by the controller **140** and causes the controller **140** to shut down some or all of the air conditioning system **100** (e.g., shut down at least the air handler **102**) as described herein.

(175) As shown in FIG. **9**, in some example embodiments, the controller **210** may receive a shutdown signal at **S910** from a remote computing device **700**, subsequently to transmitting the warning signal to the remote computing device **700** at **S904**. The remote computing device **700** may generate the shutdown signal automatically (e.g., without human intervention) in response to receiving the warning signal that is generated at **S904**. The remote computing device **700** may generate the shutdown signal in response to human user interaction with the remote computing device **700**.

(176) As shown in FIG. **9**, in some example embodiments, the controller **210** may receive a shutdown signal at **S912** from a remote computing device **700**. The remote computing device **700**

may generate the shutdown signal automatically (e.g., without human intervention) or in response to human user interaction with the remote computing device **700**. The shutdown signal may be received at **S912** independently of any warning signal generated at **S904**—while the shutdown signal may be generated at the remote computing device **700** and transmitted to the drain cleaner apparatus **200** to be received at the controller **210** at **S910** in response to the warning signal generated at **S904**, the shutdown signal that is generated at the remote computing device **700** and transmitted to the drain cleaner apparatus **200** to be received at the controller **210** at **S912** may be generated, transmitted, and received independently of any signal generated at the drain cleaner apparatus **200**.

(177) In some example embodiments, the controller **210** may generate a shutdown signal at **S908** that causes some or all of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down in response to receiving the shutdown command at **S910** and/or **S912**. In some example embodiments, the controller **210** may generate a shutdown signal at **S908** independently of any signal generated by the moisture sensor at **S902** (e.g., the controller **210** may generate a shutdown signal at **S908** in response to receiving the shutdown signal at **S912**).

(178) FIG. **10** is a schematic view of a controller of a computing device **1000** according to some example embodiments. The computing device **1000** may implement any of the computing devices, controllers, processors, or the like according to any of the example embodiments, including controller **140**, controller **210**, and any portion of remote computing device **700**.

(179) As shown in FIG. **10**, the computing device **1000** may include some or all of a processor **1020** (e.g., a CPU), a memory **1030** (e.g., a solid state drive, or SSD), a communication interface **1040** (e.g., a wireless network communication interface, which may for example implement network communication interface **224**, network communication interface **750**, network communication interface **142**, a network communication interface of the air conditioning system **100**, or the like), and a power supply **1050** that are communicatively coupled together via a bus connection **1010**. It will be understood that any type of non-transitory computer readable storage device may be used as the memory **1030** in addition or alternative to an SSD. The computing device **1000** may include additional devices, including a user interface device **1060** (e.g., “interface”) that may include a display device (e.g., an LED display screen, OLED display screen, etc.), a touchscreen display, a button interface, any combination thereof, or the like. The user interface device **1060** may be communicatively coupled to the bus connection **1010**.

(180) In some example embodiments, some or all of any of the computing device **1000** may include, may be included in, and/or may be implemented by one or more instances (e.g., articles, pieces, units, etc.) of processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), or any other device or devices capable of responding to and executing instructions in a defined manner. In some example embodiments, the processing circuitry may include a non-transitory computer readable storage device, or memory (e.g., memory **1030**), for example a solid state drive (SSD), storing a program of instructions, and a processor (e.g., processor **1020**) that is communicatively coupled to the non-transitory computer readable storage device (e.g., via a bus connection **1010**) and configured to execute the program of instructions to implement the functionality of some or all of any of the devices and/or mechanisms of any of the example embodiments and/or to implement some or all of any of the methods of any of the example embodiments. It will be understood that, as described herein, an element (e.g., processing circuitry, digital circuits, etc.) that is described as “implementing” an element (e.g., controller **210**, drain cleaner apparatus **200**, controller **140**, air conditioning system **100**, remote computing device **700**, etc.) will be understood to implement the

functionality of said implemented element and/or any other elements (e.g., the functionality of the controller **210**, the functionality of the drain cleaner apparatus **200**, the functionality of the controller **140**, the functionality of the air conditioning system, the functionality of the remote computing device **700**, etc.).

(181) FIG. **11A** is a perspective top-front-right view of a drain cleaner apparatus system **1100** according to some example embodiments. FIG. **11B** is a perspective bottom-rear-left view of the drain cleaner apparatus system **1100** of FIG. **11A** according to some example embodiments. FIG. **11C** is a perspective cross-sectional view of the drain cleaner apparatus system **1100** along cross-sectional view line XIC-XIC' of FIG. **11A** according to some example embodiments. FIG. **11D** is a plan cross-sectional view of the drain cleaner apparatus system **1100** along cross-sectional view line XIC-XIC' of FIG. **11A** according to some example embodiments. FIG. **11E** is a perspective cross-sectional view of the drain cleaner apparatus system **1100** along cross-sectional view line XIE-XIE' of FIG. **11A** according to some example embodiments. FIG. **11F** is a plan cross-sectional view of the drain cleaner apparatus system **1100** along cross-sectional view line XIE-XIE' of FIG. **11A** according to some example embodiments.

(182) FIG. **12A** is a perspective top-front-right view of the drain cleaner apparatus **200** shown in FIG. **11A** according to some example embodiments. FIG. **12B** is a plan cross-sectional view of the drain cleaner apparatus **200** along cross-sectional view line XIIB-XIIB' of FIG. **12A** according to some example embodiments. FIG. **12C** is a plan cross-sectional view of the drain cleaner apparatus **200** along cross-sectional view line XIIC-XIIC' of FIG. **12A**. FIG. **12D** is a plan top view of the of the drain cleaner apparatus **200** of FIG. **12A** according to some example embodiments.

(183) FIG. **13A** is a perspective top-front-right view of the cartridge **300** shown in FIG. **11A** according to some example embodiments. FIG. **13B** is a perspective bottom-rear-left view of the cartridge **300** shown in FIG. **13A** according to some example embodiments. FIG. **13C** is a plan cross-sectional view of the cartridge **300** along cross-sectional view line XIIIC-XIIIC' of FIG. **13A** according to some example embodiments. FIG. **13D** is a plan cross-sectional view of the cartridge **300** along cross-sectional view line XIIID-XIID' of FIG. **13A** according to some example embodiments.

(184) It will be understood that the drain cleaner apparatus **200** shown in FIGS. **11A-12D** may include any of the elements of any of the example embodiments of the drain cleaner apparatus shown in any of the drawings and/or described herein. It will be understood that the cartridge **300** shown in FIGS. **11A-11F** and **13A-13D** may include any of the elements of any of the example embodiments of the cartridge shown in any of the drawings and/or described herein. The drain cleaner apparatus **200** may be referred to interchangeably herein as a drain cleaner base, a drain cleaner apparatus base, a drain cleaner system base, a drain cleaner base device, or the like.

(185) Referring generally to FIGS. **11A-12D**, in some example embodiments, the drain cleaner apparatus **200** includes a housing **201** include a side housing **1104** and a base housing **1106** which are coupled together to at least partially define an interior of the drain cleaner apparatus **200**. As shown, the side housing **1104** may at least partially define one or more portions of the drain cleaner apparatus **200** including, for example, the apparatus reservoir **202**, a connector interface **1110C** of the drain cleaner apparatus **200**, or the like.

(186) Referring to FIGS. **11A-12D** and further referring to FIGS. **13A-13D**, the drain cleaner apparatus **200** may be coupled (e.g., detachably coupled, reversibly coupled, etc.) with a cartridge **300** having a cartridge housing **302** enclosing a cartridge reservoir **304** holding the cleaning composition in order to establish flow communication between the cartridge reservoir **304** and the dispenser device **204** (e.g., an inlet port of the dispenser device **204**) of the drain cleaner apparatus **200**. As shown, the apparatus reservoir **1102**, also referred to herein interchangeably as a connection port structure, cartridge sleeve structure, or the like, is configured to receive and accommodate at least a portion of the cartridge **300** holding the cleaning composition when the cartridge **300** is detachably coupled with the drain cleaner apparatus **200**, such that the apparatus

reservoir **1102** may include one or more inner surfaces **1102s** that may define at least a portion of an open cylindrical enclosure **1102c** which may at least partially enclose at least the cartridge outlet **302A** of the cartridge **300** coupled to the drain cleaner apparatus **200**.

(187) As shown, the drain cleaner apparatus **200** may include a connector interface **1110** configured to couple with one or more complementary connector interfaces of the cartridge **300** to couple the cartridge **300** at the cartridge outlet **302A** with the drain cleaner apparatus **200**. The connector interface **1110** may include a connector structure **1111** configured to engage the cartridge outlet **302A** and to establish a friction fit seal with the cartridge housing **302** to enable flow communication to be established between the cartridge reservoir **304** and the dispenser device **204**. The connector structure **1111** may include an upper disc structure **1110A** having a top surface **1110U** configured to be directly exposed to the cartridge reservoir **304** when the cartridge **300** is coupled with the connector interface **1110** and a cylindrical sidewall structure **1110B** having an outer sidewall surface **1110S** and one or more O-rings **1112** extending circumferentially around the outer sidewall surface **1110S**. As further shown, the connector structure **1111** may include one or more elements at least partially defining a check valve **306**, such as the cylindrical structure **1120**, but example embodiments are not limited thereto. Each of the interfaces and/or structures **1110A**, **1110B**, **1110C**, and/or **1111** may be referred to, individually or collectively, as a connector interface of the drain cleaner apparatus **200**.

(188) The one or more complementary connector interfaces of the cartridge **300** may include, for example, connector interface **1208A** and connector interface **1208B**. Connector interface **1208A** is a bayonet connector and complementary to bayonet connector interface **1110C**. Connector interface **1208B** is an inner surface of the cartridge housing **302** at the cartridge outlet **302A** and configured to engage and establish a friction fit with an outer sidewall surface **1110S** of the connector interface **1110** and/or an O-ring **1112** extending around the outer sidewall surface **1110S**. The bayonet connector interface **1110C** may be configured to couple with the connector interface **1208A** of the cartridge **300** to establish a bayonet interface connection between the drain cleaner apparatus **200** and the cartridge **300**. As shown, the connector interface **1110C** and the connector interface **1208A** of the cartridge **300** may be complementary interfaces, including complementary bayonet connector interfaces, but example embodiments are not limited thereto and may include any type of complementary connector interfaces including, for example, complementary threaded connector interfaces.

(189) As shown, the connector interface **1110C** may be a structure (e.g., bayonet connector interface structure) at least partially defined by a surface and/or structure of the apparatus reservoir **1102**. For example, the apparatus reservoir **1102** structure may have an inner surface **1102s** at least partially defining an open cylindrical enclosure **1102c** configured to receive at least a portion of the cartridge housing **302** including the cartridge outlet **302A** and in some example embodiments further include one or more complementary connector interfaces **1208B** and/or **1208A**, where the inner surface **1102s** at least partially defines lateral sidewalls of the open cylindrical enclosure **1102c** from which the connector interface **1110C** structure (e.g., a bayonet interface structure configured to establish a bayonet connection with a complementary connector interface **1208A** of the cartridge **300**) extends into the open cylindrical enclosure **1102c**. In some example embodiments, either or both of the complementary connector interfaces **1110C/1208A** and/or **1110B/1208B** may couple (e.g., detachably couple) the cartridge **300** with the drain cleaner apparatus **200**. In some example embodiments, the complementary connector interfaces **1110B/1208A** may be configured to couple the cartridge outlet **302A** with the drain cleaner apparatus **200** to establish flow communication between the cartridge reservoir **304** and the dispenser device **204** via at least the inlet port **1136**, and the complementary connector interfaces **1110C/1208A** may secure (e.g., reversibly lock) the cartridge **300** to the drain cleaner apparatus **200**.

(190) As shown in at least FIG. **12B**, the drain cleaner apparatus **200** may include an electrical

switch device **1280** that may include a structure extending into enclosure **1102c** and configured to be engaged and moved from a switch-open position to a switch-closed position by at least a portion of the cartridge **300** when a connector interface of the cartridge **300** (e.g., connector interface **1208A**) couples with a connector interface of the drain cleaner apparatus **200** (e.g., connector interface **1110C**). The electrical switch device **1280** may be configured to close an electrical circuit that includes the controller **210** when moved to the switch-closed position, thereby enabling an electrical signal to be received at the controller **210**. The controller **210** may be configured to apply electrical power to the circuit and may be configured to determine that the cartridge **300** is coupled with the drain cleaner apparatus **200** in response to determining that the circuit including the electrical switch device **1280** is closed such that an electrical signal (e.g., an induced current) is present in the circuit). The controller **210** may be configured to selectively enable or disable actuation of the dispenser device **204** based upon whether a cartridge **300** is determined to be coupled to the drain cleaner apparatus **200** (e.g., based upon receiving an electrical signal via the circuit including the switch device **1280** to determine that the circuit is closed and thus a cartridge **300** is coupled with the drain cleaner apparatus **200** to move the switch device **1280** to the switch-closed position).

(191) Still referring to FIGS. **11A-12D**, the drain cleaner apparatus **200** may include a reservoir **1130**, also referred to herein as a dispenser reservoir, apparatus reservoir, first reservoir of the drain cleaner apparatus **200**, internal reservoir, or the like. While the reservoir **1130** is shown in FIGS. **11A-12D** to be separate from the dispenser device **240**, it will be understood that the reservoir **1130** may be referred to as being a dispenser reservoir included within the dispenser device, separately from a valve of the dispenser device **240**, such as shown in at least FIG. **4** with regard to reservoir **406** and valve **404** of the dispenser device **204** (and absent the first valve **402**, or where the first valve **402** is a check valve as described herein with reference to FIGS. **11A-12D**).

(192) As shown, the drain cleaner apparatus **200** may be configured to establish flow communication from the cartridge reservoir **304** of a coupled (e.g., detachably coupled) cartridge **300** to the reservoir **1130** of the drain cleaner apparatus **200**, where the reservoir **1130** is in flow communication between at least one connector interface of the drain cleaner apparatus **200** (e.g., the connector interface **1110**) and at least one valve **1198** of the dispenser device **204**, which may be the same as any of the valves described herein according to any of the example embodiments, for example any of the first valve **402**, the second valve **404**, or the like. The at least one valve **1198** may, for example, be a solenoid valve. The dispenser device **204** may further be understood to be configured to be in fluid communication between the connector interface **1110** (e.g., via at least the reservoir **1130** and the inlet port **1136**) and the apparatus outlet **206**. The dispenser device **204** may thus be configured to be actuated (e.g., by controller **210**) to selectively dispense an amount (e.g., a particular amount) of the cleaning composition from the cartridge reservoir **304** and through the apparatus outlet **206** (e.g., via the reservoir **1130**). The at least one valve **1198** may be configured to be controlled by the controller **210** to be actuated similarly to any of the valves of any of the example embodiments of the dispenser device **204**. The controller **210** may be configured to actuate the dispenser device **204** (e.g., actuate the at least one valve **1198**) to cause the amount of the cleaning composition to be dispensed through the apparatus outlet **206** without manual intervention.

(193) While the dispenser device **204** may include at least one valve **1198** (e.g., a solenoid valve), example embodiments are not limited thereto. In some example embodiments, the dispenser device **204** may include a pump (e.g., any known positive displacement pump) that is configured to operate for a particular period of time to move an amount of the cleaning composition **230** from the cartridge reservoir **304** and through the apparatus outlet **206** (e.g., from an inlet of the pump that is open to and/or in fluid communication with the reservoir **1130** to an outlet of the pump that is open to and/or in fluid communication with the apparatus outlet **206**, etc.), based on a control signal generated by the controller **210**.

(194) Still referring to FIGS. 11A-12D, the connector interface **1110** may include an upper disc structure **1110A** and a cylindrical sidewall structure **1110B** which may be separate parts of a single piece of material (e.g., plastic) or separate pieces of material of the connector structure **1111**, and where the cylindrical sidewall structure **1110B** may include one or more circumferential grooves configured to accommodate separate, respective O-rings **1112** or any other known seal structure. The outer sidewall surface **1110S** of the cylindrical sidewall structure **1110B** and/or the O-ring(s) **1112** may be configured to engage a complementary inner surface of the cartridge housing **302** at the cartridge outlet **302A** which defines the connector interface **1208B** of the cartridge **300**. As a result, the outer sidewall surface **1110S** of the cylindrical sidewall structure **1110B**, alone or in combination with one or more of the O-rings **1112**, establishes a fluid seal (e.g., air-tight seal) between the cylindrical sidewall structure **1110B** (and thus the connector interface **1110**) and the cartridge housing **302**, thereby minimizing or preventing leaking of cleaning composition from the cartridge reservoir **304** to an exterior of the cartridge **300** independently of being supplied through the apparatus outlet **206** by the dispenser device **204**, for example minimizing or preventing leaking of cleaning composition from the cartridge reservoir **304** into the open cylindrical enclosure **1102c**.

(195) As shown, when the connector structure **1111** and thus the connector interface **1110** couples with the connector interface **1208B** of the cartridge **300** (e.g., at the cylindrical sidewall structure **1110B** where the coupling is sealed by one or more surfaces of the cylindrical sidewall structure **1110B**, the connector interface **1208B**, and/or one or more of the O-rings **1112**), the upper disc structure **1110A** of the connector interface **1110** may be exposed directly to an interior of the cartridge reservoir **304** and at least some or any cleaning composition held in the cartridge reservoir **304**.

(196) Still referring to FIGS. 11A-12D, the connector interface **1110** may include a check valve **306** which may be configured to open in response to the connector interface **1110** coupling with one or more connector interfaces **1208A** and/or **1208B** of the cartridge **300** to establish fluid communication between the cartridge reservoir **304** and the dispenser device **204** (e.g., via the reservoir **1130**). As shown, the check valve **306** may be at least partially defined by a cylindrical structure **1120** (which may be a part of a single piece of material with at least the upper disc structure **1110A** of the connector structure **1111**) having an inner surface **1120** is defining cylindrical side surfaces of an internal cylindrical conduit **1118**, a top plate **1116** defining a top surface of the internal cylindrical conduit **1118** and having one or more ports **1114**, also referred to interchangeably as openings, extending therethrough to the cylindrical conduit **1118** and configured to be directly exposed to at least the open cylindrical enclosure **1102c** and thus to the cartridge reservoir **304** when the cartridge **300** is coupled with the connector interface **1110**, a bottom structure **1122** defining a bottom surface of the internal cylindrical conduit **1118**, a seal **1121** such as an O-ring extending around a lower portion of the bottom structure **1122**, and a spring **1117** in contact between the top plate **1116** and the bottom structure **1122**.

(197) As shown, the bottom structure **1122** may include a pin protrusion extending axially through the cylindrical conduit **1118** and which may extend through a central opening in the top plate **1116**. The bottom structure **1122**, alone or together with the seal **1121**, may be configured to engage against a ledge structure **1120L** of the cylindrical structure **1120** to selectively seal an interface between the bottom structure **1122** and the cylindrical structure **1120**. As further shown, the reservoir **1130** may be at least partially defined by a cylindrical side structure **1124** and a bottom disc structure **1126**, where the bottom disc structure **1126** may at least partially define the inlet port **1136** to the dispenser device **204** (e.g., to at least one valve **1198** thereof). As shown, the cylindrical side and bottom disc structures **1124** and **1126** may define an open cylindrical enclosure that is enclosed at a top end by the combined cylindrical structure **1120** and ledge structure **1120L** thereof and a bottom surface of the bottom structure **1122** extending through an opening space between opposing surfaces of the ledge structure **1120L**, such that inner surfaces of the structures **1124**, **1126**, **1120**, and **1122** at least partially define the reservoir **1130**. As further shown, the drain

cleaner apparatus **200** may include a fixed structure **1128** which may be coupled to the bottom disc structure **1126** and may be a part of a same single piece of material as the bottom disc structure **1126**. The fixed structure **1128** may project upwards into the reservoir **1130** under the bottom structure **1122** of the check valve **306**.

(198) Still referring to FIGS. **11A-12D**, the connector interface **1110** is configured to move axially downwards **1202** (e.g., toward the apparatus outlet **206**) in response to the cartridge **300** coupling with the drain cleaner apparatus **200** (e.g., the connector interface **1110** coupling with one or more of the connector interfaces **1208A** and/or **1208B** of the cartridge **300**), for example based on the weight of the cartridge **300** and the cleaning composition held within pushing the connector interface **1110** downwards **1202**. As shown, the outer surface **1120os** of the cylindrical structure **1120** coupled to the upper disc structure **1110A** is configured to engage and establish a seal (in some example embodiments with one or more O-rings) with the inner surface **1124s** of the cylindrical side structure **1124** at least partially defining the reservoir **1130**, thereby minimizing or preventing leakage of cleaning composition from the reservoir **1130** via the interface between surfaces **1120os** and **1124s**.

(199) As the connector interface **1110** moves downward **1202** due to the weight of the cartridge **300** and cleaning composition therein (which may directly contact the top surface of the upper disc structure **1110A** and the top plate **1116**) may push the top plate **1116** and the cylindrical structure **1120** downwards **1202** axially, where the spring **1117** may further push the bottom structure **1122** axially downwards based on the top plate **1116** pushing the top end of the spring **1117** downwards. As shown, the top plate **1116** may engage an underside of a ledge or lip structure of the upper disc structure **1110A** so that the downwards **1202** axial movement of the upper disc structure **1110A** causes the top plate **1116** to move downwards **1202** axially together with the upper disc structure **1110A**. As a result, the top plate **1116** together with the spring **1117** may cause the bottom structure **1122** and the cylindrical structure **1120** to move downwards **1202** together until a bottom surface of the bottom structure **1122** contacts (e.g., directly contacts) a top surface of the fixed structure **1128** in the reservoir **1130** interior. As the fixed structure **1128** is fixed to a surface at least partially defining the reservoir **1130** (e.g., fixed to the bottom disc structure **1126**), the contact between opposing surfaces of the bottom structure **1122** and the fixed structure **1128** may arrest downwards axial movement of the bottom structure **1122** and compress the spring **1117** while the cylindrical structure **1120**, top plate **1116**, and connector interface **1110** continue to move axially downwards **1202**, thereby causing the relative movement of the bottom structure **1122** in relation to the cylindrical structure **1120** to be upwards **1204**, opening an annular passage **1250** between the downwards-moving ledge structure **1120L** and the arrested bottom structure **1122** (and any washer or seal such as an O-ring seal **1121** configured to seal an interface between the bottom structure **1122** and the ledge structure **1120L**) fixed in place between the spring **1117** and the fixed structure **1128**. The opened annular passage **1250** may enable a flow along flow path **1192** (e.g., based on enabling fluid communication) through the cylindrical conduit **1118** to the reservoir **1130** via ports **1114** and the opened annular passage **1250**.

(200) As long as the weight of the cartridge **300** and the cleaning composition held therein on the connector interface **1110** is greater than the spring force of the spring **1117**, the top plate **1116** and the bottom structure **1122** contacting the fixed structure **1128** may compress the spring **1117** and open the annular passage **1250** to the reservoir **1130** to enable a flow of cleaning composition along the flow path **1192** from the cartridge reservoir **304** to the reservoir **1130** via the check valve **306**. When the weight of the cartridge **300** and the cleaning composition held therein on the connector interface **1110** is smaller than the spring force of the spring **1117**, the spring force of the spring **1117** may enable the spring **1117** to push the top plate **1116**, and thus the connector interface **1110** upwards **1204** axially away from the bottom structure **1122** and/or seal **1121** to close the annular passage **1250** and close the fluid communication between the cartridge reservoir **304** and the reservoir **1130**.

(201) Still referring to FIGS. 11A-12D, the connector structure **1111** may establish (e.g., define) an air volume **1132** in fluid communication with the ambient environment via the open cylindrical enclosure **1102c**, and the connector interface **1110** (e.g., the connector structure **1111**) may include an air tube **1134** extending through the connector interface to the upper disc structure **1110A** to establish fluid connection between the air volume **1132** and a top region of the cartridge reservoir **304** when the cartridge **300** is coupled to the drain cleaner apparatus **200**. The air tube **1134** may be configured to supply air into the upper portion of the cartridge reservoir **304** as cleaning composition leaves the cartridge reservoir **304** via the cartridge outlet **302A** (e.g., via the check valve **306**) to equalize pressure in the cartridge reservoir **304**, thereby preventing vacuum in the cartridge reservoir **304** and preventing loss of flow rate of the flow along flow path **1192** into the reservoir **1130**. The air tube **1134** may include a backflow prevention valve **1134v**, such as a duckbill valve which may also be interchangeably referred to as a duck mouth valve, at a distal end, where the backflow prevention valve **1134v** may be configured to reduce, minimize, or prevent flow of cleaning composition from the cartridge reservoir **304** into the air volume **1132** via the air tube **1134** while still enabling air to flow into the cartridge reservoir **304** from the air volume **1132** via the air tube **1134**.

(202) Still referring to FIGS. 11A-12D, the dispenser device **204** may include at least one valve **1198** (e.g., a solenoid valve corresponding to the second valve **404** shown in at least FIG. 4) that is configured to be controlled (e.g., selectively actuated) by the controller **210** to selectively induce a flow of cleaning composition along the flow path **1194** from the reservoir **1130** to the apparatus outlet **206**, thereby dispensing the cleaning composition from the drain cleaner apparatus **200**. The at least one valve **1198** may operate, and/or may be configured to be controlled to operate/actuate, in the same way as any of the valves described herein according to any of the example embodiments.

(203) Accordingly, as shown in at least FIGS. 11A-12D, the dispenser device **204** may include at least one valve **1198** that is configured to be selectively opened based on a control signal generated (e.g., transmitted) by the controller **210** to establish a flow path **1194** through the at least one valve **1198** to the apparatus outlet **206**, and the drain cleaner apparatus **200** may include a reservoir **1130** (e.g., dispenser reservoir) that is in flow communication between the check valve **306** and the at least one valve **404**, such that the connector interface **1110** is configured to detachably couple with the connector interface **1208A** of the cartridge **300** to establish flow communication (e.g., flow path **1192**) from the cartridge reservoir **304** to the reservoir **1130**, and the dispenser device **204** may be configured to be actuated (e.g., by controller **210**) to selectively dispense an amount of the cleaning composition from the reservoir **1130** and through the apparatus outlet **206**. The controller **210** may be configured to actuate the dispenser device **204** based on causing the at least one valve **1198** to open the flow path **1194** to enable at least a portion of the cleaning composition held in the reservoir **1130** to flow from the reservoir **1130** to the apparatus outlet **206**.

(204) While FIGS. 11A-12D show a drain cleaner apparatus **200** and cartridge **300** configured to couple via a connector interface **1110** which includes a check valve **306**, it will be understood that example embodiments are not limited thereto, and in some example embodiments different configurations of connector interfaces **1110** and/or connector structures **1111** may be present in the drain cleaner apparatus **200**. In some example embodiments, the check valve **306** may be omitted. For example, in some example embodiments the cartridge **300** may include a flexible membrane (e.g., a silicone membrane) extending transversely across the cartridge outlet **302A**, and the connector interface **1110** may include at least a puncturing structure (e.g., one or more needles) configured to puncture the membrane when the cartridge **300** is coupled with the connector interface **1110** in order to establish fluid communication between the cartridge reservoir **304** and the dispenser device **204** (e.g., via reservoir **1130**). The connector interface **1110** may include another puncturing structure (e.g., similar in function to air tube **1134**) configured to allow air to flow into an upper portion of the cartridge reservoir **304** to enable pressure equalization as cleaning

composition flows out of the cartridge reservoir **304**. The connector interface **1110** may include a protecting plate defining a recess having openings aligned with the puncturing structures and that is spring-loaded by a spring and is configured to move vertically between an upper rest position where the puncturing structures are underneath the protecting plate and external to the recess and a lower compressed position where the spring is compressed and where the puncturing structures extend through the openings in the protecting plate to be located within the recess. The protecting plate may be configured to receive the cartridge outlet **301A** into the recess such that the cartridge **300** pushes the protecting plate downwards against the spring to expose the puncturing structures to puncture the membrane of the cartridge **300** and to establish fluid communication between the cartridge reservoir **304** and the dispenser device **204** (e.g., via the reservoir **1130**). Upon removal of the cartridge **300** from the drain cleaner apparatus **200**, the protecting plate may rise, under load from the spring, back to the rest position to obscure the puncturing structures. A distal portion of the cartridge **300** including the cartridge outlet **302A** may be indented (e.g., include a notch structure or cavity) in relation to a remainder to the cartridge housing **302**, and the drain cleaner apparatus **200** may include a spring-loaded locking mechanism configured to engage and couple with the indented portion of the cartridge **300** when the cartridge **300** is inserted into the apparatus reservoir **1102** to hold the cartridge **300** coupled with the drain cleaner apparatus **200**. The locking mechanism may further be configured to lock the protecting plate in the upper rest position when the locking mechanism is in a spring-loaded rest position. The locking mechanism may be configured to move (e.g., move horizontally) against the spring to a compressed position to unlock the vertical movement of the protecting plate, based on the locking mechanism engaging a surface of a cartridge **300** being inserted into the drain cleaner apparatus **200**, thereby enabling the cartridge outlet **302A** to enter the recess of the protecting plate and push the protecting plate downwards to expose the puncturing structures. The cartridge **300** may be configured to include the indented portion that is positioned to engage the locking mechanism when the cartridge outlet **302A** is inserted into the bottom of the recess of the protecting plate and the protecting plate is moved downwards to the lower, compressed position. When the locking mechanism engages the indented portion, the locking mechanism may return from the compressed position to an at least partial rest position, where the locking mechanism engaged with the indented portion may be in locking engagement with the cartridge **300** and may lock the cartridge **300** in place in relation to the drain cleaner apparatus **200**. The drain cleaner apparatus **200** may include a release mechanism configured to release the locking mechanism from locking engagement with the cartridge **300** to enable decoupling of the cartridge **300** from the drain cleaner apparatus **200**.

(205) As further shown in FIGS. **11A-12D**, the drain cleaner apparatus **200** may include a power supply compartment **1140** which may be at least partially defined by the housing **201** (e.g., the side housing **1104**) and in which a power supply (e.g., batteries **1142** may be located and may be electrically coupled (e.g., via internal circuitry of the drain cleaner apparatus **200**) with the dispenser device **204**, the controller **210**, a network communication interface **224**, etc., or the like of the drain cleaner apparatus **200**. The drain cleaner apparatus **200** may include a power supply cover plate **1108** which may be configured to couple with the housing **201** to cover the power supply compartment **1140** and to at least partially define an outer surface of the drain cleaner apparatus **200**.

(206) Still referring to FIGS. **11A-12D**, the drain cleaner apparatus **200** may include a user interface **1182** (e.g., a button) with which a user may interact (e.g., press the button) to control operation of the drain cleaner apparatus **200**, for example to turn the drain cleaner apparatus **200** on or off (e.g., activate or deactivate the drain cleaner apparatus **200**), to cause the controller **210** to enable/activate controlling of the dispenser device **204** (e.g., one or more valves **1198** thereof) to be actuated to dispense cleaning composition at fixed intervals) and/or to cause the controller **210** disable/deactivate the dispenser device **204** from being actuated at fixed intervals. It will be understood that the controller **210** of the drain cleaner apparatus **200** may include any of the

elements of any of the example embodiments of the controller **210** as described herein and/or illustrated in any of the drawings. It will be understood that the drain cleaner apparatus **200** shown in FIGS. **11A-12D** may include any of the elements of any of the example embodiments of the drain cleaner apparatus **200** as described herein and/or illustrated in any of the drawings, including for example a network communication interface **224**.

(207) The drain cleaner apparatus **200** and/or any portion thereof (e.g., controller **210**, network communication interface **224**, etc.) may be configured to perform any of the functions described herein and/or illustrated in any of the drawings with regard to any of the example embodiments. For example, in some example embodiments the controller **210** may be configured to actuate the dispenser device **204** (e.g., at least one valve **1198** thereof) in response to a determination, by the controller **210**, of an elapse of a particular (e.g., predetermined, fixed) period of time. The controller **210** may be configured to repeatedly actuate the dispenser device **204** at a fixed time interval that is the particular period of time, based on monitoring a timer (which may be implemented by the controller **210**) that increments a timer value at a fixed frequency, actuating the dispenser device **204** in response to the timer value reaching a particular time value corresponding to the elapse of the particular period of time, and resetting the timer value to an initial timer value in response to actuating the dispenser device **204**. The controller **210** may be configured to monitor a counter (which may be implemented by the controller **210**) that increments a counter value in response to each actuation of the dispenser device **204**, and generate a depletion signal (which may be communicated to an external device via the network communication interface **224** and/or may be used to generate a visual signal by one or more light indicators **1184** such as activating a yellow LED thereof) in response to the counter value reaching a particular counter value that corresponds to at least partial depletion of a fixed reservoir (e.g., the reservoir **1130** and/or the cartridge reservoir **304**) of the cleaning composition.

(208) In some example embodiments, the controller **210** may be configured to adjust (e.g., calibrate) the particular counter value to correspond to a number of actuations corresponding to a particular volume of the cartridge reservoir **304**. For example, in some example embodiments, the cartridge reservoir **304** is configured to hold a volume of about 36 oz of cleaning composition, but example embodiments are not limited thereto; for example, the drain cleaner apparatus **200** may be configured to couple with various sizes of cartridges **300** having similar connector interfaces **1208A** and **1208B** configured to couple with the connector interface **1110** of the drain cleaner apparatus **200** but having different volumes of cartridge reservoir **304**, including a volume of 36 oz, 72 oz, or the like. The controller **210** may be configured to determine a volume of the cartridge reservoir **304** in response to receiving a command signal indicating the volume of the cartridge reservoir, and adjust the particular counter value based on the determination of the volume of the cartridge reservoir. For example, in some example embodiments the drain cleaner apparatus **200** may be configured to receive a command signal indicating the cartridge reservoir **304** volume of a coupled cartridge **300** via a command from a remote computing device **700** received via the network communication interface **224** based on human user interaction with at least a portion of an interface of the remote computing device **700** (e.g., the display screen interface **760**, which may be a touchscreen display) to cause the remote computing device **700** to inform the drain cleaner apparatus **200** of the volume of the coupled cartridge **300** and/or to command the drain cleaner apparatus **200** to adjust the particular counter value to correspond to the volume of the coupled cartridge **300**. In another example, in some example embodiments the drain cleaner apparatus **200** may be configured to receive a command signal indicating the cartridge reservoir **304** volume of a coupled cartridge **300** via a command received from a user interface **1182** of the drain cleaner apparatus **200** via user interaction therewith.

(209) In another example, in some example embodiments the drain cleaner apparatus **200** may be configured to receive a command signal indicating the cartridge reservoir **304** volume of a coupled cartridge **300** based on sensor data generated by a sensor device of the drain cleaner apparatus **200**.

The drain cleaner apparatus **200** may include a pressure sensor (e.g., any known pressure sensor) that is exposed to the reservoir **1130**, the cylindrical conduit **1118**, the upper surface of the upper disc structure **1110A** configured to be directly exposed to the cartridge reservoir **304** of a coupled cartridge **300**, or any portion of the drain cleaner apparatus **200** configured to be in fluid communication with the cartridge reservoir **304** of a coupled cartridge **300**. The pressure sensor may generate sensor data indicating a static pressure of cleaning composition at the location of the pressure sensor in the drain cleaner apparatus **200** and may communicate such sensor data to the controller **210**. The controller may be configured to process the sensor data to determine a pressure value indicated by the sensor data and may determine a corresponding volume of cleaning composition held in a cartridge reservoir **304** of a coupled cartridge **300** based on applying the sensor data and/or pressure value indicated thereby to an empirically-determined look-up table that associates sensor data and/or indicated pressure values with corresponding magnitudes of volume of cleaning composition held in the cartridge reservoir **304** of a coupled cartridge **300**. The controller **210** may be configured to monitor variations in the pressure data and/or corresponding volume indicated by the sensor data and look-up table over time. In response to a rate of change of the pressure and/or volume indicated by the sensor data that exceeds a threshold rate of change that is stored at the controller, where exceeding the threshold rate is associated with an at least partially depleted cartridge **300** being replaced with a new, more full cartridge **300** being newly coupled to the drain cleaner apparatus **200**, the controller **210** may responsively monitor a new volume indicated by the sensor data and look-up table subsequent to the rate of change of indicated volume/pressure value subsequently dropping below the threshold rate to indicate that the newly-coupled cartridge **300** is stabilized, where the new volume determined based on processing the sensor data in view of the look up corresponds to the volume of the cartridge reservoir **304**. The controller **210** may responsively adjust the particular counter value to a value corresponding to a quantity of actuations (each actuation corresponding to causing the dispenser device **204** to dispense a particular amount (e.g., volume) of cleaning composition such as 3 oz) that is at least a particular proportion of the determined volume of the new cartridge reservoir **304** (e.g., 90% of the determined volume).

(210) In some example embodiments, the drain cleaner apparatus **200** may include a network communication interface **224** that is configured to establish a network communication link with a remote device (e.g., a remote computing device). The controller **210** may be configured to cause a depletion signal to be transmitted to the remote computing device **700** via the network communication link. The controller **210** may be configured to cause the counter value to be reset to an initial counter value in response to receiving a reset signal from the remote computing device via the network communication link. It will be understood that the controller **210** and/or the network communication interface **224** may be configured to perform any of the communications and/or interactions with one or more remote computing devices **700** as described herein with regard to any of the example embodiments of the drain cleaner apparatus **200**, the remote computing device **700**, or the like, including the operations and/or interactions between the drain cleaner apparatus **200** and a remote computing device **700** via network communication link **702** as described herein with regard to at least FIG. 7, FIG. 8, FIG. 9, or the like.

(211) In some example embodiments, referring to FIGS. 11A-12D and further referring to FIG. 1, the drain cleaner apparatus **200** may be communicatively coupled, via an electrical connection **1152** extending through an opening **1150** in the housing **201** or via a wireless network connection, with a float switch of the air handler **102** of an air conditioning system **100** with a controller of the air condition system **100**, with the air handler **102** itself (e.g., with a controller **140** of the air handler **102** via a float switch signal connection interface **148** of the controller **140**), and/or with a device/apparatus configured to control the float switch and/or controller of the air conditioning system **100**.

(212) The electrical connection **1152** may comprise one or more wires (“wiring”) electrically

coupled to the controller **210** and extending out of the drain cleaner apparatus **200**, for example via opening **1150**. For example, in some example embodiments, the electrical connection **1152** may include two or more separate sets of wires, each separate set of wires including at least two wires comprising at least a portion of an electrical circuit that includes at least a portion of circuitry of the drain cleaner apparatus **200** (e.g., an electrical circuit that includes at least the controller **210**), such that the electrical connection **1152** may comprise a plurality of sets of wires that at least partially comprise separate, respective electrical circuits that each include at least a portion of circuitry of the drain cleaner apparatus **200** (e.g., the separate respective electrical circuits may include separate and/or common portions of the controller **210**). Each wire extending from the drain cleaner apparatus **200** may at least partially comprise an electrical connection, implemented by at least serially coupled wires and connections (e.g., male/female connections), between the drain cleaner apparatus **200** and one or more separate devices, as described herein.

(213) In some example embodiments, the controller **210** may be configured to detect an occurrence of an event (e.g., actuation of a float switch) based on detecting a signal in a first electrical circuit (labeled as electrical connection **1502** herein) that comprises at least one set of wires of the electrical connection **1152**, for example a first set of at least two wires extending from the drain cleaner apparatus **200** to be centrally connected to a drain cleaner float switch **1610** as described herein to establish the first electrical circuit as including at least a portion of the drain cleaner apparatus **200** and at least a portion of the drain cleaner float switch **1610**. It will be understood that, as described herein, elements “electrically connected” to other elements may be directly or indirectly electrically connected thereto (e.g., electrically connected via one or more interposing conductive elements, including one or more serially connected wires). The drain cleaner apparatus **200** may provide a power source (e.g., from a power supply of the drain cleaner apparatus **200** such as batteries **1142**) coupled to the first set of at least two wires to drive the electrical current in the first electrical circuit when said first electrical circuit is closed. The signal detected in the first electrical circuitry by the drain cleaner apparatus **200** may include a float switch signal that may be detected by the drain cleaner apparatus **200** (e.g., by controller **210**) as an initiation or inhibition of electrical current in the first electrical circuit due to closing or opening of a switch implemented by and/or included in the drain cleaner float switch **1610** to close or open the first electrical circuit. For example, the drain cleaner apparatus **200** may include a sensor such as an ammeter, current sensor, or the like configured to detect (e.g., generate signals which may be processed by the controller **210** to detect in response to) a presence and/or magnitude of electrical current in the first set of at least two wires extending from the drain cleaner apparatus **200** to at least partially comprise the first electrical circuit. The controller **210** may respond to a determination of a presence or absence of electrical current (e.g., a presence or absence of current above a threshold magnitude) in the first set of at least two wires, a change between an absence and a presence of at least a threshold electrical current in the first set of at least two wires, or the like, based on processing and/or detecting signals generated by the sensor to determine that a float switch signal is received from the drain cleaner float switch **1610** indicating that the drain cleaner float switch **1610** is actuated in response to a fluid engaging the drain cleaner float switch **1610**.

(214) In some example embodiments, the controller **210** may be configured to, in response to detecting the occurrence of the event (e.g., in response to detecting the float switch signal from the drain cleaner float switch **1610** via detecting presence, absence, change between presence and absence, etc. of an electrical current in the first electrical circuit), transmit a separate signal in a separate, second electrical circuit (labeled as electrical connection **1504** in FIG. **15A**) that comprises a separate, second set of wires of the electrical connection **1152**, for example a second set of at least two wires extending from the drain cleaner apparatus **200** to a remote device, such as the actuator apparatus **900** as described herein, an actuator **910** of the actuator apparatus **900** as described herein, or the like to establish the second electrical circuit as including at least a portion of the drain cleaner apparatus **200** and at least a portion of the actuator apparatus **900** (e.g., the

actuator **910**, a servoactuator or servomotor thereof, etc.). The drain cleaner apparatus **200** may provide a power source (e.g., from a power supply of the drain cleaner apparatus **200** such as batteries **1142**) coupled to the second set of at least two wires to drive an electrical current in the second electrical circuit when said second electrical circuit is closed. The separate signal may be “transmitted” based on the drain cleaner apparatus **200** (e.g., the controller **210**) controlling (e.g., initiating, for a particular, predetermined period of time) a flow of electrical current in the second electrical circuit (e.g., based on controller **210** operating a switch implemented by and/or included in the drain cleaner apparatus **200** to close or open the second electrical circuit, respectively, based on the controller **210** selectively and/or adjustably controlling a supply of electrical power to the actuator apparatus **900** and/or the actuator **910** via the second electrical circuit, or the like) for a particular period of time. Such controlled flow of electrical current in the second electrical circuitry may be referred to as a signal, command, signal or the like (e.g., an actuator command signal) to cause the actuator **910** of the actuator apparatus **900** to actuate an air handler float switch **160** of the air handler **102** which is held in the actuator apparatus **900**, to cause a flow of electrical current in a separate electrical circuit (labeled as electrical connection **1506** herein) between at least a portion of the air handler **102** (e.g., controller **140** thereof) and the air handler float switch **160** to be initiated or inhibited (e.g., based on the actuation of the air handler float switch **160** closing or opening the separate electrical circuit between the air handler **102** and the air handler float switch **160**). Such initiated or inhibited flow of electrical current in the separate electrical circuit that includes at least a portion of the air handler **102** (e.g., the controller **140**) and the air handler float switch **160**, where the initiation or inhibition of electrical current in the separate electrical circuit is caused by actuation of the air handler float switch **160** by the actuator **910** of the actuator apparatus **900**, may be detected and processed by a portion of the air conditioning system **100** (e.g., the air handler **102**, the controller **140**, etc.) to be a receipt and/or detection of a float switch signal “transmitted” by the air handler float switch **160** to command a shutdown of at least a portion of at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) thereby causing at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down (e.g., based on operation of the controller **140**) in response to detection (e.g., by the controller **140**) of the signal that is “transmitted” by the air handler float switch **160** due to actuation thereof by the actuator **910** of the actuation apparatus **900** based on a signal transmitted to the actuator apparatus **900** from the drain cleaner apparatus **200** via the separate, second electrical circuit. For example, the separate electrical circuit (e.g., electrical connection **1506**) may at least partially comprise at least two particular wires extending from the controller **140** (e.g., electrically connected to at least interface **148** thereof) and electrically connected with separate, respective wires extending from the air handler float switch **160** and at least partially comprising an electrical circuit that includes the air handler float switch **160**, and the air handler float switch **160** may be configured to selectively open or close the separate electrical circuit, to selectively inhibit or initiate flow of electrical current in the separate electrical circuit, based on being actuated. The air handler **102** may be configured to provide an electrical power supply to the separate electrical circuit to drive an electrical current in the separate electrical circuit when closed. The air handler **102** (e.g., the controller **140**) may include a sensor such as an ammeter, current sensor, or the like configured to detect and/or generate data which may be processed by the controller **140** to detect (e.g., generate signals which may be processed by the controller **140** to detect in response to) a presence and/or magnitude of electrical current in the at least two particular wires extending from the controller **140** (e.g., electrically connected to at least interface **148** thereof) to at least partially comprise the separate electrical circuit. The controller **140** may respond to a determination of a presence or absence of electrical current (e.g., a presence or absence of current above a threshold magnitude) in the at least two particular wires, a change between an absence and a presence of at least a threshold electrical current in the second set of at least two wires, or the like, based on processing and/or detecting signals generated by the sensor to determine that a float switch signal is received from the

air handler float switch **160** indicating that the air handler float switch **160** is actuated in response to the actuator **910** actuating the air handler float switch **160**.

(215) In some example embodiments, the controller **210** may be configured to, in response to detecting the occurrence of the event (e.g., in response to detecting the float switch signal from the drain cleaner float switch **1610** via detecting presence, absence, change between presence and absence, etc. of an electrical current in the first electrical circuit), transmit a separate signal in a separate, second electrical circuit (labeled as electrical connection **1504** in FIG. **15B**) that comprises a separate, second set of wires of the electrical connection **1152**, for example a second set of at least two wires extending from the drain cleaner apparatus **200** to a remote device, such as the air handler **102** of the air conditioning system **100**, a controller **140** of the air handler **102**, or the like to establish the second electrical circuit as including at least a portion of the drain cleaner apparatus **200** and at least a portion of the air conditioning system **100** (e.g., at least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, etc.). The drain cleaner apparatus **200** may provide a power source (e.g., from a power supply of the drain cleaner apparatus **200** such as batteries **1142**) coupled to the second set of at least two wires to drive an electrical current in the second electrical circuit when said second electrical circuit is closed. The separate signal may be “transmitted” based on the drain cleaner apparatus **200** (e.g., the controller **210**) controlling (e.g., initiating, inhibiting, adjusting a magnitude and/or frequency of, etc.) a flow of electrical current in the second electrical circuit, for example based on controller **210** operating a switch implemented by and/or included in the drain cleaner apparatus **200** to close or open the second electrical circuit, respectively, based on the controller **210** selectively and/or adjustably controlling a supply of electrical power to the portion of the air conditioning system **100** (e.g., at least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, etc.) via the second electrical circuit, or the like. Such controlled (e.g., initiated, inhibited, adjusted, etc.) flow of electrical current in the second electrical circuitry may be referred to as a signal, command, signal or the like (e.g., an electrical signal, a float switch signal, or the like) transmitted to the portion of the air conditioning system **100** (e.g., at least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, etc.). Such initiated or inhibited flow of electrical current in the second electrical circuit that includes the portion of the air conditioning system **100** may be detected and processed by the portion of the air conditioning system **100** (e.g., the air handler **102**, the controller **140**, etc.) to be a receipt and/or detection of a float switch signal “transmitted” by the drain cleaner apparatus **200** to command a shutdown of at least a portion of at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) thereby causing at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down (e.g., based on operation of the controller **140**) in response to detection (e.g., by the controller **140**) of the signal that is “transmitted” by the drain cleaner apparatus **200** (e.g., by controller **210**). For example, the second electrical circuit (e.g., electrical connection **1504** as shown in FIG. **15B**) may at least partially comprise at least two particular wires extending from the controller **140** (e.g., electrically connected to at least interface **148** thereof) that are electrically connected to separate, respective wires of the second set of at least two wires extending from the drain cleaner apparatus **200** and at least partially comprising an electrical circuit that includes at least a portion of the drain cleaner apparatus **200** (e.g., the controller **210**) and the portion of the air conditioning system **100** (e.g., at least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, etc.). The air handler **102** (e.g., the controller **140**) may include a sensor such as an ammeter, current sensor, or the like configured to detect and/or generate data which may be processed by the controller **140** to detect (e.g., generate signals which may be processed by the controller **140** to detect in response to) a presence and/or magnitude of electrical current in the at least two particular wires extending from the controller **140** (e.g., electrically connected to at least interface **148** thereof) to at least partially comprise the second electrical circuit. The controller **140** may respond to a determination of a presence or

absence of electrical current (e.g., a presence or absence of current above a threshold magnitude) in the at least two particular wires, a change between an absence and a presence of at least a threshold electrical current in the second set of at least two wires, or the like, based on processing and/or detecting signals generated by the sensor to determine that an electrical signal (e.g., a float switch signal is received, indicating that a float switch is actuated.

(216) Each set of one or more wires extending out of the drain cleaner apparatus **200** may at least partially comprise the electrical connection **1152** such that each set of wires of the electrical connection **1152** may at least partially comprise a set of serially coupled wires and connections (e.g., male/female connections) to connect a wire extending from the drain cleaner apparatus **200** to a separate wire extending from one or more separate devices to at least partially establish an electrical circuit that includes at least a portion of the drain cleaner apparatus **200** and at least a portion of the one or more separate devices, where such one or more separate devices may include, for example, at least one of the float switch apparatus **800**, the actuator apparatus **900**, the air conditioning system **100**, the air handler **102**, the controller **140** of the air handler **102**, the float switch interface connection **148** of the controller **140**, or the like. The controller **210** may be configured to cause the air conditioning system **100** to shut down (e.g., based on causing an air handler float switch of the air conditioning system **100** to actuate, based on transmitting a command signal to the controller of the air conditioning system **100** which causes the controller of the air conditioning system **100** to shut down the air conditioning system **100**. The controller **210** may be configured to cause the air conditioning system **100** to shut down in response to receiving a shutdown command signal from a remote computing device via the network communication link established by the network communication interface **224**.

(217) Still referring to FIGS. **11A-12D**, the drain cleaner apparatus **200** may include light indicators **1184** (e.g., light emitting diodes, or LEDs) which may extend through respective openings in the housing **201** (e.g., respective openings in the side housing **1104** as shown) and may be configured to provide visible indications of a status of the drain cleaner apparatus **200**. For example, referring to FIG. **11A**, the light indicators **1184** may include a left-most green LED configured to selectively emit green light, a center-left yellow LED configured to selectively emit yellow light, a center-right red LED configured to selectively emit red light, and a right-most blue LED configured to selectively emit blue light. The controller **210** may selectively activate the green LED to emit green light to indicate that the drain cleaner apparatus **200** is activated (e.g., based on human user interaction with the user interface **1182** and/or with a remote computing device **700** to cause the remote computing device **700** to command the drain cleaner apparatus **200** to activate via a network communication link **702**) and/or to indicate that the controller **210** is presently implementing a timer to enable actuating the dispenser device **204** at a fixed frequency (e.g., fixed intervals). The controller **210** may be configured to selectively activate the yellow LED to emit yellow light to indicate a depletion signal in response to a determination that a counter value implemented by the controller **210**, as described herein, reaches a particular counter value that corresponds to at least partial depletion of a fixed reservoir (e.g., the cartridge reservoir **304**) of the cleaning composition as described herein according to any of the example embodiments. It will be understood that the controller **210** may be configured to selectively deactivate operation of at least the dispenser device **204** (e.g., disable the periodic actuation of the dispenser device **204**), activate a visual indicator such as the yellow LED, and/or transmit a warning signal to a remote computing device **700** via a network communication link **702** to cause the remote computing device to generate (e.g., transmit) a warning (e.g., a graphic indication shown on the display screen interface **760**) to warn a supported human user that the cartridge reservoir **304** is at least partially depleted in response to determination that the counter value has reached or exceeded the particular counter value. The controller **210** may be configured to selectively activate the red LED to emit red light in response to a determination that the electrical circuit including the electrical switch device **1280** is open, such that the controller **210** determines that the drain cleaner apparatus **200** is not coupled

with a cartridge **300** while the drain cleaner apparatus **200** is activated. It will be understood that the controller **210** may be configured to selectively deactivate operation of at least the dispenser device **204** (e.g., disable the periodic actuation of the dispenser device **204**), activate a visual indicator such as the red LED, and/or transmit a warning signal to a remote computing device **700** via a network communication link **702** to cause the remote computing device to generate (e.g., transmit) a warning (e.g., a graphic indication shown on the display screen interface **760**) to warn a supported human user that the drain cleaner apparatus **200** has disabled operation of the dispenser device **204** due to non-coupling of the drain cleaner apparatus **200** with a cartridge **300**. a supply of power (e.g., by batteries **1142**) is at least partially depleted (e.g., a determined loss of power source supply voltage below 10% of a predetermined reference voltage magnitude). The controller **210** may be configured to selectively activate the blue LED to emit blue light to indicate that the network communication interface **224** has established an active network communication link **702** with at least one remote computing device **700**.

(218) FIG. **14A** is a perspective bottom-rear-left view of the structure connector **220** shown in FIG. **11A** according to some example embodiments. FIG. **14B** is a perspective top-front-right view of the structure connector **220** shown in FIG. **14A** according to some example embodiments. FIG. **14C** is a perspective view of the drain cleaner apparatus **200** according to some example embodiments. FIG. **14D** is a plan bottom view of the drain cleaner apparatus **200** according to some example embodiments. It will be understood that the structure connector **220** and the drain cleaner apparatus **200** shown in FIGS. **11A-12D** may include any of the elements of any of the example embodiments of the structure connector and/or the drain cleaner apparatus shown in any of the drawings and/or described herein.

(219) As shown in FIGS. **14A-14D**, the structure connector **220** may include a housing structure **228** (e.g., a plastic structure), a coupling structure **221** that is coupled (e.g., adhered via an adhesive) to the housing structure **228**, and an interface structure **226** configured to engage a complementary coupling structure **1172** of the drain cleaner apparatus **200** to couple the structure connector **220** to the drain cleaner apparatus **200** and thus enable the structure connector **220** to couple the drain cleaner apparatus **200** to the fixed structure to which the coupling structure **221** is coupled.

(220) In some example embodiments, the coupling structure **221** is or includes a magnet configured to magnetically attach the structure connector **220** to a fixed external structure, such as a metal surface of the external structure, for example a metal housing **101** of an air handler **102** as shown in FIG. **1**. Thus, the magnet coupling structure **221** may configure the structure connector **220** to be magnetically coupled to a metal external structure such as a metal housing **101** of an air handler **102**. In some example embodiments, the coupling structure **221** may include an adhesive material configured to adhere to a surface of an external structure.

(221) As shown in FIGS. **14A-14D**, in some example embodiments, the interface structure **226** may include a flange or bracket structure configured to slidably engage with a complementary, downwards-opening complementary coupling structure **1172** (e.g., complementary flange or bracket structure) at least partially defining a slot or cavity **1402** in the drain cleaner apparatus **200** housing **201** that is configured to accommodate at least a portion of the structure connector **220** due to relative downwards motion of the drain cleaner apparatus **200** in relation to the structure connector **220** (e.g., downwards sliding engagement of the complementary coupling structure **1172** with the interface structure **226** of the structure connector **220** so that at least a closed top portion of the complementary coupling structure **1172** engages a top portion of the interface structure **226** to transfer a load or weight of the drain cleaner apparatus **200** and any cartridge **300** coupled thereto to the structure connector **220**. As a result of the structure connector **220** being coupled (e.g., magnetically coupled) to a fixed external structure via the coupling structure **221** (e.g., magnet) being coupled to the fixed external structure, the drain cleaner apparatus **200** and any cartridge **300** coupled thereto (e.g., the drain cleaner apparatus system **1100**, which may be referred to

interchangeably herein as a drain cleaner system) may at least partially rest upon the structure connector **220** to be held in place in relation to the external structure (e.g., to at least partially transfer a load or weight of the drain cleaner apparatus system **1100** to the fixed external structure via the structure connector **220**. It will be understood that the drain cleaner apparatus **200** and the cartridge **300** coupled (e.g., connected, detachably connected, etc.) thereto may collectively partially or entirely comprise the drain cleaner apparatus **1100**, which may be referred to interchangeably herein as a drain cleaner system.

(222) It will be understood that the structures of the interface structure **226** and the complementary coupling structure **1172** may be different from the example embodiments shown in FIGS. **11A-11F** and **14A-14D**. As shown, the interface structure **226** of the structure connector **220** may be a protruding tab (e.g., male, or flange) connector structure and the complementary coupling structure **1172** may be a complementary slot (e.g., female) connector structure configured to slidably engage the interface structure **226** to receive the structure connector **220** into the cavity **1402**, but example embodiments are not limited thereto. For example, in some example embodiments, the interface structure **226** of the structure connector **220** may be a slot (e.g., female) connector structure and the complementary coupling structure **1172** may be a complementary protruding tab (e.g., male, or flange) connector structure configured to slidably engage the interface structure **226** to receive the structure connector **220** into the cavity **1402**. In some example embodiments, the drain cleaner apparatus **200** may include an interlock structure configured to lock the structure connector **220** together with the drain cleaner apparatus **200**. In some example embodiments, the structure connector **220** may be configured to be detachably coupled to the drain cleaner apparatus **200** or may be a fixed part of the drain cleaner apparatus, omitting the interface structure **226** while the drain cleaner apparatus **200** omits the complementary coupling structure **1172**, that is configured to not be detached from the drain cleaner apparatus **200**.

(223) FIG. **15A** is a schematic view of a system **2000** including a drain cleaner apparatus system **1100** that further includes a drain cleaner apparatus **200** and a cartridge **300**, a float switch apparatus **800**, and an actuator apparatus **900**, according to some example embodiments. As shown, the drain cleaner apparatus **200** may be communicatively coupled to at least one remote computing device **700** via a network communication link **702** and configured to operate based at least in part upon communication with the remote computing device **700** as described herein with reference to any of the example embodiments, but example embodiments are not limited thereto. It will be understood that the drain cleaner apparatus **200** and the air handler **102** shown in FIG. **15A** may include any of the elements of any of the example embodiments of the drain cleaner apparatus **200** and/or the air handler **102** shown in any of the drawings and/or described herein. It will be understood that the float switch apparatus **800** shown in FIG. **15A** may include any of the elements of any of the example embodiments of the float switch apparatus **800** shown in any of the drawings and/or described herein. It will be understood that the actuator apparatus **900** shown in FIG. **15A** may include any of the elements of any of the example embodiments of the actuator apparatus **900** shown in any of the drawings and/or described herein.

(224) Referring to FIG. **15A**, in some example embodiments, the apparatus outlet **206** of the drain cleaner apparatus **200** may be coupled to a first end of a dispenser conduit **290** (e.g., a tube, hose, or the like), where the dispenser conduit **290** has an opposite second end that is positioned in fluid communication with the condensate drain line **124** (e.g., extending through and/or coupled with the opening **125** of the condensate drain line **124** via a clip connector, adhesive, or any known coupling device), such that the apparatus outlet **206** of the drain cleaner apparatus **200** is in fluid communication with the condensate drain line **124** through at least the dispenser conduit **290**. The dispenser conduit **290** may be a flexible tube (e.g., a vinyl tube), such that the dispenser conduit **290** may enable the drain cleaner apparatus **200** to be adjustably positioned in various fixed positions in relation to the opening **125** of the condensate drain line **124**, thereby improving flexibility of the drain cleaner apparatus **200** to provide cleaning composition **230** dispensation to

the condensate drain line in various environments having different arrangements of structures to which the drain cleaner apparatus **200** may be coupled (e.g., a surface of housing **101** to which the drain cleaner apparatus **200** may be coupled via structure connector **220**) in relation to the opening **125** of the condensate drain line **124**.

(225) Still referring to FIG. **15A**, the air handler **102** of the air conditioning system **100** may include an air handler float switch **160** (which may be any known float switch), and the air handler **102** may be configured to shut down in response to actuation of the air handler float switch **160**. For example, the air handler **102** may include a controller **140** according to any of the example embodiments described and/or illustrated herein, where the controller **140** is configured to shut down the air handler **102** in response to actuation of the air handler float switch **160**, as the air handler float switch **160** may be configured to communicate a float switch actuation signal to the controller **140** via an electrical connection **1506** between the air handler float switch **160** and the controller **140**.

(226) The electrical connection **1506** may comprise serially coupled (e.g., serially connected) wires and connections (e.g., male/female connections) to connect wiring (e.g., two or more wires) extending from the air handler float switch **160** to a separate wiring or circuitry of the air handler **102**, for example wiring (e.g., two or more separate wires) extending from a float switch signal connection interface **148** of the controller **140** or an electrical connector at the controller **140** that comprises a float switch signal connection interface **148**. The controller **140** may be configured to receive float switch signals from the float switch **160** at the float switch signal connection interface **148**. The controller **140** may be configured to cause at least a portion of the air conditioning system **100** (e.g., the air handler **102**) to shut off (e.g., shut down) in response to receiving a signal (e.g., a float switch signal) at the float switch signal connection interface **148** via which the controller **140** may be electrically connected to the air handler float switch **160** through the electrical connection **1506**.

(227) In some example embodiments, the electrical connection **1506** may include wires (also referred to herein interchangeably as wiring) extending from the air handler float switch **160** (which may be considered wiring of the air handler float switch **160**) and which may be connected to wires (e.g., wiring) and/or circuitry of the air handler **102**, which may be further connected to the controller **140** (e.g., the wiring extending from the air handler float switch **160** may extend to the a float switch signal connection interface **148** or may be connected to wiring and/or a connector of the air handler **102** that is further connected to the float switch signal connection interface **148** of the controller **140**), to establish the electrical connection **1506**. In some example embodiments, the electrical connection **1506** may be established based on connecting wiring extending from the air handler float switch **160** to corresponding wiring extending from the controller **140** (e.g., wiring extending from the float switch signal connection interface **148**) in the air handler **102**, for example based on connection of complementary (e.g., male/female) connectors of the respective connected wiring extending from the air handler float switch **160** and the controller **140**. In some example embodiments, the electrical connection **1506** may be established based on connecting wiring extending from the air handler float switch **160** to the float switch signal connection interface **148** of the controller **140**. In some example embodiments, the electrical connection **1506** may be established based on connecting an electrical connector at a distal end of wiring extending from the air handler float switch **160** to a complementary electrical connector of the air handler **102** that is electrically connected, via internal wiring and/or circuitry, to the float switch signal connection interface **148** of the controller **140**.

(228) Referring to FIG. **15A**, in some example embodiments, the electrical connection **1506** may comprise a particular set of at least two wires of the air conditioning system **100** which are electrically connected (directly or indirectly) to the controller **140** (e.g., via being electrically connected to the float switch signal connection interface **148**) and extend from the air conditioning system **100** (e.g., from air handler **102**) and which may each be connected (e.g., directly or

indirectly or indirectly, as part of a serial connection of wires and/or interfaces/connectors therebetween) to separate, respective wires extending from the air handler float switch **160** to establish the electrical connection **1506** as a float switch electrical circuit that includes at least a portion of the air conditioning system **100** (e.g., at least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, or the like) and at least a portion of the air handler float switch **160**. The air handler **102** may provide a power source (e.g., from a power supply of the air handler **102**, such as a connection to mains power) coupled to the particular set of at least two wires to drive the electrical current in the float switch electrical circuit of the electrical connection **1506** when said float switch electrical circuit is closed. The air handler float switch **160** may actuate to close or open the float switch electrical circuit of the electrical connection **1506** and thus enable the initiation or inhibition, respectively, of electrical current therein. The signal detected in the float switch electrical circuit of the electrical connection **1506** by at least a portion of the air conditioning system **100** (e.g., a signal received at the controller **140** from the air handler float switch **160** via the electrical connection **1506**) may include a float switch signal that may be detected by at least a portion of the air conditioning system, such as the air handler **102** (e.g., by controller **140**) as an initiation, inhibition, and/or change in magnitude of electrical current in the float switch electrical circuit of the electrical connection **1506** due to closing or opening of the switch implemented by and/or included in the air handler float switch **160** to close or open the float switch electrical circuit of the electrical connection **1506**. For example, the air handler **102** (e.g., the controller **140**) may include a sensor such as an ammeter, current sensor, or the like configured to detect (e.g., generate signals which may be processed by the controller **140** to detect in response to) a presence and/or magnitude of electrical current in the float switch electrical circuit of the of the electrical connection **1506**. The controller **140** may respond to a determination of a presence or absence of electrical current (e.g., a presence or absence of current above a threshold magnitude) in the float switch electrical circuit of the electrical connection **1506**, a change in magnitude of electrical current in the float switch electrical circuit of the electrical connection **1506** (e.g., a change of at least a threshold magnitude), or the like, based on processing and/or detecting signals generated by the sensor to determine that a float switch signal is received from the air handler float switch **160**, via the float switch electrical circuit of the electrical connection **1506**, indicating that the air handler float switch **160** is actuated. The air conditioning system **100** (e.g., the air handler **102**, the controller **140**, etc.) may be configured to shut down (e.g., shut off) at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) in response to determining (e.g., detecting) that the float switch signal is received from the air handler float switch **160**, via the float switch electrical circuit of the electrical connection **1506** (e.g., based on processing signals generated by the sensor), indicating that the air handler float switch **160** is actuated.

(229) The air handler float switch **160** may be normally configured to be positioned in fluid communication with the condensate drain line **124** and/or a drip pan **122** of the air handler **102**. However, as shown, the system **2000** may be configured to actuate **980** the air handler float switch **160**, to cause the air handler **102** to shut down, without the air handler float switch encountering condensate (e.g., in response to a separate float switch being actuated, in response to a command received at the drain cleaner apparatus **200**, or the like), thereby enabling improved control over air handler **102** operation and thus of air conditioning system **100** operation.

(230) Still referring to FIG. **15A**, the system **2000** may include, in addition to the drain cleaner apparatus **200**, an actuator apparatus **900** configured to be electrically coupled to the drain cleaner apparatus **200** (e.g., via electrical connection **1504**, which may at least partially comprise electrical connection **1152** to the controller **210** as shown in FIGS. **11A-12D**). In some example embodiments, the actuator apparatus **900** may be referred to interchangeably as a “air handler float switch holster,” a “float switch holster,” a “holster,” or the like. As shown, the actuator apparatus **900** may include an actuator **910** (e.g., a servoactuator or servomotor-driven piston, linear actuator,

or the like), and the actuator apparatus **900** may be configured to position the air handler float switch **160** in relation to the actuator, such that the actuator apparatus **900** is configured to cause the actuator **910** to actuate **980** the air handler float switch **160** (e.g., based on causing a float of the air handler float switch **160** to move in relation to a remainder of the air handler float switch **160**) in response to receiving an actuator command signal from the drain cleaner apparatus **200** (e.g., via electrical connection **1504**).

(231) The controller **210** of the drain cleaner apparatus **200** may be configured to transmit the actuator command signal to the actuator apparatus **900** (e.g., via electrical connection **1504**) to cause the actuator **910** to actuate **980** the air handler float switch **160**. The controller **210** may transmit the actuator command signal in response to receiving a command at the drain cleaner apparatus **200**. Such a command may include a signal (e.g., an electrical signal, also referred to herein as a float switch signal) received from a separate float switch that is in fluid communication with the condensate drain line **124** and electrically coupled to the drain cleaner apparatus **200** (e.g., the controller **210**) via an electrical connection **1502** (where electrical connections **1502** and **1504** collectively comprise the electrical connection **1152** described herein although potentially including separate electrical wire connections).

(232) Electrical connections **1502** and **1504** may each comprise a set of serially coupled wires and connections (e.g., male/female connections) to connect a wire extending from the drain cleaner apparatus **200** to a separate wire extending from one of the float switch apparatus **800** or the actuator apparatus **900**. It will be understood that in some example embodiments the electrical connections **1502** and/or **1504** may be wireless network communication links between respective network communication interface devices of the drain cleaner apparatus **200** and at least one of the float switch apparatus **800** or the actuator apparatus **900**.

(233) Referring to FIG. **15A**, in some example embodiments, the electrical connection **1502** may comprise a first set of at least two wires of the drain cleaner apparatus **200** (at least partially comprising electrical connection **1152**) which are electrically connected (directly or indirectly) to the controller **210** and extend from the drain cleaner apparatus **200** (e.g., via opening **1150**) and which may each be connected (e.g., directly or indirectly, as part of a serial connection of wires and/or interfaces there between) to separate, respective wires extending from the drain cleaner float switch **1610** of the float switch apparatus **800** to establish the electrical connection **1502** as a first electrical circuit that includes at least a portion of the drain cleaner apparatus **200** and at least a portion of the drain cleaner float switch **1610**. The drain cleaner apparatus **200** may provide a power source (e.g., from a power supply of the drain cleaner apparatus **200** such as batteries **1142**) coupled to the first set of at least two wires to drive the electrical current in the first electrical circuit of the electrical connection **1502** when said first electrical circuit is closed. The drain cleaner float switch **1610** may actuate, in response to a fluid in the condensate drain line **124** engaging the drain cleaner float switch **1610**, to close or open the first electrical circuit of the electrical connection **1502** and thus enable initiation or inhibition, respectively, of electrical current in the first electrical circuit of the electrical connection **1502**. The signal detected in the first electrical circuit by the drain cleaner apparatus **200** (e.g., a signal received at the drain cleaner apparatus **200** from the drain cleaner float switch **1610** via the electrical connection **1502**) may include a float switch signal that may be detected by the drain cleaner apparatus **200** (e.g., by controller **210**) as an initiation, inhibition, and/or change in magnitude of electrical current in the first electrical circuit of the electrical connection **1502** due to closing or opening of the switch implemented by and/or included in the drain cleaner float switch **1610** to close or open the first electrical circuit of the electrical connection **1502**. For example, the drain cleaner apparatus **200** (e.g., the controller **210**) may include a sensor such as an ammeter, current sensor, or the like configured to detect (e.g., generate signals which may be processed by the controller **210** to detect in response to) a presence and/or magnitude of electrical current in the first set of at least two wires extending from the drain cleaner apparatus **200** and thus of electrical current in the first electrical

circuit of the of the electrical connection **1502**. The controller **210** may respond to a determination of a presence or absence of electrical current (e.g., a presence or absence of current above a threshold magnitude) in the first electrical circuit of the electrical connection **1502**, a change in magnitude of electrical current in the first electrical circuit of the electrical connection **1502**, or the like, based on processing and/or detecting signals generated by the sensor to determine that a float switch signal is received from the drain cleaner float switch **1610**, via the first electrical circuit of the electrical connection **1502**, indicating that the drain cleaner float switch **1610** is actuated in response to a fluid engaging the drain cleaner float switch **1610** in the condensate drain line **124**.

(234) Still referring to FIG. **15A**, in some example embodiments, the electrical connection **1504** may comprise a second set of at least two wires of the drain cleaner apparatus **200** (at least partially comprising electrical connection **1152**) which are electrically connected (directly or indirectly) to the controller **210** and extend from the drain cleaner apparatus **200** (e.g., via opening **1150**) and which may each be connected (e.g., directly or indirectly, as part of a serial connection of wires and/or interfaces there between) to separate, respective wires extending from the actuator apparatus **900** (e.g., from actuator **910**) to establish the electrical connection **1504** as a second electrical circuit that includes at least a portion of the drain cleaner apparatus **200** and at least a portion of the actuator apparatus **900** (e.g., at least the actuator **910** thereof). The drain cleaner apparatus **200** may provide a power source (e.g., from a power supply of the drain cleaner apparatus **200** such as batteries **1142**) coupled to the second set of at least two wires to drive an electrical current in the second electrical circuit of the electrical connection **1504** when said second electrical circuit is closed. The drain cleaner apparatus **200** (e.g., controller **210**) may be configured to generate and transmit an electrical signal (e.g., electrical current), also referred to herein as an actuator command signal, to the actuator apparatus **900** via the second electrical circuit of the electrical connection **1504** (e.g., based on controller **210** operating a switch implemented by and/or included in the drain cleaner apparatus **200** to close or open the second electrical circuit of the electrical connection **1504**, respectively, based on the controller **210** selectively and/or adjustably controlling a supply of electrical power to the actuator apparatus **900** and/or the actuator **910** via the second electrical circuit of the electrical connection **1504**, or the like) for a particular period of time. Such controlled flow of electrical current in the second electrical circuitry may be referred to as a signal, command, signal or the like (e.g., an actuator command signal) to cause the actuator **910** of the actuator apparatus **900** to actuate the air handler float switch **160** of the air handler **102** which is held in the actuator apparatus **900**, to cause a flow of electrical current in a separate electrical circuit (e.g., float switch electrical circuit) of the electrical connection **1506** between at least a portion of the air handler **102** (e.g., controller **140** thereof) and the air handler float switch **160** to be initiated, inhibited, or adjusted in magnitude (e.g., based on the actuation of the air handler float switch **160** closing or opening the float switch electrical circuit of the electrical connection **1506**). Such initiated, inhibited, or adjusted flow of electrical current in the float switch electrical circuit of the electrical connection **1506** that includes at least a portion of the air handler **102** (e.g., the controller **140**) and the air handler float switch **160**, where the initiation or inhibition of electrical current in the separate electrical circuit is caused by actuation of the air handler float switch **160** by the actuator **910** of the actuator apparatus **900**, may be detected and processed by a portion of the air conditioning system **100** (e.g., the air handler **102**, the controller **140**, etc.) to be a receipt and/or detection of a float switch signal “transmitted” by the air handler float switch **160** to command a shutdown of at least a portion of at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) thereby causing at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down (e.g., based on operation of the controller **140**) in response to detection (e.g., by the controller **140**) of the signal that is “transmitted” by the air handler float switch **160** due to actuation thereof by the actuator **910** of the actuation apparatus **900** based on a signal transmitted to the actuator apparatus **900** from the drain cleaner apparatus **200** via the separate, second electrical circuit. For example, the float switch electrical circuit of the electrical

connection **1506** may at least partially comprise at least two particular wires extending from the controller **140** (e.g., electrically connected to at least interface **148** thereof) and at least partially comprising an electrical circuit that includes the air handler float switch **160**, and the air handler float switch **160** may be configured to selectively open or close the separate electrical circuit, to selectively inhibit or initiate flow of electrical current in the separate electrical circuit, based on being actuated.

(235) Accordingly, it will be understood that the actuator apparatus **900** may be configured to function similarly to the bypass device **506** described with reference to FIG. 5, where the actuator apparatus **900** may be configured to operate as a bypass device to actuate the air handler float switch **160** to cause at least a portion of the air conditioning system, including at least the air handler **102**, to shut down (e.g., shut off), based on a signal generated by the drain cleaner apparatus **200** (e.g., based on a signal generated by the controller **210** as described herein), which may include shutting down at least one of the air mover **108**, compressor **150**, and/or air mover **154**.

(236) In addition or alternative, it will be understood that the controller **210** of the drain cleaner apparatus **200**, and thus the drain cleaner apparatus **200** itself, may be configured to function similarly to the controller **210** and/or drain cleaner apparatus **200** described with reference to FIG. 5, wherein the controller **210** is configured to cause the air handler float switch **160** to actuate to cause some or all of the air conditioning system **100** to shut down (e.g., based on operation of the controller **140** in response to float switch **160** actuation).

(237) It will be understood that the controller **210** may be configured to generate the signal that is communicated to the actuator apparatus **900** as described herein, to cause the actuator **910** of the actuator apparatus **900** to operate (e.g., actuate) so that the actuator apparatus **900** actuates the air handler float switch **160** to cause the air handler **102** to shut down, based on the float switch signal generated by a separate flow switch, such as a float switch of the float switch apparatus **800**. However, example embodiments are not limited thereto, and in some example embodiments the controller **210** may be configured to generate the signal that is communicated to the actuator apparatus **900** as described herein, to cause the actuator **910** of the actuator apparatus **900** to operate (e.g., actuate) so that the actuator apparatus **900** actuates the air handler float switch **160** to cause the air handler **102** to shut down independently to actuation of any flow switch due to presence of a fluid such as condensate at any float switch. For example, in some example embodiments, the controller **210** may be configured to generate the signal that is communicated to the actuator apparatus **900** as described herein, to cause the actuator **910** of the actuator apparatus **900** to operate (e.g., actuate) so that the actuator apparatus **900** actuates the air handler float switch **160** to cause the air handler **102** to shut down in response to the controller receiving and processing a command signal received at the drain cleaner apparatus **200**. In some example embodiments, the command signal may be received at the drain cleaner apparatus **200** via a signal received at the network communication interface **224** from a remote computing device **700** via a network communication link **702** as described herein with regard to at least FIG. 7, and the drain cleaner apparatus **200** may be configured to cause the air handler float switch **160** to be actuated by the actuator apparatus **900** to cause the air handler **102** to shut down based on a remote computing device **700** transmitting a command to the drain cleaner apparatus **200** which, when received by the network communication interface **244** and transmitted to the controller **210**, where the controller **210** responds to receipt and processing of the command by transmitting the signal to the actuator apparatus **900** via the electrical connection **1504**. As a result, the drain cleaner apparatus **200** may be configured to cause the air handler **102** to shut down independently of a float switch being actuated due to presence of condensate in the condensate drain line **124**. For example, a user supported by the remote computing device **700** may desire to cause the air conditioning system **100** to shut down, and the drain cleaner apparatus **200** and/or the actuator apparatus **900** may be configured to implement such a shutdown via the network communication link **702** and the

electrical connection **1504** with the actuator apparatus **900** in which the air handler float switch **160** is positioned, thereby enabling improved control over the operation at least the air handler and/or the air conditioning system **100** as a whole via remote control.

(238) The above operation of the drain cleaner apparatus **200** to control the actuator apparatus **900** to actuate **980** the air handler float switch **160** may proceed as shown in FIG. **28**. FIG. **28** is a flowchart illustrating operation of the system **2000** according to some example embodiments. Referring to FIG. **28**, at **S2802**, the drain cleaner float switch **1610** of the float switch apparatus **800** that is coupled to the condensate drain line **124** may actuate in response to a presence of a fluid such as condensate in the condensate drain line. The drain cleaner float switch **1610** may include a bimetal switch device and may actuate the bimetal switch device in response to a float of the drain cleaner float switch **1610** moving (e.g., rising) with a level of a surface of condensate in the condensate drain line. At **S2804**, the drain cleaner float switch **1610**, and thus the float switch apparatus **800**, may transmit an electrical signal (e.g., float switch signal) to the drain cleaner apparatus **200** (e.g., the controller **210**) via electrical connection **1502** to indicate that the drain cleaner float switch **1610** is actuated. At **S2806**, the electrical signal transmitted by the drain cleaner float switch **1610** is received at the controller **210** of the drain cleaner apparatus **200** and processed to determine that the drain cleaner float switch **1610** has actuated. At **S2808**, the controller **210** responds to determining that the drain cleaner float switch **1610** has actuated by transmitting an electrical signal and/or command (e.g., actuator command signal) to the actuator apparatus **900** via electrical connection **1504**. The controller **210** may further, at **S2808**, generate and transmit a warning signal in response to determining that the drain cleaner float switch **1610** has actuated. The warning signal may be transmitted to a remote computing device **700** via network communication link **702** to cause the remote computing device **700** to provide a warning to a supported human user (e.g., via a display screen interface **760**) informing the supported human user that the drain cleaner float switch **160** is actuated and/or that at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) is being shut down (e.g., shut off). At **S2810**, the actuator apparatus **900** causes the actuator **910** thereof to actuate **980** in order to move a float of the air handler float switch **160** positioned in relation to the actuator **910** by the actuator apparatus **900**, thereby causing the air handler float switch **160** to actuate. As shown in at least FIGS. **25A-25B**, the air handler float switch **160** may include a bimetal switch device **2510** and may actuate the bimetal switch device **2510** in response to a float of the air handler float switch **160** moving (e.g., rising) due to operation of the actuator **910** in the actuator apparatus **900**. At **S2812**, the air handler float switch **160** may transmit an electrical signal (e.g., float switch signal) to the air handler (e.g., controller **140**) via electrical connection **1502** to indicate that the air handler float switch **160** is actuated. At **S2814**, the air handler **102** (e.g., the controller **140** thereof) may selectively shut down at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) in response to determining that the air handler float switch **160** is actuated.

(239) In some example embodiments, the controller **210** may be configured to transmit the electrical signal to the actuator apparatus **900** at **S2808** to cause the actuator **910** to actuate **980** independently of a signal from the drain cleaner float switch **1610**, for example in response to receiving a command signal from the remote computing device **700** via network communication link **702** based on human user interaction with the remote computing device **700**, thereby enabling the drain cleaner apparatus **200** to provide remote human user control over air conditioning system **100** operation. In such example embodiments, operations **S2802** to **S2806** may be omitted.

(240) In some example embodiments, subsequently to operation **S2808**, the controller **210** may transmit a subsequent electrical signal to the actuator apparatus **900** via electrical connection **1504** to cause the actuator **910** to “de-actuate” from an actuated position to a non-actuated position, to cause the float of the air handler float switch **160** to move (e.g., drop) from an actuated position to a rest, non-actuated position and thus to cause the air handler float switch **160** to become de-actuated (e.g., reset). Such transmission may occur subsequently to **S2814**. Such transmission may be

performed by the controller **210** in response to a determination at the controller **210** that a particular period of time has elapsed after the transmission of the actuator command signal at **S2808**. Such transmission may be performed by the controller **210** in response to receiving a command (e.g., a reset command) from a remote computing device **700** via a network communication link (e.g., based on human user interaction with the remote computing device **700**). The air handler **102** (e.g., controller **140**) may be configured to re-start at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) in response to the air handler float switch **160** being de-actuated. As a result, the system **2000** (e.g., at least the drain cleaner apparatus **200**) may enable improved control over operation of the air conditioning system **100** based on being configured to enable shut down and/or re-start of at least a portion of the air conditioning system **100**, including being configured to enable remote control of the air conditioning system **100** (e.g., shut down and/or re-start) via human user interaction with the remote computing device **700**.

(241) It will be understood that the signal transmitted to the actuator apparatus **900** by the drain cleaner apparatus **200** at **S2808** may be an electrical current which causes an actuator motor (e.g., servomotor) of the actuator **910** to operate for a particular period of time in order to cause an actuator piston of the actuator **910** to move a certain distance in order to cause a float of the air handler float switch **160** to move in relation to a remainder of the air handler float switch **160**, thereby actuating the air handler float switch **160**. The magnitude (e.g., current and/or voltage magnitude) and duration of the current may be stored at the controller **210** (e.g., at a memory thereof) and the controller **210** may control a supply of electrical power to the actuator **910** via the electrical connection **1504** at the magnitude and duration indicated via information stored at the controller **210** in order to actuate the actuator **910**.

(242) The controller **210** may be configured to cause the actuator **910** (e.g., an actuator piston driven by a servomotor and rotary gear as described herein) to actuate **980** to cause an actuator piston thereof to move from a non-actuated position to an actuated position and remain at the actuated position for a particular period of time, where the particular period of time is associated with causing the float of the air handler float switch **160** to remain at a certain position for a sufficient period of time to ensure that the air handler float switch **160** is actuated to cause the air handler **102** to shut off, based on transmitting an electrical signal that causes the actuator **910** to move the float of the float switch to the certain position and remain at the certain position for at least the sufficient period of time. The controller **210** may further be configured to cause the actuator **910** (e.g., actuator piston thereof) to, after remaining at the actuated position for at least the particular period of time, subsequently return to a non-actuated or “rest” position to return the float of the air handler float switch **160** to a non-actuated or “rest” position to enable a re-set, re-initialization and/or re-start of operation of at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) subsequently to being at least partially shut down due to actuation of the air handler float switch **160**, and thereby enabling improved control of the air handler **102** and/or air conditioning system **100**. The controller **210** may be configured to transmit a separate signal (e.g., restart signal) to the actuator apparatus **900** via electrical connection **1504** to cause the actuator **910** to move from the actuated position to the non-actuated position in response to a determination that a particular period of time (information indicating the duration of the particular period of time being stored at the controller **210** and accessed thereby) has elapsed since a signal was transmitted to the actuator apparatus **900** from the controller **210** to cause the actuator **910** to actuate **980** the air handler float switch **160**). The controller **210** may be configured to transmit a separate signal (e.g., restart signal) to the actuator apparatus **900** via electrical connection **1504** to cause the actuator **910** to move from the actuated position to the non-actuated position in response to receiving a reset signal from a remote computing device **700** via network communication link **702**. The controller **210** may be configured to transmit a separate signal (e.g., restart signal) to the actuator apparatus **900** via electrical connection **1504** to cause the actuator **910** to move from the actuated position to the non-actuated position in response to receiving a rest signal based on user interaction with a user

interface **1182** of the drain cleaner apparatus **200**, some combination thereof, or the like.

(243) It will be understood that the system **2000** may enable the air handler **102**, and thus at least a portion of the air conditioning system **100** to be controlled (e.g., shut down) based on presence of condensate in the condensate drain line **124** without requiring the air handler float switch **160** to be coupled to and/or positioned within the condensate drain line **124** (e.g., the air handler float switch **160**, being positioned in relation to the actuator **910** by the actuator apparatus **900**, may be entirely outside the condensate drain line **124** and the opening **125** thereof). As a result, the drain cleaner apparatus **200** may be configured to reduce, minimize, or prevent the likelihood of the dispensing of cleaning composition **230** into the condensate drain line **124** affecting operation of the air handler float switch **160** (and thus the air handler **102** and air conditioning system **100**) based on reducing, minimizing, or preventing contact between the air handler float switch **160** and the cleaning composition **230** supplied to the condensate drain line **124** by the drain cleaner apparatus **200** and thus reducing, minimizing, or preventing inadvertent operation and/or actuation of the air handler float switch **160** due to such contact. The float switch apparatus **800**, as described herein, may be configured to provide a float switch (e.g., drain cleaner float switch **1610** as described herein) which enables the air handler float switch **160** to be actuated in response to presence of condensate or other fluids in the condensate drain line (e.g., due to the electrical connections between the float switch apparatus **800** and drain cleaner apparatus **200** via electrical connection **1502**, the electrical connection between the drain cleaner apparatus **200** and the actuator apparatus **900**, and the positioning of the air handler float switch **160** in relation to the actuator **910** by the actuator apparatus **900**) where the float switch apparatus **800** may be configured to reduce, minimize, or prevent the likelihood of the float switch thereof (e.g., drain cleaner float switch **1610**) being actuated due to contact with cleaning composition dispensed to the condensate drain line **124** by the drain cleaner apparatus based on the float switch apparatus **800** being configured to position the float switch thereof in the condensate drain line **124** spaced apart from a position in the condensate drain line at which the cleaning composition **230** is supplied into the condensate drain line **124** interior (e.g., offset from a central axis of the float switch apparatus **800**). As a result, the float switch apparatus **800** may be configured to enable reliable operation of system **2000** to dispense cleaning composition **230** to the condensate drain line **124** and control the operation of the air handler **102** via control of the air handler float switch **160** via actuator apparatus **900** despite potential variations in the shape or structure of the air handler float switch **160** which might otherwise affect the likelihood of the air handler float switch **160** being actuated by the cleaning composition **230** being applied in to the condensate drain line **124** and to provide reliable access and supply of the cleaning composition **230** into the condensate drain line **124** via opening **125** while further allowing a float switch to be positioned into the condensate drain line **124** through the same opening **125** with reduced, minimized, or prevented obstruction of the opening **125** which might reduce, minimize, or prevent cleaning composition **230** supply into the condensate drain line **124** via opening **125** and further reduce, minimize, or prevent the likelihood of inadvertent operation, corrosion, wear, damage or the like of the float switch positioned in the condensate drain line **124** by the supplied cleaning composition while still enabling reliable operation of the float switch (e.g., drain cleaner float switch **1610** as described herein), thereby improving overall performance and reliability of the system **2000** and the air conditioning system **100**.

(244) It will be understood that in some example embodiments the float switch apparatus **800** may be omitted from system **2000**.

(245) While the above description of the drain cleaner apparatus **200** together with the actuator apparatus **900** causing the air handler **102** to shut down based on actuation of the air handler float switch **160** involves the drain cleaner apparatus **200** transmitting a signal to the actuator apparatus **900** to actuate the air handler float switch **160** in response to the drain cleaner apparatus **200** receiving a command from a remote computing device **700** via a wireless network communication link (e.g., **702** as described in FIG. 7), example embodiments are not limited thereto. For example,

in some example embodiments, the controller **210** may generate and transmit the signal to the actuator apparatus **900**, to cause the actuator apparatus **900** to actuate the air handler float switch **160**, in response to receiving a command signal at the drain cleaner apparatus **200** via manual interaction with a user interface of the drain cleaner apparatus **200** (e.g., a human pressing a user interface **1182** of the drain cleaner apparatus **200** that is a button).

(246) FIG. **15B** is a schematic view of a system **2000** including a drain cleaner apparatus system **1100** that further includes a drain cleaner apparatus **200** and a cartridge **300** and a float switch apparatus **800**, according to some example embodiments.

(247) In some example embodiments, the actuator apparatus **900** may be omitted from the system **2000**, and the drain cleaner apparatus **200** may be electrically connected to the air handler **102** (e.g., controller **140** thereof via being electrically connected to the float switch signal connection interface **148** of the controller **140**) without an interposing actuator apparatus **900** and/or air handler float switch **160**. For example, the air handler float switch **160** shown in FIG. **15A** may, as shown in FIG. **15B**, be disconnected from the air handler **102**, such that the electrical connection **1506** is omitted, and the drain cleaner apparatus **200** may be electrically connected to the air handler **102** (e.g., to the float switch signal connection interface **148** of the controller **140**) in place of the air handler float switch **160**. For example, wires and/or connections extending from the drain cleaner apparatus **200** may be connected to wires, circuitry, and/or connections of the air handler **102** to establish electrical connection **1504** to electrically connect the drain cleaner apparatus **200** to the float switch signal connection interface **148** of the controller **140**, where the controller **140** is configured to respond to signals (e.g., float switch signals) received at the float switch signal connection interface **148** by causing at least a portion of the air conditioning system **100** (e.g., the air handler **102**) to shut off.

(248) As a result, the electrical connection **1504** as shown in FIG. **15B** may comprise serially coupled wires and connections (e.g., male/female connections) to connect one or more wires extending from the drain cleaner apparatus **200** to separate one or more wires or circuitry of the air handler **102**, for example wiring extending from a float switch signal connection interface **148** of the controller **140** or an electrical connector at the controller **140** that comprises a float switch signal connection interface **148**. The controller **140** may be configured to cause the air handler **102** to shut off in response to receiving a signal (e.g., a float switch signal) from the drain cleaner apparatus **200** at the float switch signal connection interface **148** via electrical connection **1504**.

(249) As a result of the drain cleaner apparatus **200** being electrically connected to the air handler **102** via the electrical connection **1504**, the drain cleaner apparatus **200** (e.g., the controller **210** thereof) may be configured to generate and transmit an electrical signal to the air handler **102** (e.g., to controller **140** via the float switch signal connection interface **148**) via the electrical connection **1504** to cause at least a portion of the air conditioning system **100** (e.g., the air handler **102**) to shut off, despite the omission of the air handler float switch **160** from the air handler **102**. The controller **140**, receiving the electrical signal from the drain cleaner apparatus **200** via the float switch signal connection interface **148** and electrical connection **1504**, may process (e.g., interpret) the received electrical signal as if the electrical signal were received from an actuated air handler float switch **160** and may respond accordingly to cause at least a portion of the air conditioning system **100** (e.g., the air handler **102**) to shut off.

(250) As shown in FIG. **15B**, and as described above with reference to FIG. **15A**, the drain cleaner apparatus **200** may be electrically coupled to the float switch apparatus **800** via electrical connection **1502**, and the float switch apparatus **800** may include a drain cleaner float switch **1610** that may be configured to transmit a float switch signal to the drain cleaner apparatus **200** via the electrical connection **1502** in response to the drain cleaner float switch **1610** being actuated due to fluid (e.g., condensate backflow) in the condensate drain line **124**. In some example embodiments, the drain cleaner apparatus **200** (e.g., the controller **210**) may be configured to, in response to receiving a float switch signal from the drain cleaner float switch **1610** of the float switch apparatus

800 via electrical connection **1502**, generate and transmit the electrical signal to the air handler **102** (e.g., the controller **140** via float switch signal connection interface **148**) via electrical connection **1504** to cause the electrical signal to be received at the controller **140** via the float switch signal connection interface **148**, thereby causing the controller **140** to cause at least a portion of the air conditioning system **100** (e.g., the air handler **102**) to shut off in response to actuation of the drain cleaner float switch **1610** of the float switch apparatus **800**.

(251) In some example embodiments, based on electrically connecting the drain cleaner apparatus **200** to the air handler **102** (e.g., to the float switch signal connection interface **148** of the controller **140**) and configuring the drain cleaner apparatus **200** to generate and transmit an electrical signal to the air handler **102** via the electrical connection **1504** to cause at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to be shut down (e.g., shut off) in response to the drain cleaner apparatus **200** receiving a float switch signal from the drain cleaner float switch **1610** of the float switch apparatus **800** via the electrical connection **1502**, the drain cleaner apparatus **200** may be configured to perform additional operations in response to the drain cleaner float switch **1610** being actuated, beyond causing at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down.

(252) In some example embodiments, the drain cleaner apparatus **200** (e.g., the controller **210**) may be configured to, in response to receiving a float switch signal from the float switch **1610** of the float switch apparatus **800** via electrical connection **1502**, and in addition to transmitting the electrical signal to the air handler **102** via the electrical connection **1504**, control the dispenser device **204** to cause an amount of cleaning composition **230** to be dispensed (e.g., from the cartridge reservoir **304** and/or reservoir **1130**) through the apparatus outlet **206** to the condensate drain line **124**. Such a float switch-responsive dispensation of the cleaning composition **230** by the drain cleaner apparatus **200** may be performed independently of the drain cleaner apparatus **200** (e.g., the controller **210**) operating the dispenser device **204** to dispense an amount of cleaning composition **230** based on operation of a timer as described herein. For example, where an actuation of the drain cleaner float switch **1610** of the is due to an obstruction (e.g., clogging) of the condensate drain line **124** due to buildup of one or more various substances (e.g., mold, algae, mildew, bacteria, and/or fungi) within the condensate drain line **124** to cause condensate backflow in the condensate drain line **124**, the float switch-responsive dispensation of the cleaning composition **230** by the drain cleaner apparatus **200** in response to receiving the float switch signal from the drain cleaner float switch **1610** due to such actuation may reduce and/or remove the obstruction (e.g., by cleaning, chelating, breaking down, etc. the one or more various substances at least partially comprising the obstruction). As a result, the drain cleaner apparatus **200** may be configured to at least partially mitigate backflow of condensate in the condensate drain line **124**, facilitate drainage of the condensate through the condensate drain line **124**, and reduce or prevent the risk of damage that might result from condensate backflow through the condensate drain line opening **125** due to the obstruction.

(253) The float switch-responsive dispensation of the cleaning composition **230** may not cause any re-setting of the aforementioned timer that is implemented by the drain cleaner apparatus **200** (e.g., by the controller **210**) to repeatedly actuate the dispenser device **204** at a fixed time interval or may cause the timer to re-set. The float switch-responsive dispensation of the cleaning composition **230** may cause the aforementioned counter that is implemented by the drain cleaner apparatus **200** (e.g., by the controller **210**) to increment a counter value in response to each actuation of the dispenser device **204** to be incremented to represent a dispensation of cleaning composition **230** from the drain cleaner apparatus system **1100** as a result of the float switch-responsive dispensation of the cleaning composition **230**.

(254) In some example embodiments, based on the drain cleaner apparatus **200** being configured to further implement a float-switch responsive dispensing of cleaning composition by the dispenser device **204** in response to receiving the float switch signal from the drain cleaner float switch **1610**,

the drain cleaner apparatus **200** may be configured to attempt a corrective action in response to indications of condensate drain line **124** obstruction causing condensate backflow to actuate the drain cleaner float switch **1610**, thereby potentially reducing, mitigating, or correcting the problem causing the condensate backflow and providing an active solution in addition to causing at least a portion of the air conditioning system **100** to at least partially shut down and to report the shutdown to a human user supported by a remote computing device **700**, thereby improving operational performance of the air conditioning system **100** based on providing an ability to responsively mitigate condensate backflow in the condensate drain line beyond simply shutting down at least a portion of the air conditioning system **100**.

(255) In some example embodiments, the drain cleaner apparatus **200** (e.g., the controller **210**) may be configured to, in response to receiving a float switch signal from the float switch **1610** of the float switch apparatus **800** via electrical connection **1502**, in addition to transmitting the electrical signal to the air handler **102** via the electrical connection **1504**, transmit a warning signal to a remote computing device **700** via a network communication link **702** to cause the remote computing device to generate (e.g., transmit) a warning (e.g., a graphic indication shown on the display screen interface **760**) to warn a supported human user that at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) is being shut down (e.g., shut off).

(256) In some example embodiments, based on the drain cleaner apparatus **200** being configured to further enable a human user supported by a remote computing device **700** to be warned that at least a portion of the air conditioning system **100** is being shut down due to the actuation of the drain cleaner float switch **1610** (e.g., in response to condensate backflow in the condensate drain line **124**), the drain cleaner apparatus **200** may be configured to provide improved communication of the state of operation of the air conditioning system **100** to a remote human user and enabling more rapid human response to shutdown of the air conditioning system (e.g., manual cleaning of the condensate drain line **124**, requesting and/or performing maintenance on the air conditioning system **100**, re-setting and re-starting the air conditioning system **100**, etc.), thereby reducing or minimizing air conditioning system **100** downtime and thereby reducing or minimizing excessive warming of a structure being cooled by the air conditioning system **100**.

(257) Referring to FIG. **15B**, in some example embodiments, the electrical connection **1504** may comprise a second set of at least two wires of the drain cleaner apparatus **200** (at least partially comprising electrical connection **1152**) which are electrically connected (directly or indirectly) to the controller **210** and extend from the drain cleaner apparatus **200** (e.g., via opening **1150**) and which may each be connected (e.g., directly or indirectly, as part of a serial connection of wires and/or interfaces there between) to separate, respective wires extending from the air conditioning system **100** (e.g., from air handler **102**, from the controller **140**, from the float switch signal connection interface **148**, etc.) to establish the electrical connection **1504** as a second electrical circuit (e.g., where the electrical circuit of the electrical connection **1502** is referred to as a first electrical circuit). The drain cleaner apparatus **200** (e.g., controller **210**) may be configured to generate and transmit an electrical signal (e.g., electrical current), also referred to herein as a float switch signal, to the air conditioning system **100** (e.g., controller **140** via interface **148**) via the electrical connection **1504**, for example in response to the controller **210** determining that a float switch signal is received from the drain cleaner float switch **1610** via the first electrical circuit of the electrical connection **1502**. The electrical signal may be “transmitted” based on the drain cleaner apparatus **200** (e.g., the controller **210**) controlling (e.g., initiating, inhibiting, adjusting a magnitude and/or frequency of, etc.) a flow of electrical current in the second electrical circuit of the electrical connection **1504**, for example based on controller **210** operating a switch implemented by and/or included in the drain cleaner apparatus **200** to close or open the second electrical circuit of the electrical connection **1504**, respectively, based on the controller **210** selectively and/or adjustably controlling (e.g., initiating, inhibiting, adjusting a magnitude thereof, or the like) a supply of electrical power to the portion of the air conditioning system **100** (e.g., at

least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, etc.) via the second electrical circuit of the electrical connection **1504**, or the like. Such controlled (e.g., initiated, inhibited, adjusted, etc.) flow of electrical current in the second electrical circuitry may be referred to as a signal, command, signal or the like (e.g., an electrical signal, a float switch signal, or the like) transmitted to the portion of the air conditioning system **100** (e.g., at least a portion of the air handler **102**, the controller **140**, the float switch signal connection interface **148**, etc.). Such initiated, inhibited, or adjusted flow of electrical current in the second electrical circuit of the electrical connection **1504** that includes the portion of the air conditioning system **100** may be detected and processed by the portion of the air conditioning system **100** (e.g., the air handler **102**, the controller **140**, etc.) to be a receipt and/or detection of a float switch signal “transmitted” by the drain cleaner apparatus **200** to command a shutdown of at least a portion of at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) thereby causing at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) to shut down (e.g., based on operation of the controller **140**) in response to detection (e.g., by the controller **140**) of the signal that is “transmitted” by the drain cleaner apparatus **200** (e.g., by controller **210**). For example, the air handler **102** (e.g., the controller **140**) may include a sensor such as an ammeter, current sensor, or the like configured to detect and/or generate data which may be processed by the controller **140** to detect (e.g., generate signals which may be processed by the controller **140** to detect in response to) a presence and/or magnitude of electrical current in the second electrical circuit of the electrical connection **1504** (e.g., an electrical current at interface **148**). The controller **140** may respond to a determination of a presence or absence of electrical current (e.g., a presence or absence of current above a threshold magnitude) in the second electrical circuit of the electrical connection **1504**, a change in magnitude of electrical current in the second electrical circuit of the electrical connection **1504**, or the like, based on processing and/or detecting signals generated by the sensor to determine that an electrical signal (e.g., a float switch signal generated and transmitted by the drain cleaner apparatus **200** via electrical connection **1504**) is received. The air conditioning system **100** (e.g., the air handler **102**, the controller **140**, etc.) may be configured to shut down (e.g., shut off) at least a portion of the air conditioning system **100** (e.g., at least the air handler **102**) in response to determining (e.g., detecting, based on processing signals generated by the sensor) that the float switch signal is received (e.g., received at interface **148**) as a result of the drain cleaner apparatus **200** generating the electrical signal that is transmitted to the air conditioning system **100** and received (e.g., at interface **148**) via electrical connection **1504**.

(258) FIG. **16A** is a perspective top-front-right view of a float switch apparatus **800** according to some example embodiments. FIG. **16B** is a perspective bottom-rear-left view of the float switch apparatus **800** of FIG. **16A** according to some example embodiments. FIG. **16C** is a perspective cross-sectional view of the float switch apparatus **800** along cross-sectional view line XVIC-XVIC' of FIG. **16A** according to some example embodiments. FIG. **16D** is a plan cross-sectional view of the float switch apparatus **800** along cross-sectional view line XVIC-XVIC' of FIG. **16A** according to some example embodiments. FIG. **16E** is a plan top view of the float switch apparatus **800** of FIG. **16A** according to some example embodiments.

(259) It will be understood that the float switch apparatus **800** shown in FIGS. **16A-16E** may include any of the elements of any of the example embodiments and of the devices, apparatuses, or the like shown in any of the drawings and/or described herein.

(260) Referring to FIGS. **16A-16E**, in some example embodiments, the float switch apparatus **800** is configured to be coupled to the condensate drain line **124** (e.g., coupled to the opening **125** of the condensate drain line **124**). The float switch apparatus **800** may include a drain cleaner float switch **1610**. The drain cleaner float switch **1610** may include float switch circuitry **1618** (e.g., switch device, a bimetal switch device, processing circuitry, electrical circuitry, an electrical switch, or the like configured to generate a float switch signal in response to movement of the float **1612** along the shaft **1614**) in a housing including a shaft **1614** and a stop **1616**, and the drain cleaner float

switch **1610** may include a float **1612** configured to move axially along the longitudinal axis of the shaft **1614** to actuate the drain cleaner float switch **1610** to cause the circuitry **1618** to responsively generate a float switch signal. It will be understood that the drain cleaner float switch **1610** is not limited to the structure shown in FIGS. **16A-16D** and may include any known float switch (e.g., bimetal float switch).

(261) The drain cleaner float switch **1610** (e.g., the circuitry **1618** thereof) may be configured to be electrically coupled (e.g., electrically connected, which may include directly or indirectly electrically connected) to the drain cleaner apparatus **200** (e.g., via electrical connection **1502**, which may include one or more wires extending from the drain cleaner float switch **1610**, through the open enclosure **1608**, and out of the float switch apparatus **800** to electrically couple with the controller **210** of the drain cleaner apparatus **200**, for example based on coupling with a separate one or more wires extending from the drain cleaner apparatus **200**. The float switch apparatus **800** may be configured to couple with the opening **125** of the condensate drain line **124** to position the drain cleaner float switch **1610** in the condensate drain line **124** so that the drain cleaner float switch **1610** may be positioned within the condensate drain line **124** and configured be actuated to transmit a float switch signal to the drain cleaner apparatus **200** in response to a presence of fluid in the condensate drain line **124** which may cause the float **1612** to move (e.g., rise) axially in relation to the shaft **1614** to cause the drain cleaner float switch **1610** to actuate.

(262) Referring back to FIG. **15A**, the controller **210** of the drain cleaner apparatus **200** may be configured to transmit the actuator command signal to the actuator apparatus **900** to cause the actuator **910** to actuate **980** the air handler float switch **160** (to cause the air handler **102** to shut down) in response to the controller **210** receiving the float switch signal from the drain cleaner float switch **1610**.

(263) Referring again to FIGS. **16A-16E**, the float switch apparatus may include a support housing **1602** configured to couple with an opening **125** of the condensate drain line **124**. The support housing **1602** may include a cylindrical structure **1606** configured to extend into the condensate drain line **124** and may be configured to establish a friction fit with an inner surface to the condensate drain line **124** to hold the float switch apparatus **800** in place. The support housing **1602** may include an upper lip or ledge structure **1604** which may be configured to be outside the condensate drain line **124** and engaging the outer end of the opening **125** of the condensate drain line **124** in order to restrict the distance that the float switch apparatus **800** extends into the condensate drain line **124** and to improve ease of coupling and decoupling of the float switch apparatus **800** with the opening **125** of the condensate drain line **124**. As shown, the support housing **1602** may define a cup structure defining an interior open cylindrical enclosure **1608** and configured to fit into the opening **125** of the condensate drain line **124**. The support housing **1602** may comprise a single piece of material (e.g., plastic) defining the structures **1604** and **1606**. At least a portion of the drain cleaner float switch **1610**, for example at least the shaft **1614**, may be integrated into the support housing **1602** so that, for example, at least the shaft **1614** and the support housing **1602** are separate portions of the same piece of material (e.g., plastic material), but example embodiments are not limited thereto. It will be understood that the drain cleaner float switch **1610** may be referred to as being attached to the support housing **1602**, either as a separate element coupled via adhesion, friction fit, coupling, or the like or based on at least a part of the drain cleaner float switch **1610** being a part of a same piece of material defining at least a portion of the support housing **1602**.

(264) In some example embodiments, the float switch apparatus **800** may be configured to enable the drain cleaner apparatus **200** to supply the cleaning composition into the condensate drain line **124**, for example based on the support housing **1602** being configured to enable the dispenser conduit **290** to extend through the support housing **1602** to be in fluid communication with the condensate drain line **124** interior, but example embodiments are not limited thereto. Referring to FIGS. **16A-16E**, in some example embodiments, the float switch apparatus **800** may include a

supply conduit **1620**, which may be rigid (e.g., plastic) or flexible tube and may extend through the support housing **1602**. As shown, the supply conduit **1620** may have opposite first and second ends **1622** and **1624**. The first end **1622** of the supply conduit **1620** may be configured to be coupled with the second end **292** of the dispenser conduit **290** (e.g., via a clip connector, via friction fit between opposing surfaces of the first end **1622** and the second end **292**, or the like). The second end **1624** of the supply conduit **1620** may be configured to be in fluid communication with the condensate drain line **124** when the float switch apparatus **800** is coupled to the condensate drain line **124**, such that the supply conduit **1620** is configured to establish the fluid communication of the apparatus outlet **206** of the drain cleaner apparatus **200** with the condensate drain line **124** interior through the dispenser conduit **290** and the supply conduit **1620** coupled thereto. As shown, the supply conduit **1620** may penetrate through a thickness **1602t** of the support housing **1602** to extend from the open enclosure **1608** to the same lower side of the cup formed by the support housing **1602** as the drain cleaner float switch **1610** that is on an opposite side of the support housing **1602** from the open enclosure **1608**. In some example embodiments, the supply conduit **1620** and the support housing **1602** may be separate portions of a single piece of material (e.g., plastic material, which as described herein may include polyvinyl chloride or PVC material). (265) Still referring to FIGS. **16A-16E**, the drain cleaner float switch **1610** and the supply conduit **1620** are offset from a central axis **1690** of the support housing **1602**, for example offset in a direction that is perpendicular to the central axis **1690**. As shown, the drain cleaner float switch **1610** (e.g., at least the shaft **1614**) and the supply conduit **1620** may extend in parallel with each other and in parallel with the central axis **1690**. As a result, the float switch apparatus **800** may be configured to minimize or prevent any effect of the supply of cleaning composition to the condensate drain line **124** via the supply conduit **1620** on the operation (e.g., actuation) of the drain cleaner float switch **1610**.

(266) FIG. **17A** is a perspective top-front-right view of an actuator apparatus **900** according to some example embodiments. FIG. **17B** is a perspective bottom-rear-left view of the actuator apparatus **900** of FIG. **17A** according to some example embodiments. FIG. **17C** is a perspective bottom-rear-right view of the actuator apparatus **900** of FIG. **17A** according to some example embodiments.

(267) FIG. **18A** is a perspective top-front-right view of an actuator apparatus **900** according to some example embodiments. FIG. **18B** is a perspective cross-sectional view of the actuator apparatus **900** along cross-sectional view line XVIII B-XVIII B' of FIG. **18A** according to some example embodiments. FIG. **18C** is a plan cross-sectional view of the actuator apparatus **900** along cross-sectional view line XVIII B-XVIII B' of FIG. **18A** according to some example embodiments. FIG. **18D** is a perspective cross-sectional view of the actuator apparatus **900** along cross-sectional view line XVIII D-XVIII D' of FIG. **18A** according to some example embodiments. FIG. **18E** is a plan cross-sectional view of the actuator apparatus **900** along cross-sectional view line XVIII D-XVIII D' of FIG. **18A** according to some example embodiments.

(268) FIG. **19A** is a perspective top-front-right view of an actuator apparatus **900** according to some example embodiments. FIG. **19B** is a perspective cross-sectional view of the actuator apparatus **900** along cross-sectional view line XIX B-XIX B' of FIG. **19A** according to some example embodiments. FIG. **19C** is a perspective cross-sectional view of the actuator apparatus **900** along cross-sectional view line XIX C-XIX C' of FIG. **19A** according to some example embodiments.

(269) FIG. **20** is a perspective view of elements of the actuator apparatus **900** according to some example embodiments.

(270) FIG. **21A** is a perspective view of a containment apparatus **1730** according to some example embodiments. FIG. **21B** is a perspective cross-sectional view of the containment apparatus **1730** along cross-sectional view line XXIB-XXIB' of FIG. **21A** according to some example embodiments. FIG. **21C** is a perspective cross-sectional view of the containment apparatus **1730**

along cross-sectional view line XXIC-XXIC' of FIG. 21A according to some example embodiments. FIG. 22 is a perspective view of outer shells 1732 and hinge connection 1734 of a containment apparatus according to some example embodiments. FIG. 23A is a perspective view of an adaptor sleeve structure 1736 of a containment apparatus according to some example embodiments. FIG. 23B is a perspective cross-sectional view of the adaptor sleeve structure 1736 along cross-sectional view line XXIIIB-XXIIIB' of FIG. 23A according to some example embodiments. FIG. 24A is a perspective view of an adaptor sleeve structure 1736 of a containment apparatus according to some example embodiments. FIG. 24B is a perspective cross-sectional view of the adaptor sleeve structure 1736 along cross-sectional view line XXIVB-XXIVB' of FIG. 24A according to some example embodiments.

(271) FIG. 25A is a plan cross-sectional view of the actuator apparatus 900 along cross-sectional view line XVIIIB-XVIIIB' of FIG. 18A in which an air handler float switch 160 is positioned according to some example embodiments. FIG. 25B is a plan cross-sectional view of the actuator apparatus 900 along cross-sectional view line XVIIID-XVIIID' of FIG. 18A in which an air handler float switch 160 is positioned according to some example embodiments.

(272) It will be understood that the actuator apparatus 900 and any elements thereof shown in FIGS. 17A-25A and the air handler float switch 160 shown in FIGS. 25A to 25B may include any of the elements of any of the example embodiments and of the devices, apparatuses, or the like shown in any of the drawings and/or described herein.

(273) Referring to FIGS. 17A-19C, the actuator apparatus 900 may include a body housing 1702 including a side housing 1706, a bottom housing 1708, and a top housing 1728. In the illustrated example embodiments, the top housing 1728 and the side housing 1706 may be separate portions of a single piece of material (e.g., plastic material), but example embodiments are not limited thereto. As further shown, the actuator apparatus 900 may include a lid 1704 having a tab protrusion 1714 and which may be coupled to the body housing 1702 to cover the top housing 1728. The lid 1704 may be configured to be hingeably coupled to the body housing 1702 via a hinge connection 1712, but example embodiments are not limited thereto.

(274) Still referring to FIGS. 17A-19C, the actuator apparatus 900 may include the actuator 910 within the interior defined by the body housing 1702 (e.g., the side housing 1706, the bottom housing 1708, the top housing 1728, etc. As shown, the actuator 910 may include a servomotor 1762 coupled to a rotary gear 1764 (e.g., spur gear) (e.g., a servoactuator) and an actuator piston 1766 including at least a piston structure 1767 having linear gear teeth 1768 configured to engage the gear teeth of the rotary gear 1764. As shown, the servomotor 1762 may be configured to rotate the rotary gear 1764 in a clockwise or counterclockwise direction to cause the actuator piston 1766 to move along a first axis 1802 (e.g., move up or down linearly). The actuator apparatus 900 may include a support housing 1798 configured to fix the servomotor 1762 and rotary gear 1764 in place and to further laterally limit motion (perpendicular to the first axis 1802) of the piston structure 1767 of the actuator piston 1766 to ensure meshing of the respective gear teeth of the actuator piston 1766 and the rotary gear 1764.

(275) Still referring to at least FIGS. 17A-19C, the actuator apparatus 900 may include one or more connector interfaces configured to couple (e.g., attach) the actuator apparatus 900 to an external structure. For example, as shown, the actuator apparatus 900 may include two separate magnets 1770 within the interior of the actuator apparatus 900, isolated from direct exposure to the exterior of the actuator apparatus 900 by at least the side housing 1706, that serve as connector interfaces configured to magnetically couple (e.g., attach) the actuator apparatus 900 to a metal surface of an external structure (e.g., a metal surface of the housing 101 of the air handler 102) separately from the drain cleaner apparatus 200, thereby enabling the actuator apparatus 900 and the drain cleaner apparatus 200 to be adjustably and variably attached to one or more external structures at least partially independently of each other, thereby enabling improved flexibility of arrangement of the apparatuses in different environments.

(276) Still referring to FIGS. 17A-19C and further referring to FIGS. 25A-25B, the actuator apparatus **900** may be configured to receive an air handler float switch **160** of an air handler **102** as shown in FIG. 15A and to hold the air handler float switch **160** in place (e.g., hold at least the shaft parts **2502** and **2504** of the air handler float switch **160** in a fixed position) in relation to the actuator piston **1766**. As shown, the air handler float switch **160** may include upper and lower shaft parts **2502** and **2504**, float **2508**, and circuitry **2510** (e.g., a switch device, a bimetal switch device, processing circuitry, electrical circuitry including an electrical switch, etc.) configured to generate a float switch signal and transmit the float switch signal via the electrical connection **1506** to the air handler **102** (e.g., to the controller **140** thereof) to cause the air handler **102** to shut down (e.g., to cause the controller **140** to initiate shutdown of the air handler **102**), but it will be understood that example embodiments are not limited thereto. The actuator apparatus **900** may be configured to actuate the air handler float switch **160** based on causing a float **2508** of the air handler float switch **160** to move in relation to a remainder of the air handler float switch **160** (e.g., in relation to the shaft parts **2504** and **2502** and the circuitry **2510**) along the first axis **1802** based on the actuator piston **1766** being caused by the servomotor **1762** to move along the first axis **1802** to actuate the air handler float switch **160** to cause the circuitry **2510** to responsively generate a float switch signal. It will be understood that the air handler float switch **160** is not limited to the structure shown in FIGS. 25A to 25B and may include any known float switch (e.g., bimetal float switch). It will be understood that the actuator **910** may include other types of actuators, servoactuators, or the like, including for example a linear actuator.

(277) Still referring to FIGS. 17A-19C and FIGS. 25A-25B, the actuator apparatus **900** may include a cup structure **1780** that is coupled to the actuator **910**, for example coupled (e.g., directly or indirectly coupled) to an upper end of the actuator piston **1766**. For example, as shown in FIGS. 17A-19C and 25A-25B, the actuator piston **1766** may include a first spring **1772** coupled to the upper end of the piston structure **1767** and further coupled between the piston structure **1767** and the cup structure **1780**, such that the cup structure **1780** may be coupled to the upper end of the piston structure **1767** via a first spring **1772**. In some example embodiments, the cup structure **1780** may be understood to be part of the actuator piston **1766**. For example, in some example embodiments the first spring **1772** may be omitted and the cup structure **1780** may be coupled to (e.g., directly or indirectly coupled to, a part of a same piece of material as, etc.) an upper end of the piston structure **1767** so that the actuator piston **1766** excludes any first spring **1772** between the cup structure **1780** and the piston structure **1767** and the actuator piston **1766** includes the cup structure **1780** and the piston structure **1767**. It will be understood that any signal as described herein may be an electrical signal.

(278) The cup structure **1780** may define an open enclosure **1785** that is configured to accommodate at least a portion (e.g., lower shaft part **2504**) of the air handler float switch **160** and may define an upper surface **1782** or ridge configured to directly engage an underside (e.g., lower surface **2508s**) of the float **2508** of the air handler float switch **160**. The cup structure **1780** may be configured to engage the float **2508** to cause the float **2508** to move axially along the first axis **1802** based on movement of the actuator piston **1766** along the first axis **1802**, as the axial movement of the actuator piston **1766** may be transferred to the cup structure **1780** (e.g., via the piston structure **1767**, and in some example embodiments further via the first spring **1772**) and thus further transferred to the float **2508**. As described herein, the remainder of the air handler float switch **160** may be held in place while the float **2508** is free to be moved by the cup structure **1780**, so the movement of the float **2508** by the cup structure **1780** may cause the air handler float switch **160** to actuate in response thereto.

(279) Still referring to FIGS. 17A-19C and FIGS. 25A-25B, the actuator apparatus **900** may include a conduit structure **1750** having an inner surface **1750s** defining a conduit space **1792** extending along the first axis **1802** and having opposite first and second openings **1752** and **1754**. In some example embodiments as shown, the conduit structure may be a cylindrical structure that

may be a separate portion of a same piece of material as at least the top housing **1728** of the body housing **1702**, and the first opening **1752** may be an opening in the top housing **1728**, but example embodiments are not limited thereto. As shown, the conduit structure **1750** may extend along the first axis **1802** into the interior of the actuator apparatus **900** from the first opening **1752** at the top housing **1728** so that the second opening **1754** is an opening into an interior of the actuator apparatus **900** that is proximate to the actuator **910**.

(280) Referring to FIGS. **17A-19C**, **20**, and FIGS. **25A-25B**, the conduit structure **1750** may be configured to receive the air handler float switch into the conduit space **1792** (e.g., at least an upper region of the conduit space **1792** extending between the first opening **1752** and the inner ledge structure **1784** described below) through the first opening **1752**. The conduit structure **1750** may be further configured to receive at least the cup structure **1780** into the conduit space **1792** (e.g., at least a lower region of the conduit space **1792** extending between the second opening **1754** and the inner ledge structure **1784**) through the second opening **1754**. The inner ledge structure **1784**, extending into the conduit space **1792** from the inner surface **1750s** of the conduit structure **1750** and extending circumferentially around the inner surface **1750s**, may define the upper and lower regions of the conduit space **1792**. The cup structure **1780** may include protrusions **1796** that extend laterally underneath the ledge structure **1784** to vertically at least partially overlap the ledge structure **1784**. The outer diameter of the cup structure itself **1780** may be smaller than an inner diameter of an opening defined by the ledge structure **1784** so that the cup structure **1780** may be configured to at least partially move axially between the lower and upper regions of the conduit space **1792**, and the protrusions **1796** may be configured to engage the ledge structure **1784** to restrict the axial movement of the cup structure **1780** into the upper region of the conduit space **1792**.

(281) Still referring to FIGS. **17A-19C**, **20**, and FIGS. **25A-25B**, the enclosure **1785** of the cup structure **1780** and the first spring **1772** may be configured to enable the cup structure to move axially to compensate for varying lengths and/or shapes of the air handler float switch **160**. An air handler float switch **160** having a longer lower shaft part **2504** may contact the bottom surface of the cup structure **1780** partly defining the enclosure **1785** and push the cup structure **1780** downwards to compress the first spring **1772** while maintaining engagement between surfaces **1782** and **2508s**. A float switch **160** having a shorter lower shaft part **2504** may result in the cup structure **1780** moving axially at least partially into the upper region of the conduit space **1792** based on the spring force exerted by the first spring **1772** to maintain engagement between surfaces **1782** and **2508s**. As further shown, the actuator apparatus **900** may include a second spring **1774** coupled to an opposite end of the piston structure **1767** of the actuator piston **1766** in relation to the first spring **1772** and configured to be compressed between the piston structure **1767** of the actuator piston **1766** and the bottom housing **1708**, the second spring **1774** supported and held in place by support structure **1776**. The first and second springs **1772** and **1774** may collectively balance and/or adjust the axial position of the piston structure **1767** of the actuator piston **1766** in relation to the remainder of the actuator **910** under the cup structure **1780**. As a result of the above, the actuator apparatus **900** may be configured to accommodate air handler float switches **160** having various shapes, particularly various lengths along the first axis **1802**, based on compression or expansion of the at least one of the first or second springs **1772** or **1774** and the resultant axial movement of the cup structure **1780** independent of the operation of the actuator piston **1766** by the servomotor **1762** and the rotary gear **1764**.

(282) Still referring to FIGS. **17A-25B**, the actuator apparatus **900** may include a containment apparatus **1730** that may couple with the air handler float switch **160** and may further couple with a portion of the actuator apparatus **900** (e.g., the conduit structure **1750**) so that the containment apparatus **1730** holds the air handler float switch **160** in place (e.g., in place in relation to the actuator **910** and/or the cup structure **1780**, at least partially within the conduit space **1792** in relation to at least the cup structure **1780**, etc.). The containment apparatus **1730** may be

interchangeably referred to herein as a float switch holster, a float switch holster apparatus, a float switch holster device, a float switch holster assembly, a float switch holder, a float switch holder apparatus, a float switch holder device, a float switch holder assembly, a holder, a holster, a float switch cup, a float switch sheath, a float switch adaptor apparatus, a float switch adaptor device, a float switch adaptor assembly, an adaptor, or the like. While the containment apparatus **1730** cooperates with at least a portion of the actuator apparatus **900** (e.g., the conduit structure **1750**) to hold the air handler float switch **160** in place, the actuator **910** may be configured to cause the cup structure **1780** to move axially along the first axis **1802** to engage the float **2508** of the air handler float switch **160** and cause the float **2508** to move upwards along the first axis **1802** in relation to the remainder of the air handler float switch **160**, thereby actuating the air handler float switch **160**. The containment apparatus **1730** may include one or more outer surfaces (e.g., outer surface **1732os**) configured to engage in a friction fit with one or more opposing surfaces of the actuator apparatus **900** (e.g., inner surface **1750s** of the conduit structure **1750**) in order to hold the containment apparatus **1730** and the coupled air handler float switch **160** (e.g., the upper and lower shaft parts **2502** and **2504**) in place despite the actuator **910** causing the float **2508** to move upwards.

(283) In some example embodiments, the containment apparatus **1730** and the conduit structure **1750** may collectively define one or more support structures **1760** configured to position the air handler float switch **160** in relation to the actuator **910**. It will be understood that the one or more support structures **1760** may include any one or more structures configured to hold the air handler float switch **160** in place while remaining electrically coupled to the air handler **102** via electrical connection **1506** and where the actuator **910** is configured to actuate the air handler float switch **160**.

(284) In some example embodiments, the containment apparatus **1730** may be omitted from the actuator apparatus **900**, such that the one or more support structures **1760** may exclude the containment apparatus **1730**. For example, in some example embodiments the actuator apparatus **900** may include a locking mechanism, a latch mechanism, or the like (e.g., a spring-loaded locking mechanism) configured to engage the air handler float switch **160** held in the conduit space **1792** at least partially defined by the conduit structure **1750** in order to hold the air handler float switch **160** in place in relation to the actuator **910**, thereby reducing, minimizing, or preventing upwards movement of the air handler float switch **160** (e.g., at least upward movement of the shaft part and circuitry (e.g., switch circuitry) thereof) in response to the actuator **910** causing a float of the air handler float switch **160** to move upwards. In some example embodiments, the conduit structure **1750** may include a structure, including for example a ledge structure **1784** that is configured to structurally support a weight of the air handler float switch **160** resting on the structure at least partially in the conduit space **1792**, where the actuator **910** is configured to apply an upwards force to the float of the air handler float switch **160** that does not transmit sufficient force to the remainder of the air handler float switch **160** to overcome the weight of the air handler float switch **160**, such that the weight of the air handler float switch **160** keeps the air handler float switch in place in the actuator apparatus **900** despite the actuator **910** causing the float of the air handler float switch **160** to move in relation to the actuator apparatus **900**.

(285) As shown in FIGS. **17A-25B**, the containment apparatus **1730** may include at least two outer shells **1732** that are configured to reversibly couple together (e.g., reversibly open and close) in order to enable reversibly enclosing and/or defining an inner conduit **1790** between opposing inner surfaces **1732is** of the outer shells **1732** and in which at least an upper portion (e.g., upper shaft part **2502**) of the air handler float switch **160** may be held by the containment apparatus **1730**.

(286) As shown, the outer shells **1732** may be coupled together via a hinge connection **1734** which includes a pin extending through hinge connection structures of the outer shells **1732**, so that the outer shells **1732** may open and close in a clamshell manner, to thereby open and close the inner conduit **1790** in a clamshell manner. As shown, the outer shells **1732** may collectively define a cup

structure having outer surfaces **1732os** configured to engage in a friction fit with the inner surface **1750s** of the conduit structure **1750** to hold the containment apparatus **1730** in place to resist upwards motion in response to the cup structure **1780** moving the float **2508** upwards.

(287) As further shown, the cup structure formed by the closed outer shells **1732** may have an inner conduit **1790** that is defined by the respective, opposing inner surfaces **1732is** of the outer shells. As shown, the containment apparatus **1730** is configured to enclose (e.g., at least horizontally and/or radially enclose) at least a portion of the air handler float switch **160** within the inner conduit **1790**.

(288) Still referring to FIGS. **17A-25B**, the containment apparatus **1730** may include an adaptor sleeve structure **1736** configured to be held in place at least partially within the inner conduit **1790** when the two outer shells **1732** are coupled together (e.g., closed in a clamshell manner to define the inner conduit **1790** as a cylindrical conduit). The adaptor sleeve structure **1736** may be interchangeably referred to herein as an adaptor sleeve, an adaptor sleeve device, or the like. The adaptor sleeve structure **1736** may include a central shaft structure **2301** extending through the inner conduit **1790** and opposing upper and lower flange structures **2302** and **2304** configured to extend over respective upper and lower ends of the inner conduit **1790** to hold the adaptor sleeve structure **1736** in place in relation to the inner conduit **1790**. As shown, the adaptor sleeve structure **1736** may have one or more inner surfaces **1736is** that define an inner conduit **1740** configured to accommodate at least an upper portion (e.g., upper shaft part **2502**) of the air handler float switch **160** and to engage said upper portion to hold the air handler float switch **160** in place in relation to the containment apparatus **1730**. As a result, the adaptor sleeve structure **1736** may be understood to at last partially fill an annular space between the portion of the air handler float switch **160** (e.g., the upper shaft part **2502**) and the inner surfaces **1736is** defining the inner conduit **1790** to hold the air handler float switch **160** in place in relation to the inner conduit **1790**.

(289) The adaptor sleeve structure **1736** may comprise a flexible material, such as silicone, rubber, or the like and may be configured to grip outer surfaces of at least the portion of the air handler float switch **160** engaged by the adaptor sleeve structure **1736** to hold the air handler float switch **160** in place. As further shown, the adaptor sleeve structure **1736** may define an upper conduit **1738** that is configured to enable one or more wires comprising the electrical connection **1506** to extend from the air handler float switch **160** (e.g., the circuitry **2510** thereof) out of the containment apparatus **1730** and further out to the actuator apparatus **900** to the air handler **102**.

(290) Referring to at least FIGS. **23A-24B**, the adaptor sleeve structure **1736** may be a single piece of flexible material having a seam **2310** which may be opened to expose the interior of the conduits **1740** and **1738** to enable at least a portion of the air handler float switch **160** (e.g., upper shaft part **2502**) to be inserted into the exposed conduit **1740** and to enable at least a portion of one or more wires at least partially comprising the electrical connection **1506** between the air handler float switch **160** and the air handler **102** to be inserted into at least the exposed conduit **1738** without requiring disconnection of the one or more wires from either the air handler float switch **160** or the air handler **102**, and the flexible material comprising the adaptor sleeve structure **1736** may then be extended to enclose the portion of the air handler float switch **160** and the portion of the one or more wires to close the seam **2310** as shown in at least FIGS. **23A**, **24A**, and **25A-25B**. Based on enabling the air handler float switch **160** to be inserted into at least the exposed conduit **1738** without requiring disconnection of the one or more wires from either the air handler float switch **160** or the air handler **102**, installation and positioning of the air handler float switch **160** into the actuator apparatus **900** to configure the actuator apparatus **900** to actuate **980** the air handler float switch **160** may be simplified, and reliability of the electrical connection **1506** may be protected from interruption or damage due to such disconnection.

(291) In some example embodiments, the portion of the air handler float switch **160** may be inserted into and/or removed from the conduit **1740** of the adaptor sleeve structure **1736** via the opening in the bottom flange **2304**.

(292) Still referring to FIGS. 17A-25B, the body housing **1702** (e.g., the bottom housing **1708** as shown) may define an opening **1710** into the interior of the actuator apparatus **900** through which an electrical connection **1504** (e.g., including one or more wires) between the actuator apparatus **900** (e.g., the actuator **910**) and the drain cleaner apparatus **200** may extend to electrically couple at least the actuator **910** with the drain cleaner apparatus **200** and to configure the actuator **910** to be controlled by a controller **210** of the drain cleaner apparatus **200**. As further shown, the lid **1704** may have a ridge **1720** defining a gap **1722** therethrough and the top housing **1728** may have a ridge **1724** defining a gap **1726** therethrough, where the ridges **1720** and **1724** are configured to align the gaps **1722** and **1726** to collectively define an opening **1718** when the lid **1704** is coupled to the body housing **27016** to cover the top housing **1728**. The opening **1718** may enable communication between a space between the top housing **1728** and the lid **1704** and an exterior of the actuator apparatus **900** when the lid **1704** is closed, thereby enabling an electrical connection **1506** (e.g., including one or more wires) between the air handler float switch **160** held in position within the actuator apparatus **900** and the air handler **102** to extend from the air handler float switch **160**, out of the containment apparatus **1730** via the upper conduit **1738**, and further out of the actuator apparatus **900** via the opening **1718** to electrically couple at least the air handler float switch **160** with the air handler **102** (e.g., the controller **140** thereof) even when the lid **1704** is closed and to configure the air handler float switch **160** to transmit a float switch signal to the air handler **102** to cause the air handler **102** to shut off in response to the air handler float switch **160** being actuated by the actuator **910** of the actuator apparatus **900**.

(293) Referring now to FIGS. 23A-24B, the adaptor sleeve structure **1736** may have different shapes to accommodate different types, shapes, etc. of air handler float switches **160** therein. For example, as shown in FIGS. 23A-23B, the adaptor sleeve structure **1736** may include a lower flange structure **2304** with a cutout **2308** configured to accommodate larger-diameter portions of an air handler float switch **160** below the upper shaft part **2502** and with an upper flange structure **2302** having a partial cutout **2306** which may improve routing of one or more wires of the electrical connection **1506** out of the upper conduit **1738** and thus out of the containment apparatus **1730**. Additionally, the inner surfaces **1736is** of the adaptor sleeve structure **1736** may define an at least partially conical inner conduit **1740** to accommodate a particularly-shaped upper shaft part **2502** of the air handler float switch **160**. In another example, as shown in FIGS. 24A-24B, the adaptor sleeve structure **1736** may have upper and lower flange structures **2302** and **2304** and an inner surface **1736is** that collectively define a cylindrical inner conduit **1740** extending entirely between the top and bottom ends of the adaptor sleeve structure **1736** to thereby accommodate a differently-shaped air handler float switch **160** than the adaptor sleeve structure **1736** shown in at least FIGS. 23A-23B.

(294) Referring back to FIG. 15A, while FIG. 15A illustrates a system **2000** which includes the drain cleaner apparatus **200**, the actuator apparatus **900** and the float switch apparatus **800**, it will be understood that example embodiments are not limited thereto. For example, in some example embodiments the float switch apparatus **800** may be omitted from the system **2000**, and/or the drain cleaner apparatus **200** (e.g., the controller **210** thereof) may be configured to transmit the actuator command signal to the actuator apparatus, via electrical connection **1504**, in response to receiving an actuator command via a user interface (e.g., **1182**) of the drain cleaner apparatus **200**, a network communication link to a remote computing device via a network communication interface **224** of the drain cleaner apparatus **200**, or the like.

(295) FIG. 26A is a perspective top-front-left view of a drain cleaner apparatus system **2600** according to some example embodiments. FIG. 26B is a perspective bottom-rear-left view of the drain cleaner apparatus system **2600** of FIG. 26A according to some example embodiments. FIG. 26C is a perspective bottom-rear-left view of the drain cleaner apparatus system **2600** of FIG. 26A according to some example embodiments. FIG. 26D is a perspective view of an actuator holster structure **2602** according to some example embodiments. FIG. 26E is a perspective cross-sectional

view of the actuator holster along cross-sectional view line XXVIE-XXVIE' in FIG. 26D according to some example embodiments.

(296) It will be understood that the drain cleaner apparatus **200**, cartridge **300**, structure connector **220**, and the like shown in FIGS. 26A-26C may include any of the elements of any of the example embodiments of the drain cleaner apparatus **200**, cartridge **300**, structure connector **220**, and the like shown in any of the drawings and/or described herein.

(297) Referring to FIGS. 26A-26E, in some example embodiments, at least a portion of the actuator apparatus **900**, for example at least the actuator **910**, may be accommodated in an actuator holster structure **2602** which can be physically coupled to the drain cleaner apparatus **200** (e.g., engaged with at least a portion of the housing **201** of the drain cleaner apparatus **200**), where the actuator holster structure **2602** may further accommodate an air handler float switch in a position to be engaged and actuated by at least the actuator **910** further held in the actuator holster structure **2602**. As a result, the actuator **910** may be configured to actuate **980** an air handler float switch **160**, so that the air handler float switch **160** transmits a signal via electrical connection **1506** to cause the air handler **102** to shut down, based on a signal received at the actuator **910** via an electrical connection **1504** with the drain cleaner apparatus **200**.

(298) As shown the actuator holster structure **2602** may include a structure having an inner surface **2610s** defining a cylindrical conduit **2610** extending between opposite upper and lower ends **2612** and **2614**. One or both of the upper end **2612** or the lower end **2614** may be an opening exposing the cylindrical conduit **2610** to an exterior of the actuator holster structure **2602** or a closed end of the cylindrical conduit **2610**. As shown, at least a portion of the actuator **910** may be accommodated in a lower portion of the cylindrical conduit **2610**, and at least a portion of the air handler float switch **160** may be held in place in the upper portion of the cylindrical conduit **2610** so that the actuator holster structure **2602** holds the air handler float switch **160** in place in relation to the actuator **910**, to enable the actuator **910** to actuate **980** the air handler float switch **160**.

(299) As further shown, the actuator holster structure **2602** may include a connector structure **2616** that is configured to engage a complementary connector structure **2618** of the drain cleaner apparatus **200**. The connector structure **2616** may include a male connector structure (e.g., a male flange structure, tab, etc.) and the complementary connector structure **2618** may include a female connector structure (e.g., a female flange structure, slot, etc.) configured to slidably engage with the connector structure **2616** to couple the actuator holster structure **2602** with the drain cleaner apparatus **200**. For example, the connector structures **2616** and **2618** may be configured to establish a friction fit between the drain cleaner apparatus **200** and the actuator holster structure **2602** to hold at least the actuator **910** and the air handler float switch **160** in place in relation to the drain cleaner apparatus **200**. As shown, the actuator holster structure **2602**, including the connector structure **2616** and the various structures defining the inner surface **2610s** and the upper and lower ends **2612** and **2614** may be separate parts of a single piece of material (e.g., plastic material). As further shown, the complementary connector structure **2618** may be partially or entirely defined by one or more parts of the housing **201** of the drain cleaner apparatus **200**, such as being defined by separate parts of the base housing **1106** and the side housing **1104**.

(300) FIG. 27 is a perspective top-front-right view of a drain cleaner apparatus system **2700** according to some example embodiments. It will be understood that the drain cleaner apparatus **200** shown in FIG. 27 may include any of the elements of any of the example embodiments of the drain cleaner apparatus shown in any of the drawings and/or described herein. It will be understood that the cartridge **300** shown in FIG. 27 may include any of the elements of any of the example embodiments of the cartridge shown in any of the drawings and/or described herein.

(301) Referring to FIG. 27, in some example embodiments, the drain cleaner apparatus **200** may be configured to couple with cartridges **300** having various different cartridge reservoir **304** volumes. For example, while the cartridge **300** shown in at least FIGS. 11A-11E and 13A-13D includes a cartridge reservoir **304** having a first particular volume (e.g., 36 oz), the cartridge **300** shown in

FIG. 127 may have a different, second particular volume (e.g., 72 oz), and the drain cleaner apparatus 200 may be configured to couple (e.g., detachably couple) with either of the cartridges 300 having a first or second volume. As described herein, the controller 210 of the drain cleaner apparatus 200 may be configured to adjust a particular counter value for continuing the dispenser device 204 actuations associated with depletion of the cartridge reservoir 304 to accommodate the drain cleaner apparatus 200 being coupled to different-volume cartridges 300, such that the drain cleaner apparatus 200 may be configured to interchangeably couple with various cartridges 300 having different cartridge reservoir 304 volumes.

(302) Example embodiments have been disclosed herein; it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

Claims

1. A system for dispensing a cleaning composition into a condensate drain line of an air handler of an air conditioning system, the system comprising: a first apparatus configured to dispense the cleaning composition, the first apparatus including an apparatus outlet in fluid communication with an exterior of the first apparatus, a pump configured to operate to selectively dispense an amount of the cleaning composition through the apparatus outlet, a power supply configured to supply power supply electrical power, the power supply including a battery and a wired power connection, and a controller configured to operate the pump based on the power supply electrical power from the power supply to cause the amount of the cleaning composition to be dispensed through the apparatus outlet; and a dispenser conduit having a first end and a second end that is opposite to the first end, the dispenser conduit coupled at the first end to the apparatus outlet and in fluid communication at the second end with the condensate drain line, such that the apparatus outlet of the first apparatus is in fluid communication with an interior of the condensate drain line through at least the dispenser conduit, wherein the wired power connection is electrically coupled to the air handler and is configured to receive air handler electrical power from the air handler.
2. The system of claim 1, wherein the power supply includes a charging circuit, the charging circuit configured to recharge the battery from the wired power connection based on the air handler electrical power received from the air handler at the wired power connection.
3. The system of claim 1, wherein the power supply is configured to enable the battery to supply the power supply electrical power, such that the controller operates the pump based on the power supply electrical power supplied from the battery, based on an absence of the air handler electrical power being received from the air handler at the wired power connection.
4. The system of claim 1, wherein the dispenser conduit is a flexible tube that is configured to enable the first apparatus to be adjustably positioned in relation to the condensate drain line, the system further includes a second apparatus coupled to the condensate drain line, the second apparatus including a rigid cylindrical structure extending at least into an opening of the condensate drain line, and the flexible tube has a first flexible tube end and an opposite second flexible tube end, the first flexible tube end coupled to the apparatus outlet of the first apparatus, the opposite second flexible tube end coupled to the second apparatus, such that the opposite second flexible tube end is in fluid communication with the interior of the condensate drain line through the second apparatus, and the apparatus outlet of the first apparatus is in fluid communication with the condensate drain line through at least the flexible tube and the rigid cylindrical structure.
5. The system of claim 1, further comprising: a container that is external to a housing of the first apparatus, the housing at least partially enclosing the controller and the pump, the container having a reservoir configured to hold the cleaning composition, wherein a port that is exposed at an outer surface of the housing of the first apparatus is coupled to the container that is external to the

housing of the first apparatus such that the reservoir of the container is in fluid communication with the pump.

6. A method, comprising: providing a first apparatus configured to dispense a cleaning composition, the first apparatus including an apparatus outlet in fluid communication with an exterior of the first apparatus, a pump configured to operate to selectively dispense an amount of the cleaning composition through the apparatus outlet, a power supply configured to supply power supply electrical power, the power supply including a battery and a wired power connection, and a controller configured to operate the pump based on the power supply electrical power from the power supply to cause the amount of the cleaning composition to be dispensed through the apparatus outlet; coupling a first end of a dispenser conduit to the apparatus outlet and positioning a second end of the dispenser conduit in fluid communication with a condensate drain line of an air conditioning system, such that the apparatus outlet of the first apparatus is in fluid communication with an interior of the condensate drain line through at least the dispenser conduit; and electrically coupling the wired power connection of the first apparatus to an air handler of the air conditioning system, the wired power connection configured to receive air handler electrical power from the air handler.

7. The method of claim 6, wherein the power supply includes a charging circuit, and the method further includes the charging circuit recharging the battery from the wired power connection based on the air handler electrical power received from the air handler at the wired power connection.

8. The method of claim 6, further comprising supplying the power supply electrical power from the battery such that the controller operates the pump based on the power supply electrical power supplied from the battery based on an absence of the air handler electrical power being received from the air handler at the wired power connection.

9. The method of claim 6, wherein the dispenser conduit is a flexible tube that is configured to enable the first apparatus to be adjustably positioned in relation to the condensate drain line, the flexible tube having a first flexible tube end and an opposite second flexible tube end, and the method further includes coupling a second apparatus with the condensate drain line, the second apparatus including a rigid cylindrical structure, the coupling the second apparatus with the condensate drain line including causing the rigid cylindrical structure to extend at least into an opening of the condensate drain line, and coupling the first flexible tube end to the apparatus outlet of the first apparatus and coupling the opposite second flexible tube end to the second apparatus, such that the opposite second flexible tube end is in fluid communication with the interior of the condensate drain line through the second apparatus, and the apparatus outlet of the first apparatus is in fluid communication with the condensate drain line through at least the flexible tube and the rigid cylindrical structure.

10. The method of claim 6, further comprising: coupling a port that is exposed at an outer surface of a housing of the first apparatus to a container that is external to the housing of the first apparatus and has a reservoir configured to hold the cleaning composition, such that the reservoir of the container is in fluid communication with the pump, the housing at least partially enclosing the controller and the pump.

11. A system for dispensing a cleaning composition into a condensate drain line of an air handler of an air conditioning system, the system comprising: a first apparatus configured to dispense the cleaning composition, the first apparatus including an apparatus outlet in fluid communication with an exterior of the first apparatus, a pump configured to operate to selectively dispense an amount of the cleaning composition through the apparatus outlet, and a controller configured to operate the pump to cause the amount of the cleaning composition to be dispensed through the apparatus outlet; a dispenser conduit having a first end and a second end, the dispenser conduit coupled at the first end to the apparatus outlet of the first apparatus, wherein the dispenser conduit is a flexible tube that is configured to enable the first apparatus to be adjustably positioned in relation to an opening of the condensate drain line; and a second apparatus coupled with the condensate drain

line, the second apparatus including a rigid cylindrical structure extending at least into the opening of the condensate drain line, the second apparatus coupled with the second end of the dispenser conduit to establish fluid communication between the second end of the dispenser conduit and an interior of the condensate drain line through the second apparatus, such that the apparatus outlet of the first apparatus is in fluid communication with the condensate drain line through at least the flexible tube and the rigid cylindrical structure, wherein the flexible tube has a first flexible tube end and an opposite second flexible tube end, the first flexible tube end coupled to the apparatus outlet of the first apparatus, the opposite second flexible tube end coupled to the second apparatus, such that the opposite second flexible tube end is in fluid communication with the interior of the condensate drain line through the second apparatus, and the apparatus outlet of the first apparatus is in fluid communication with the condensate drain line through at least the flexible tube and the rigid cylindrical structure.

12. The system of claim 11, further comprising: a container that is external to a housing of the first apparatus, the housing at least partially enclosing the controller and the pump, the container having a reservoir configured to hold the cleaning composition, wherein a port that is exposed at an outer surface of the housing of the first apparatus is coupled with the container that is external to the housing of the first apparatus such that the reservoir of the container is in fluid communication with the pump.

13. A method, comprising: providing a first apparatus configured to dispense a cleaning composition, the first apparatus including an apparatus outlet in fluid communication with an exterior of the first apparatus, a pump configured to operate to selectively dispense an amount of the cleaning composition through the apparatus outlet, and a controller configured to operate the pump to cause the amount of the cleaning composition to be dispensed through the apparatus outlet; coupling a second apparatus with a condensate drain line of an air handler of an air conditioning system, the second apparatus including a rigid cylindrical structure, the coupling the second apparatus with the condensate drain line including causing the rigid cylindrical structure to extend at least into an opening of the condensate drain line; and coupling a first flexible tube end of a flexible tube to the apparatus outlet of the first apparatus and coupling an opposite second flexible tube end of the flexible tube to the second apparatus, such that the opposite second flexible tube end of the flexible tube is in fluid communication with an interior of the condensate drain line through the second apparatus, and the apparatus outlet of the first apparatus is in fluid communication with the condensate drain line through at least the flexible tube and the rigid cylindrical structure.

14. The method of claim 13, further comprising: coupling a port that is exposed at an outer surface of a housing of the first apparatus with a container that is external to the housing of the first apparatus and has a reservoir configured to hold the cleaning composition, such that the reservoir of the container is in fluid communication with the pump, the housing at least partially enclosing the controller and the pump.
